CHAPTER 1

THE PROBLEM AND ITS SETTINGS

Background of the Study

Technology changes rapidly every year, and its pace cannot be controlled. These changes, including the emergence of Artificial Intelligence (AI), affected every industry in different ways; therefore, adaptation became necessary to keep up with these developments. Based on my 15 years of experience in the BPO industry, it was observed that HR departments often used manual, spreadsheet-based methods to record, monitor, and evaluate applicants.

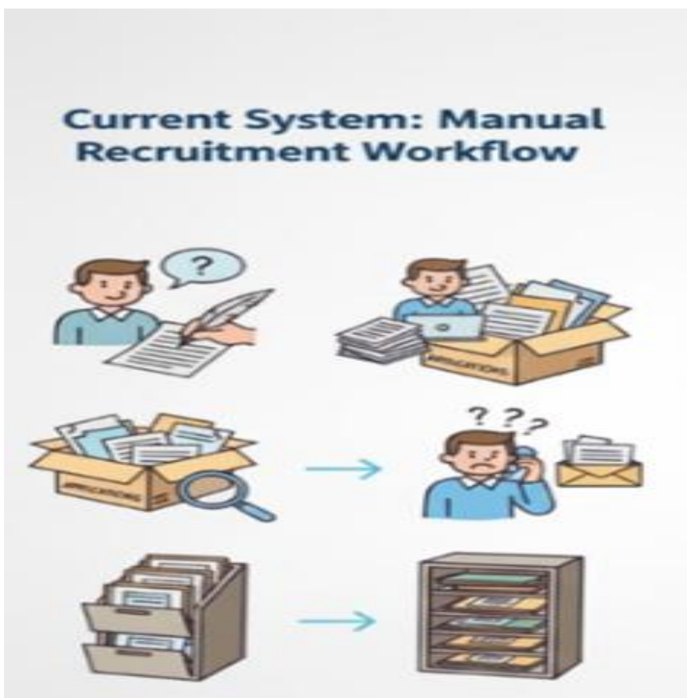


Figure 1. Current manual system on our HR dept

Before the new system, recruitment relied on multiple Excel and Google Sheets files managed by different HR staff, which sometimes caused inconsistent data entry, duplicate records, and made it hard to combine information for reports and decisions. When applicants came on-site, they filled out paper forms, and HR staff re-encoded them. Applicants from online sources like Indeed or job ads were also processed through recruitment email inboxes, and HR staff had to manually enter their data into tracking files.

This manual and scattered process led to several problems. Without a central database, it was hard to keep applicant information accurate, consistent, and secure. The high number of applicants in the BPO industry also caused delays in screening and scheduling interviews. Because there was no single monitoring system, HR had difficulty in tracking applicant status, interviews, deployment, and recruiters' performance in real time. As a result, HR managers struggled to create reports and spot hiring trends needed to improve efficiency.

Recent findings demonstrated that digital HRM systems significantly increased HR efficiency in organizations undergoing digital transformation (Mahmoud et al., 2025), which further supported the need to replace manual recruitment workflows with automated, data-driven systems. There was a greater need for a streamlined, data-driven HR recruitment process. Recent research also highlighted that digital HR transformation went beyond simple digitization of tasks—it required structural changes, modernized workflows, and enhanced information systems to support more efficient decision-making (Bansal, 2023). These emerging trends showed that many organizations were adopting role-based access systems, interactive dashboards, and secure web applications to increase accountability, transparency, and efficiency in HR operations.

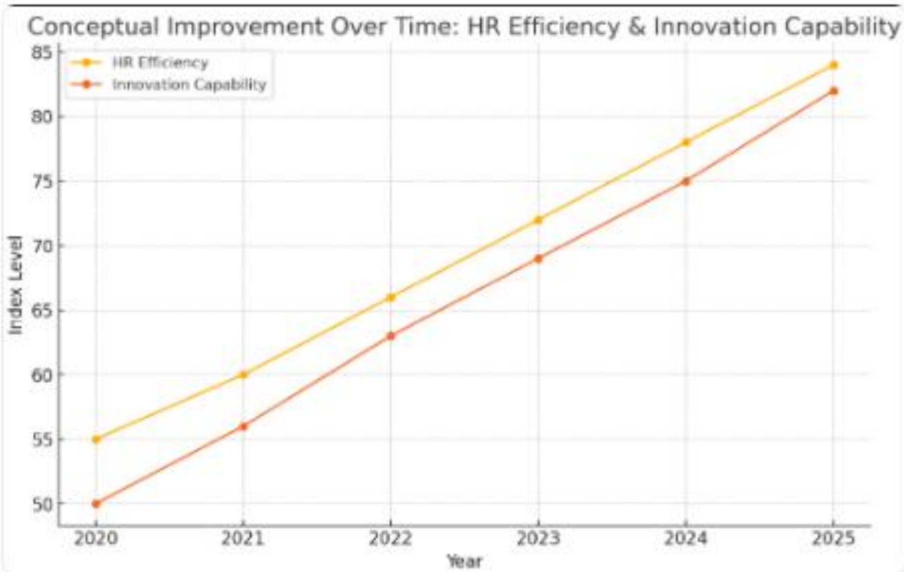


Figure 2. Conceptual improvement over time using digital application

To address these challenges, this research set out to design and build a Three-Tier Web-Based Recruitment, Interviewer dashboard and Applicant Management System (R-AMS) with an Applicant User Web Form and an HR Recruitment Management System, both connected to a centralized database. The intention of the project was to create an innovative, secure, and efficient recruitment tool that reduced manual work, kept data accurate, and helped HR staff make timely, informed decisions. The study also sought to qualitatively understand user experience, workflow improvements, and system feasibility within an actual HR operational environment.

Objectives of the Study

The general objective of this study was to design and develop a secure, database-driven Web-based Recruitment System consisting of an Applicant Web form, HR Administration Dashboard for end-to-end applicant management, and a dedicated Interviewer Dashboard for streamlined candidate evaluation. This enabled HR personnel to manage, update, extract, and visualize recruitment information within a role-based access environment using modern web technologies.

Specifically, this study aimed to:

- Developed an Intelligent Applicant Web Form using HTML, CSS, and JavaScript that captured applicant data and implemented an automated pre-screening algorithm to calculate initial qualification scores based on education, experience, and skills in real-time.
- Developed an HR Management System using JavaScript, PHP REST APIs, and MySQL with full CRUD functionalities for applicant records, interview schedules, and recruitment statuses to centralize data management.
- Implemented a Role-Based Access Control (RBAC) system using PHP sessions to strictly separate the environments for HR Managers, HR Staff, and Interviewers, ensuring that users were securely redirected to their specific dashboards upon login.
- Developed a Dynamic HR Dashboard featuring an "Intelligent Smart Filter" mechanism that allowed users to perform multi-criteria filtering and integrated a search function for specific fields, with bulk actions such as "Select All" for efficient record handling.
- Developed a customizable HR table interface per user that allowed users to add or remove specific fields based on their preferences.

- Implemented an Appointment Scheduling and Notification Module using PHP, JavaScript, and MySQL that prompted HR personnel about upcoming or due interviews, ensuring timely candidate processing.
- Developed a Secure Interviewer Dashboard that provided assigned personnel with a restricted, real-time view of assigned interview schedules, secure access to candidate resumes, and a standardized interface for submitting ratings (1–5 stars) and feedback.
- Developed a Robust Bulk Import Module utilizing PapaParse.js that parsed CSV files, validated data integrity, and automatically applied pre-screening scores to imported records to maintain consistency with individual or manual entries submitted through the recruitment form.
- Integrated Advanced Data Extraction Features that allowed HR users to generate reports through Custom CSV Template Mapping, enabling the system to adapt to various reporting formats without requiring code modifications.
- Designed a Custom Analytics Suite using Chart.js that visualized key performance indicators (KPIs) such as deployment trends, top application sources, recruiter performance, and other relevant metrics, with interactive date-range filtering.
- Implemented Strong Security Protocols, including SQL injection protection through prepared statements, encrypted password hashing (e.g., password_hash), secure session management, and role-based access restrictions.
- Developed a Resume Management System that supported file uploads (PDF/Docs) to Google Drive, ensuring flexible document viewing for interviewers.

- Integrated a Comprehensive Audit Trail that logged all critical system actions (such as logins, updates, exports, and bulk uploads) to ensure accountability and data integrity.
- Evaluated the System based on ISO/IEC 25010 software quality standards, specifically focusing on functional suitability, performance efficiency, usability, and security through user acceptance testing with HR personnel.

Technologies Included:

Category	**Technologies / Tools**
Frontend Frameworks & Design	• HTML • TailwindCSS (Utility-first styling framework) • Google Fonts ("Inter" typeface) • FontAwesome (Iconography for buttons and dashboards)
Client-Side Scripting & Libraries	• JavaScript (Core logic) • SweetAlert2 (Interactive modal alerts and confirmations) • Chart.js (Data visualization) • Chart.js Adapter Date-fns (Time-series support for charts)

	• ChartDataLabels (Data labels inside charts) • Litepicker (Date range selection) • PapaParse.js (CSV parsing for bulk uploads) • Choices.js (Enhanced dropdown menus)
Backend Development	• PHP (Server-side scripting) • MySQLi & PDO (Database drivers for secure SQL queries)
Database Management	• Server Intel Xeon - MySQL (Relational Database Management System) • SQL (Structured Query Language for complex joins and queries)
Security Mechanisms	• **Bcrypt (password_hash)** (Password encryption) • Prepared Statements (SQL Injection prevention) • HTML Entity Escaping (XSS prevention)
API & Communication	• RESTful API Architecture (Endpoint structure)

	• Fetch API (Asynchronous HTTP requests) • JSON (Data interchange format)
Development Environment	• XAMPP / WAMP (Local Web Server - Apache/MySQL) • Visual Studio Code (Integrated Development Environment) • Google Chrome DevTools (Debugging and testing)

Scope and Limitations of the Study

The study focused on the design and development of a secure, database-driven Web-Based Recruitment and Applicant Management System for the Human Resources (HR) department of the organization. The system operated within the company’s local intranet environment, addressing the need to automate the end-to-end recruitment workflow and centralize applicant records, and replacing manual data handling. The application was intended for use by HR Managers, HR Staff, and designated Interviewers(Managers) within the company. It was designed to be compatible with standard office workstations and modern web browsers (Chrome, Edge, Firefox, Safari), providing a secure and consistent user experience strictly within the internal network infrastructure to prioritize data confidentiality.

The application served as a centralized command center for recruitment operations, consolidating applicant data, updating the status pipeline, interview schedules, and evaluation metrics. It ensured that authorized personnel had real-time access to critical hiring information and automated tools to guide decision-making processes.

Key features of the system included:

- An automated pre-screening algorithm that instantly calculated qualification scores and statuses based on education, experience, and skills upon application submission.
- A Role-Based Access Control (RBAC) mechanism that strictly separated the administrative HR dashboard from the Interviewer portal.
- An intelligent dashboard with "Smart Filtering" and visual analytics to track KPIs such as deployment trends, applicant volume, and recruiter performance in real-time.
- A Customizable HR table interface per user, which allowed the user to add or remove specific fields based on preference.
- A robust Bulk Import and Custom Export module, utilizing PapaParse.js to validate CSV uploads and map database records to user-defined reporting templates.
- A dedicated Interviewer Portal that allowed evaluators to view assigned schedules, securely access resumes, and submit standardized 5-star ratings and feedback.

The development of the application utilized modern web technologies including HTML, TailwindCSS, and JavaScript for the interactive front-end, PHP for backend REST API logic, and MySQL for centralized database management. Security measures such as MIME-type sniffing and encrypted session management were integrated to ensure data integrity. The study was evaluated by purposively selected respondents composed of HR personnel, recruitment managers, and IT experts using the ISO/IEC 25010 software quality model to assess functional suitability, performance efficiency, usability, and security.

Limitations of the Study Despite its extensive functionality, the study had specific boundaries:

- **Network Accessibility:** The system was designed exclusively for the Local Area Network (Intranet) and was not accessible via the public internet, prioritizing internal security over remote access.
- **Algorithmic vs. AI Capabilities:** While the system included an automated scoring algorithm based on logical rules (Boolean/Weighted logic), it did not utilize Machine Learning (ML) or Generative AI for parsing unstructured text from resumes or predicting candidate success.
- **Notification Scope:** Alerts for upcoming interviews and tasks were in-application dashboard notifications only. The system did not integrate with external third-party communication APIs (e.g., SMS gateways or SMTP email servers) due to network isolation policies.
- **Browser Dependency:** Compatibility testing was limited to modern web browsers (Chrome, Edge) available on the company's standard workstations.
- **Device Accessibility:** The system was designed and optimized for desktop and laptop workstations operating within the company's local intranet. It was not available for use on mobile devices such as smartphones or tablets.

- **Recruitment Focus:** The system functioned solely as an Applicant Tracking System (ATS) and did not include post-hiring modules such as Payroll, Timekeeping, or Employee Performance Evaluation.

CHAPTER 2

CONCEPTUAL FRAMEWORK

This chapter presented the review of related literature and studies that provided the theoretical, conceptual, and empirical foundation of the proposed Recruitment and Applicant Management System (R-AMS). It discussed relevant concepts on digital HR transformation, HR analytics, data-driven recruitment, and digital system adoption. The chapter also outlined the conceptual model, the tools and technologies used in system development, the software testing framework, and the operational definition of terms.

Review of Related Literature

Digital Transformation in Human Resource Management

Digital Transformation in Human Resource Management refers to the integration of digital technologies, automated workflows, data-driven tools, and modernized infrastructures into HR operations to replace slow, fragmented, and manual processes. It represents a strategic shift in how HR teams manage data, interact with applicants, and execute recruitment decisions. As Bansal (2023) explains in *A Study of Human Resource Digital Transformation (HRDT)*, HR digitalization involves the modernization of infrastructure, workflow enhancement, and organizational adaptation to digital systems. HR departments that transition from paper-based or spreadsheet-dependent processes to digital environments experience significant improvements in data accuracy, reduction of redundancy, cost saving for less paper, faster processing times, and greater interdepartmental coordination.

Mahmoud, Ali, Alrifae, and Abu Eitah (2025), in their article *The Impact of Digital HRM System and Digital Transformation on HR Efficiency*, emphasize that digital HR systems improve organizational agility by automating repetitive tasks, centralizing applicant data, and enabling analytic tools. Their findings highlight the importance of dashboards, real-time summaries, and systematic interfaces in minimizing human error and increasing operational accuracy. Understanding these concepts formed part of the researcher's **knowledge requirements**, as familiarity with digital HRM frameworks was essential in designing modules for the Recruitment and Applicant Management System (R-AMS). The researcher needed to study recruitment workflows, common bottlenecks in traditional HR practices, and the theoretical foundation behind automation to ensure that R-AMS accurately reflects modern HR operations.

Digital transformation literature also reinforces the importance of technical competencies. To implement the digital processes highlighted by these studies, the researcher/developer had to learn **software development tools and technologies** such as PHP for backend data processing, JavaScript for interactive dashboards and interface behavior, MySQL for centralized storage, and TailwindCSS for responsive UI design. These technologies operationalize the digital concepts described in HR transformation research. For instance, dashboards recommended by Mahmoud et al. (2025) were translated into real system components through **Chart.js**, while centralized data storage was achieved through relational database structures in MySQL. The bulk upload improvement seen in digital HR systems required the implementation of **PapaParse.js**, enabling the system to handle large volumes of applicant data consistent with real-world recruitment needs.

The insights from Setyawati et al. (2021) in *Web-Based Management Information System for Services* further underline the significance of centralization and reliable data architecture. Their findings show that digital platforms improve visibility, accuracy, and responsiveness—key performance elements examined under **ISO/IEC 25010**, particularly the characteristics of *functional suitability*, *performance efficiency*, *and reliability*. Understanding these quality attributes was part of the researcher's preparation, ensuring that the R-AMS would meet accepted standards of software quality. The emphasis on real-time analytics and responsive interfaces aligns with ISO 25010's requirements for *usability*, *operability, and user assistance*, supporting an HR-friendly system experience.

As companies expand their workforce and applicant pools, the demand for systems capable of managing large datasets becomes increasingly critical. Digital HRM systems address this challenge by consolidating applicant records, generating real-time summaries, and accelerating decision-making. The proposed R-AMS responds to these digital modernization trends by integrating secure data storage, real-time analytics panels, guided workflows, and role-based access management. These features reflect both the conceptual foundations presented in HR digital transformation literature and the practical software engineering principles required for building a reliable and scalable HR system.

Through the lens of existing research, digital transformation is not merely a technical enhancement but a strategic organizational direction. It requires an understanding of HR processes, technical development skills, and awareness of software quality standards. Thus, the development of R-AMS is grounded in this broader movement toward technologically enabled recruitment workflows—prioritizing automation, centralized control, analytics-supported decision-making, and compliance with software quality.

Pre-Screening Through Skills-Based Application Forms

Pre-screening is an initial filtering stage in recruitment where applicants are evaluated based on basic qualifications and skills before advancing to interviews. This process uses structured form fields such as skill checklists, education, and relevant experience to determine whether an applicant meets the minimum job requirements. According to a 2021 study on *Recruitment and Selection: The Relationship between Recruitment and Selection with Organizational Performance*, organizations utilizing application form criteria such as qualifications, certifications, and skills achieve better alignment between applicant competencies and job needs. This reinforces the relevance of a simple, skills-based pre-screening method in improving recruitment effectiveness.

The integration of pre-screening within the R-AMS allowed the system to automatically categorize applicants as "Highly Recommended," "Qualified,", "Underqualified" or "Review Required" based on predefined skill requirements. This automated segregation of applicants enabled HR personnel to quickly focus on those who match specific job criteria, reducing manual workload and time spent reviewing unqualified applicants. However, pre-screening does **not** replace traditional recruitment methods such as interviews and detailed assessments. Instead, it acts as a supportive tool to shortlist large applicant volumes and ensures that further evaluation is reserved for those with appropriate baseline skills.

By structuring applicant data into sortable and filterable fields, the pre-screening feature enhances usability, reliability, and functional suitability as outlined in the ISO/IEC 25010 quality model. Through this approach, the R-AMS ensures that recruitment decisions remain fair and evidence-based while still preserving the essential human judgment required in the final hiring stages.

HR Analytics and Data-Driven Recruitment

HR Analytics, often called people analytics, is essentially the practice of turning raw workforce data into a strategic asset. In the context of recruitment, this means moving beyond gut feelings and looking at hard evidence: evaluating where hires come from, predicting staffing needs, and measuring how effective recruiters actually are. Krishna and Verma (2025), in *Human Resource Analytics in the Digital Transformation Era*, point out that this analytical approach is what modernizes HR functions. By transforming unstructured data into actionable insights, analytics allow teams to identify skill gaps and fix process bottlenecks based on quantifiable facts rather than manual observation.

For this study, simply knowing how to code was not enough; the researcher had to grasp the logic behind the metrics. It was crucial to understand how applicant flow correlates with timelines and performance data. The challenge lay in understanding the recruitment lifecycle itself—from the moment an applicant enters the system to their final confirmation—to ensure the analytics reflected actual HR workflows. Without this conceptual foundation, building a dashboard would have resulted in generic charts that failed to align with the real-life KPI indicators used by the department.

Adopting a data-driven approach gives HR departments a clear view of their "blind spots," such as hidden bottlenecks or ineffective sourcing channels. Ruiz, Marín, and Benítez (2024) argue in *Digital Human Resource Strategy* that organizations leveraging these analytics operate significantly better because their functions become transparent, measurable, and outcome-based. Their findings suggest that when HR teams have access to real-time metrics, it significantly enhances HR teams' ability to allocate resources, manage applicant volume, and evaluate recruiter productivity.

To transform these analytical concepts into system features, the researcher/developer needed to master specific **software technologies**. The creation of dashboards required understanding in **Chart.js**, a JavaScript visualization library used to generate interactive KPI graphics. Knowledge of **JavaScript** and the **Fetch API** was required to retrieve filtered datasets from the backend, while **Litepicker** enabled flexible date-range filtering for time-based analyses. Proficiency in **MySQL** was necessary to construct optimized queries that return aggregated data, such as counts per status or recruiter performance metrics. These technologies were essential to implement the data-driven recruitment practices described in the academic literature.

The integration of HR analytics into R-AMS also aligns with the **ISO/IEC 25010** software quality framework. The use of dashboards supports the characteristic of *functional suitability*, ensuring the system provides the analytical functions HR personnel require. Real-time updates and interactive charts contribute to *performance efficiency* and *usability*, as users can access well-structured information without delays or complexity. Accurate data visualization further enhances *reliability* by reducing misinterpretation and providing consistent output. By grounding the analytics features of R-AMS in ISO 25010 principles, the system ensures accuracy, relevance, and operational value for HR end-users.

The proposed R-AMS fully integrates these data-driven recruitment principles through its use of analytics dashboards, KPI cards, and visualizations. Chart.js components present applicant count, Deployment progress, and platform utilization, while filtering tools allow HR personnel to refine data by date range or recruitment stage. These features provide immediate insight into recruitment performance and workload distribution, supporting evidence-based decision-making. Through this alignment with HR analytics literature, the R-AMS functions as a modern tool designed for organizational efficiency, accuracy, and strategic HR management.

Challenges and Enablers of HR Digital Systems

While digital HR systems are powerful, they don't work by magic. Rahadi et al. (2025) point out that these tools often flop if the staff isn't trained properly or if the office culture resists the shift to digital. This insight was crucial for the study—it meant that simply building a system wasn't enough. I had to understand how HR personnel actually interact with screens and what specific interface barriers might stop them from adopting the new technology.

On the data side, sticking to manual methods is risky. In a review of digital transformation, Bratamanggala et al. (2023) found that organizations relying on spreadsheets almost always deal with double entries and messy reporting. Reading this confirmed that a centralized database wasn't just a "nice-to-have"—it was necessary. These limitations reinforced the researcher’s need to understand centralized database structures, data accuracy principles, and workflow optimization to ensure that the R-AMS resolves the inefficiencies described in the literature.

Addressing these challenges also required the researcher to study specific **software and technical tools** essential for building a stable digital HR system. Backend technologies such as PHP and MySQL were needed to maintain consistent and secure records, while JavaScript and TailwindCSS supported the development of an intuitive and user-friendly interface. Features like role-based access, activity logs and validation function weren't just added for extra functionality; they were direct responses to the data errors and inconsistencies warned about in previous studies.

These literature findings directly align with the **ISO/IEC 25010** software quality model. Issues related to user adoption reflect usability attributes such as learnability and operability, while data inconsistency problems relate to reliability and functional suitability. Understanding these quality characteristics helped the researcher integrate features that support accuracy, stability, and user protection.

With this, the R-AMS was designed with simplified navigation, secure role-based access, centralized data storage, and real-time dashboards. These features address the major challenges identified in the literature—such as fragmented records and limited reporting—making the system a practical, user-centered solution for modern HR digital environments.

Application Standard Requirements

The R-AMS was a web-based HR system accessible through standard browsers such as Chrome, Edge, Firefox or Safari. It worked perfectly on desktops and laptops across workstations. As an intranet-based system, it operated within the company's secure local network, ensuring that applicant data remained internal, protected, and compliant with institutional privacy policies.

Tools and Technologies for Developing the System

- **Visual Studio Code (VS Code)** was a lightweight and powerful source-code editor used in developing the R-AMS. It operated efficiently across Windows-based workstations and provided built-in support for JavaScript, PHP, HTML, and CSS/Tailwind. As a modern development environment, Visual Studio Code enables software developers to write, debug, and manage source code effectively due to its extensibility and integrated tools. Visual Studio Code has different uses and benefits, such as:
    - Collaborating and coding remotely
    - Fixing errors as you code
    - Comparing changes in your code
    - Managing source control through built-in Git integration
- **PHP** (Hypertext Preprocessor) was an open-source, server-side scripting language used to develop the backend APIs, authentication services, and data processing components of the system. Originally known as “personal home page,” it became recognized as a recursive acronym for “hypertext preprocessor.” PHP allowed the system to communicate with the MySQL database, process form inputs, handle authentication logic, and produce dynamic content rendered on the web interface. Aspects of PHP that set it apart from other languages:

  - Scripting Language: PHP was interpreted at runtime without the need for compilation.
  - Server-Side Processing: PHP scripts ran on a web server and returned the result to the browser as HTML.
  - Open-Source: Freely available and widely supported by the programming community.
  - Object-Oriented: Supported OOP concepts for reusable and modular code.
  - Fast Performance: Memory-efficient and capable of handling heavy workloads; benchmarks showed it could perform significantly faster than several interpreted languages.
  - Simple Syntax: Easy to learn, making it accessible for building applications of varying complexity.
  - Well Supported: Compatible with major databases (MySQL, SQLite) and web servers (Apache, IIS), and supported by frameworks such as Laravel and CodeIgniter. PHP was also a loosely typed language, and had a flexible variable declaration using eight primary data types. This flexibility helped developers to build dynamic and scalable backend logic.

- **JavaScript** was the primary language used for front-end logic in the R-AMS. It made the system interactive and responsive by modifying webpage content in real time based on user actions. JavaScript enhanced the dynamic behavior of the interface through DOM manipulation, enabling changes such as table updates, form validation, modal windows, and interactive dashboard elements. JavaScript enabled several essential client-side functions:

- Dynamic DOM manipulation for realtime UI updates
- Communication with backend APIs via the Fetch API
- Rendering of charts, summaries, tables, and modals
- Handling of data validation and error prompts
- Processing and previewing of bulk upload files

- **TailwindCSS** was a utility-first CSS framework used to style and layout the system interface. It provided predefined classes for spacing, colors, typography, shadows, grids, and responsiveness. Unlike traditional CSS, TailwindCSS allowed styling directly within the HTML structure, enabling faster development and consistent design across modules. The responsive utilities ensured the system displayed properly across workstations used by HR personnel.

- **PapaParse.js** was a JavaScript library integrated into the bulk-upload feature of the R-AMS. It enabled fast and accurate parsing of CSV files directly within the browser, allowing HR staff to validate data before saving it into the database. PapaParse supported error detection such as incorrect formats, missing fields, or invalid structures—preventing corrupted data from being uploaded. PapaParse provided the following key functions:
    - Parsing large CSV files with high performance
    - Detecting and reporting formatting errors
    - Previewing parsed applicant data
    - Validating fields based on the system's data schema

- **Chart.js** was a JavaScript library used to generate the system's analytics and dashboard visualizations. It supported interactive charts such as bar graphs, line charts, and pie charts, which helped HR personnel visualize applicant flow, recruitment trends, and performance indicators. Chart.js was used in the system to:

  - Display recruitment KPIs
  - Generate visual summaries of applicant status
  - Monitor recruiter productivity
  - Provide real-time graphical analytics

- **Litepicker** was a lightweight JavaScript date-range picker integrated into the dashboard filtering module. It simplified the selection of dates and supported various formats (with predefined dates), allowing HR users to filter reports or analytics based on specific time periods. Litepicker helped the system by enabling:
  - Easy date and range selection
  - Smooth filtering of dashboard views
  - Accurate retrieval of time-based recruitment data
- **FontAwesome** brought the user interface to life by replacing standard text labels with intuitive, scalable icons. It was used throughout the Applicant Web Form and HR Dashboard to make navigation and status tracking easier to understand at a glance, giving the system a polished, professional look. FontAwesome was used in the system to:
  - Make actions instantly recognizable (e.g., using a "Trash" icon instead of the word "Delete")
  - Visually represent applicant progress (e.g., Clocks for pending, Checks for complete)
  - Declutter the layout by replacing long text descriptions with simple symbols

  - Ensure icons looked sharp and consistent on any screen resolution

- **SweetAlert2** upgraded the user experience by replacing harsh, default browser alerts with smooth, interactive popup modals. It played a vital role in preventing errors by forcing users to double-check critical actions—like archiving a record or saving feedback—before they committed to them. SweetAlert2 helped the system by:
  - Providing clear, "Are you sure?" confirmation dialogs to prevent accidental data loss
  - Showing non-intrusive "Toast" notifications (success messages) that didn't block the user's view
  - Displaying user-friendly error messages if the system ran into an issue
  - Showing a visual loading spinner so users knew the system was working

- **RESTful API Architecture** acted as the organized communication bridge between the system's frontend and backend. It structured the server-side logic clearly, ensuring that when the dashboard asked for data (like applicant records), the server knew exactly how to respond. RESTful API Architecture helped the system by:
  - Creating a standard way for the app to ask for data (GET) or send updates (POST)
  - Keeping the database logic separate from the user interface, which improved security
  - Making the system easier to maintain since each action had its own specific "address"
  - Ensuring data was exchanged quickly and reliably between the client and server

- **Fetch API** was the engine behind the system's smooth performance, allowing the dashboard to talk to the server in the background. Because of this, HR staff could save changes, upload files, or filter lists instantly without dealing with annoying and slow full-page reloads. The Fetch API was used in the system to:
  - Load data silently in the background (AJAX) so the user kept working uninterrupted
  - Submit forms and files instantly without the screen flashing or refreshing
  - Update charts and tables in real-time as soon as new data came in
  - Catch connection errors and tell the user if something went wrong
- **JSON (JavaScript Object Notation)** served as the common language that let the frontend and backend understand each other. Beyond just moving data, it also cleverly packaged complex lists—like employment history or references—so they could be stored neatly in a single database slot. JSON helped the system by:
  - Securely carrying data between the server and the browser
  - Bundling multiple pieces of information (like a list of children) into one manageable text field
  - Helping the system read API responses and immediately show them on the screen
  - Ensuring that PHP and JavaScript could share data without compatibility issues
- The system was hosted on an Intel Xeon server running a MySQL database. MySQL was a relational, SQL-based database management system storing all applicant profiles, logs, recruiter data, schedules, and system-generated activities. The Intel Xeon

server provided stability and processing strength required for handling large volumes of HR records, ensuring that data remained accurate, secure, and consistently retrievable. MySQL offered the following advantages:

- Fast query execution for large datasets
- Transaction-safe storage for sensitive information
- Scalability for growing recruitment operations
- Strong integration with PHP APIs and backend logic

Software Testing

Software testing is the critical process of evaluating the Recruitment and Applicant Management System to verify if it functioned exactly as intended. It covered executing the system's components, such as the Applicant Web Form, HR Dashboard, and Interviewer Portal, to identify any gaps between the actual behavior and the expected requirements. By testing the automated pre-screening algorithms, database interaction, and access controls, the study ensured that the final result was free of critical defects and ready for deployment within the company intranet.

Importance of Software Testing

Software testing serves as a vital checkpoint in the Software Development Life Cycle (SDLC), ensuring that the system functioned exactly as designed. It was no longer just about spotting and patching errors; instead it had become a central part of Quality Assurance, ensuring that the software remained secure, dependable, and flexible enough to meet evolving business demands (Indium Software, 2023).

Key Benefits of Software Testing Recent literature from 2020 onwards highlighted four primary advantages of implementing a rigorous testing framework:

- **Cost-Effectiveness:** Testing early in the development process, often called the "Shift-Left" strategy, was one of the most effective ways to keep a project within budget. If a team waited until the software was deployed to fix a bug, Coreflex Solutions (2024) estimated it could cost 100 times more than if they had caught it during the initial design. Basically, investing time in testing prevented expensive repairs and financial losses later on (PractiTest, 2024; Sapizon, 2022).
- **Security Assurance:** As cyber threats are getting more sophisticated, security testing was no longer optional—it was a requirement. HikeQA (2025) pointed out that this testing was the only way to ensure the system followed data protection laws and kept records safe. For recruitment platforms specifically, ImpactQA (2024) warned that skipping this step was dangerous; strict testing was necessary to stop unauthorized access and prevent data breaches.
- **Product Quality (ISO 25010):** The industry relies on the ISO/IEC 25010 standard to measure how "good" a piece of software actually was. According to Helpware (2025), following this standard ensured the system wasn't just functional, but also easy to maintain and efficient. It forced developers to look at specific qualities, like Functional Suitability (verifying the system did what it promised) and Performance Efficiency (making sure the system ran smoothly).
- **Customer Satisfaction:** Ultimately, users will only trust a system if it works reliably. Indium Software (2023) noted that a smooth, error-free experience was what made users loyal to a product. On the other hand, if an application is slow or crashes

frequently, users will likely abandon it immediately, which could cause serious damage to the organization's reputation.

Software testing was classified into three categories namely: Functional testing, Non-Functional testing or Performance testing, and Maintenance testing as shown in Figure 3 (Hamilton, 2021).

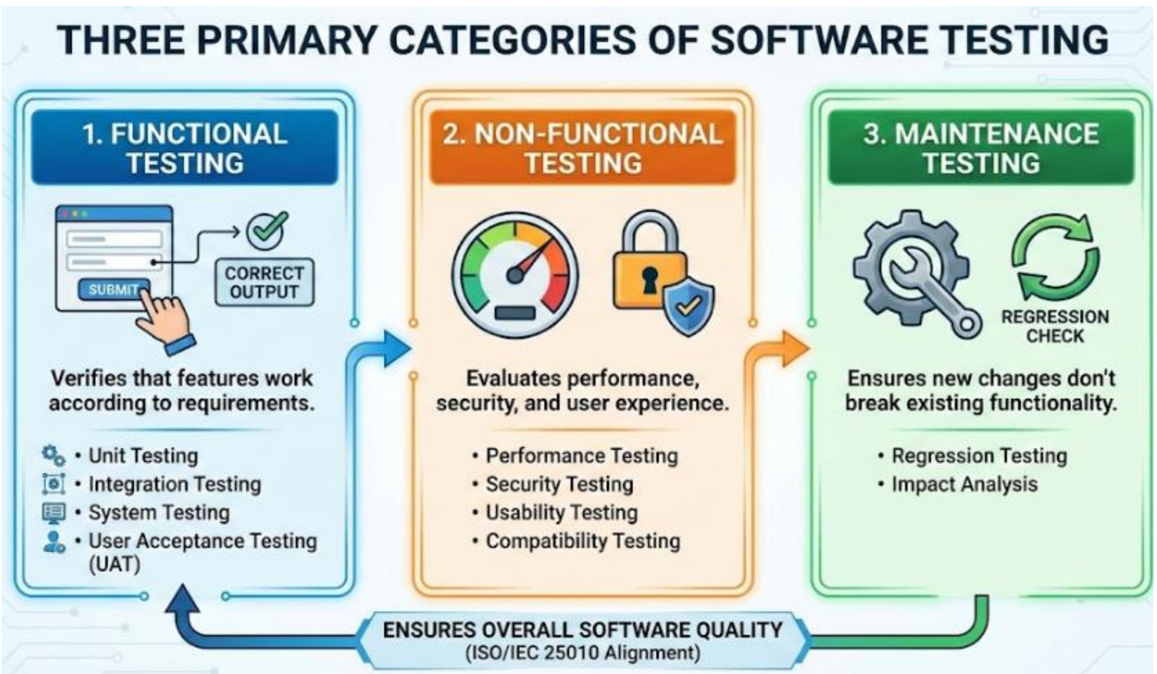


Figure 3. Primary Categories of Software testing

Software Quality Evaluation

ISO 25010, titled "Systems and software engineering – Systems and software Quality Requirements and Evaluation – System and software quality models," was a software quality standard. It described models consisting of characteristics and sub-characteristics for both software product quality and software quality in use, providing practical guidance on their application (Britton, 2021). The quality model was the cornerstone of a product quality evaluation system. The quality model determined which quality characteristics would be considered when evaluating the properties of a software product (ISO 25000, 2021).

SOFTWARE PRODUCT QUALITY								
FUNCTIONAL SUITABILITY	PERFORMANCE EFFICIENCY	COMPATIBILITY	INTERACTION CAPABILITY	RELIABILITY	SECURITY	MAINTAINABILITY	FLEXIBILITY	SAFETY
FUNCTIONAL COMPLETENESS FUNCTIONAL CORRECTNESS FUNCTIONAL APPROPRIATENESS	TIME BEHAVIOUR RESOURCE UTILIZATION CAPACITY	CO-EXISTENCE INTEROPERABILITY	APPROPRIATENESS RECOGNIZABILITY LEARNABILITY OPERABILITY USER ERROR PROTECTION USER ENGAGEMENT INCLUSIVITY USER ASSISTANCE SELF-DESCRIPTIVENESS	FAULTLESSNESS AVAILABILITY FAULT TOLERANCE RECOVERABILITY	CONFIDENTIALITY INTEGRITY NON-REPUDIATION ACCOUNTABILITY AUTHENTICITY RESISTANCE	MODULARITY REUSABILITY ANALYSABILITY MODIFIABILITY TESTABILITY	ADAPTABILITY SCALABILITY INSTALLABILITY REPLACEABILITY	OPERATIONAL CONSTRAINT RISK IDENTIFICATION FAIL SAFE HAZARD WARNING SAFE INTEGRATION

Figure 4. Software Product Quality

The ISO/IEC 25010 Product Quality Model standard serves as a comprehensive framework for evaluating software excellence. As illustrated in the figure above, it assesses the quality of a system not just by its code, but by how well it satisfies eight critical dimensions, including functionality, performance, reliability, and security.

- **Functional Suitability evaluates** whether the software actually provides the functions users need to complete their tasks. It ensures that the system’s features align with the user's expectations.
    - **Functional Completeness:** Verifies that the system covers the entire scope of work. It asks: Does the software have all the necessary features to achieve the user’s specific goals?
    - **Functional Correctness: Focuses** on accuracy and precision. It ensures that the system provides the right results (e.g., correct calculations or data retrieval) every time.

  - **Functional Appropriateness: Measures** how well the functions facilitate the accomplishment of tasks. It checks if the features are actually useful and relevant to what the user is trying to do.
- **Reliability defines** the system's ability to maintain a certain level of performance over a specific period. It answers the fundamental question: Can the user trust the system to be available and stable when they need it?
  - **Maturity:** The degree to which the system meets reliability needs under normal, everyday operations without frequent glitches.
  - **Availability: Ensures** the system **is** operational and accessible exactly when the user needs to access it.
  - **Fault Tolerance:** The system's ability to keep running smoothly even if a hardware or software error occurs (e.g., preventing a total crash due to a minor bug).
  - **Recoverability**: If a failure does occur, this measures how quickly and effectively the system can restore its data and return to a functional state.
- **Portability** assesses how easily the system can be transferred from one environment to another. This is crucial for ensuring the software works across different hardware, servers, or operating scenarios.
  - **Adaptability:** The ability of the system to adjust effectively to different or evolving hardware and software environments without requiring major changes.
  - **Installability: Measures** how easy it is to install or uninstall the software in its intended environment.

- **Replaceability:** The degree to which this system can seamlessly replace another specified software product for the same purpose, ensuring a smooth transition for the organization.

Related Studies

"Design and Implementation of a Web-Based E-Recruitment System" by Author Abdulkareem (2023) addressed the persistent issue of traditional recruitment where managing piles of physical resumes led to data loss and significant delays in hiring. To resolve this, the study developed a centralized web portal that allowed applicants to submit their data digitally, ensuring that no candidate information was misplaced. It stated that utilizing a web-based platform brought a substantial effect on organizational efficiency, specifically by reducing the "Time-to-Hire" metric. The use of a centralized database helped build up important historical data on the administrator side, allowing HR to track previous applicants easily. In addition, this study provided a solution to the redundancy problem, as the system automatically checked for duplicate applications, ensuring that the company's resources were focused only on unique, valid candidates.

The "Smart HR: An Automated Candidate Screening and Ranking System" (2022) emphasized how qualified candidates were often overlooked in mass hiring because HR staff could not physically read every resume thoroughly. To address this situation, the study developed an application which devised a Keyword Matching Algorithm that aided recruiters by automatically scanning applicant inputs against the job description. The added feature of Automated Status Updates was used to run the application and inform applicants of their progress via email notifications. It was proven in this study that the use of automation technology could help HR professionals make shortlisting decisions much easier and more objective, removing human bias from the initial stages. The Smart HR system used variations of available web technologies and logic that communicated between the applicant interface and the admin dashboard: PHP (Hypertext Preprocessor) was used by the researchers as the core backend language, which allowed the system to process form data and execute the matching logic dynamically. MySQL Database allowed their web application to manage the vast records of applicants and synchronize the status updates in real-time. The application also utilized JavaScript (AJAX) to ensure that the dashboard updated instantly, without requiring the page to be reloaded, allowing recruiters to view new applicants the moment they submitted.

"Visualization of Recruitment Data for Strategic Decision Making" by Author Li and Zhang (2024) developed a dashboard module which served as a medium that helped HR managers visualize their workforce trends. With the help of Data Visualization APIs, it helped track the volume of applicants per department and eventually helped managers allocate resources where they were needed most. It was clearly expounded in this study that the use of visual analytics played a vital role in modern business, which unexpectedly could help users identify bottlenecks in their hiring process. The application also utilized a Filtering Module to allow users to sort data by date range or specific job positions to get a granular view of performance. The Visualization System performed as built with the help of modern frontend libraries used for its dashboard to be built based on these components: Chart.js API used a canvas rendering method for precise graphical representation of its system data, allowing numbers to be converted into line graphs and pie charts. Fetch API was used by the researchers to request data asynchronously from the server, ensuring that the charts loaded immediately. Date-Range Pickers were also used to filter the reports, allowing the user to generate analytics for specific quarters or months. The

Export Module allowed an advantage for its data to be downloaded into CSV or PDF formats, which could be used for offline reporting and presentation to upper management.

Conceptual Model of the Study

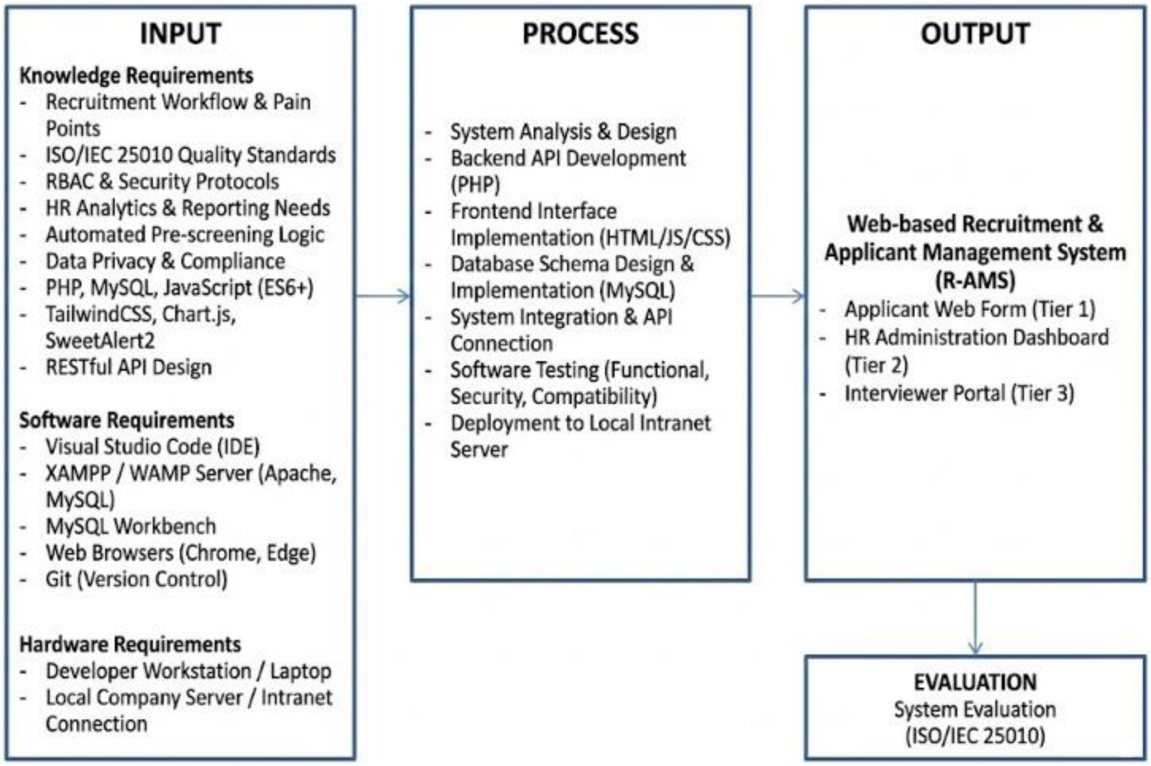


Figure 5. Conceptual Model of the study

The conceptual model served as the primary blueprint for the development of the R-AMS. The input phase was split into three main categories: knowledge requirements, software requirements, and hardware requirements, which collectively acted as the foundation for development. Knowledge requirements contained the foundational studies and concepts necessary to build the project, including recruitment workflows, automated pre-screening logic, HR analytics, and data privacy protocols. Software requirements covered the specific programming languages and tools needed to code the system, such as PHP, MySQL, JavaScript, Tailwind/CSS, and local server environments like XAMPP. On the other hand, hardware requirements referred to the physical equipment, such as developer workstations and company servers, required to build and eventually host the application.

The process phase represented the actual execution of the software development life cycle, which included system design and analysis, backend and frontend coding, testing, and continuous improvement. This phase heavily involved structural methods, such as utilizing flowcharts and database schemas based on the gathered input requirements, to map out how applicant data would flow into the HR dashboard. To ensure the overall quality, security, and stability of the software application, the ISO/IEC 25010 software evaluation instrument was used to thoroughly test and evaluate the system's performance before deployment.

The final output was the Web-based Recruitment and Applicant Management System (R-AMS). Instead of operating as a single-function tool, the completed project served as a centralized digital hub divided into three interconnected modules: the Applicant Web Form, the HR Administration Dashboard, and a dedicated Interviewer Portal. By linking these platforms together, the system successfully automated the tedious pre-screening phase and provided real-time analytics. Most importantly, it streamlined the overall hiring workflow because it directly bridged the communication gap between the HR personnel sorting the applications and the department heads who actually conducted the interviews.

Operational Definition of Terms

- **R-AMS** refers to the Recruitment and Applicant Management System, the web-based digital platform developed in this study to automate candidate screening and streamline the hiring workflow.
- **Applicant** refers to an individual who submitted their personal information, skills, and work history through the system's web form to apply for a vacant job position.
- **HR Personnel** refers to the human resource staff members (Staff or Manager) who managed the recruitment process & reviewed applicant data.
- **Interviewer** refers to the department heads (Manager) or designated assessing staff (HR Staff) who used a separate, restricted portal within the system to evaluate candidates and submit final feedback.
- **Pre-screening** refers to the initial, automated evaluation phase where the system immediately filters and scores incoming applications based on predefined skills and educational requirements.
- **Highly Recommended** refers to the pre-screen generated status for applicants who perfectly matched or exceeded the essential skills and qualifications needed for a specific role.
- **Qualified** refers to the applicant pre-screen status assigned to candidates who met the basic, minimum requirements to perform the job, but might have lacked certain advanced or preferred skills.
- **Underqualified** refers to the pre-screen status of applicants who fell short of the essential criteria, often due to a lack of necessary work experience or educational attainment.

- **Review Required** refers to the pre-screen applicant whose data needs manual checking by an HR staff member, usually triggered by incomplete entries or unusual skill combinations.
- **Archived** refers to the secure storage status of applicants who were rejected, do not pass the interviews, or withdrew their applications, kept in the database for future reference.
- **HR Dashboard** refers to the centralized graphical interface used by HR to view applicant analytics, key performance indicators, and recent system activities at a glance.
- **HR Analytics** refers to the practice of evaluating recruitment data, such as time-to-hire and applicant volume, to make evidence-based and strategic hiring decisions.
- **KPI** refers to Key Performance Indicators, the specific, quantifiable metrics displayed on the dashboard that show how effectively the recruitment process is operating.
- **RBAC** refers to Role-Based Access Control, a security framework ensuring that users, such as HR staff versus Interviewers, can only access the specific section.
- **PHP** refers to Hypertext Preprocessor, the backend scripting language used to process form submissions, execute the screening logic, and handle server-side operations.
- **MySQL** refers to the structured relational database system used to safely store and manage all applicant records, interview notes, and registered user accounts.
- **JavaScript** refers to the programming language used on the frontend to make the web pages highly interactive, such as updating dashboard charts without reloading the screen.

- **TailwindCSS** refers to a modern, utility-first styling framework used to design the user interface and ensure the application responds seamlessly across different desktop and mobile screens.
- **Fetch API** refers to a modern web technology that allows the dashboard to request and send data to the server in the background smoothly, preventing annoying page refreshes.
- **JSON** refers to JavaScript Object Notation, a lightweight data format used to bundle and transmit complex information, like an applicant's work history.
- **RESTful API** refers to the organized backend architecture that allows different parts of the system, like the web form and the dashboard, to communicate securely and efficiently.
- **Chart.js** refers to the specific JavaScript library utilized to generate the interactive visual analytics, such as bar graphs and pie charts, found on the HR dashboard.
- **SweetAlert2** refers to the tool used to replace standard browser alerts with customizable, interactive pop-up confirmations to prevent accidental data deletion.
- **Litepicker** refers to the lightweight date-range tool used in the dashboard that allows users to quickly select specific timeframes when filtering recruitment reports.
- **Bulk Upload** refers to a system feature that enables HR personnel to import multiple applicant records into the database simultaneously using CSV files, drastically cutting down on manual data entry.

- **Smart Filter** refers to the advanced search mechanism within the dashboard that allows users to instantly narrow down large lists of applicants by combining multiple criteria, such as date ranges, job positions, and specific recruiters.
- **System Logs** refers to the automated audit trail that records user actions, system errors, and data modifications to ensure accountability, security, and easier troubleshooting.
- **Applied** was the default starting status representing a candidate who had just successfully submitted their application through the web form or from bulk upload.
- **Failed Speedtest** refers to the status of applicants who do not meet the minimum typing test requirement.
- **Initial Interview** refers to the stage where an applicant is currently scheduled for, or is undergoing, their first-level assessment (L1) with an HR representative.
- **Failed L1 Interview** is the status assigned to candidates who failed the initial Interview.
- **Final Interview** indicates the critical stage where a candidate has passed the HR screening and is forwarded to the department head or operations manager.
- **Failed L2 Interview** refers to applicants who successfully reach the final stage of the hiring process but fail the final evaluation.
- **For BGV** stands for Background Verification, marking the phase where the company actively checked the candidate’s character references, criminal records, and previous employment history.
- **Job Offer** represents the stage where a successful candidate was officially presented with their compensation package, benefits, and contract terms for review.

- **Processing Requirements** indicates that a candidate has accepted the job offer and is currently in the process of gathering and submitting their mandatory pre-employment documents (e.g., medical clearances, government IDs).
- **Complete Requirements** is the status assigned once the HR department has verified and accepted all of the candidate's necessary pre-employment documents.
- **Onboarding** refers to the orientation phase where the newly hired employee is officially integrated into the company, introduced to workplace policies, and set up with their tools.
- **Pooling** designated the status for candidates who passed all interviews and assessments but were placed on a standby list until a specific project or job vacancy became available.
- **Deployed** is the ultimate success status in the system, indicating that the hired employee has officially started working on their assigned account or department.
- **Withdrawn** refers to a status applied when a candidate voluntarily pulled out of the recruitment process.
- **Declined Offer** indicates that a candidate successfully passed all evaluations and received a formal job offer, but chooses not to accept the position.
- **Interviewer Dashboard** refers to a dedicated, secure interface specifically designed for department heads and assessing managers. This portal allows them to review the profiles of candidates scheduled for final assessments, input their interview notes, and submit their final hiring verdicts without needing full access to the broader HR administrative system.

- **Requisition Modal** refers to an interactive pop-up window within the application that enables HR personnel to officially add/edit a new job opening or manpower addition. This user interface feature allows them to submit necessary details—such as the requisition ID, Project name, approved date, and needed headcount—without navigating away from their current web page.
- **Recruiters** are the human resource professionals actively utilizing the system to source candidates, facilitate the initial screening and interview stages, and guide applicants through the entire hiring pipeline until they are successfully deployed or archived.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology used in developing the Web-Based Recruitment and Applicant Management System (R-AMS) for XBP Global Company. It discusses the project design, system architecture, interface layouts, database design, development model, operation and testing procedures, and the ISO/IEC 25010-based evaluation process. The methodology ensured that the system was developed systematically and adhered to software engineering standards aligned with the company's recruitment workflows.

Project Design

This study developed a web-based Recruitment and Applicant Management System (R-AMS) for XBP Global Company to make the entire hiring process faster and more organized. The system could be accessed through any workstation within the company's intranet, allowing HR staff—such as recruiters, hiring managers, and Department Manager—to securely view, enter, update, and analyze applicant information through role-based dashboards. By centralizing all application data in one place, the platform helped reduce manual errors, speed up processing, and improve tracking of recruitment activities.

Structurally, the R-AMS operated across three distinct but interconnected layers. The first layer was the Applicant Web Form (Tier 1), which served as the primary entry point for candidates to submit their personal profiles and credentials. Once submitted, the data flowed directly into the HR Recruitment Management Module (Tier 2). This module acted as the main command center, allowing HR personnel to efficiently manage candidate records, process bulk application uploads, generate visual analytics, and track all system activities through detailed audit logs. Finally, the system featured a dedicated Interviewer Dashboard (Tier 3), providing department heads with a focused, restricted space to review shortlisted candidates assigned to them and submit their final evaluations.

To ensure strict data privacy and controlled access, all three of these components were securely hosted within the company's local network. The following sections present the System Architecture Diagram, Use Case Diagram, Data Flow Diagrams, Database Design, and UI Layouts for both the applicant interface and the HR system. These diagrams outline how the system worked, how data moved through each module, and how users interacted with the different features during development.

System Architecture Diagram

The Recruitment and Applicant Management System (R-AMS) included three major system components: the Applicant User Web Form (Tier 1), the HR Recruitment Management System (Tier 2), and the Interviewer dashboard (Tier 3). The System Design section presents the Context Diagram and Top-Level Data Flow Diagram, which illustrate how system users interact with the main modules and how data flows across the application.

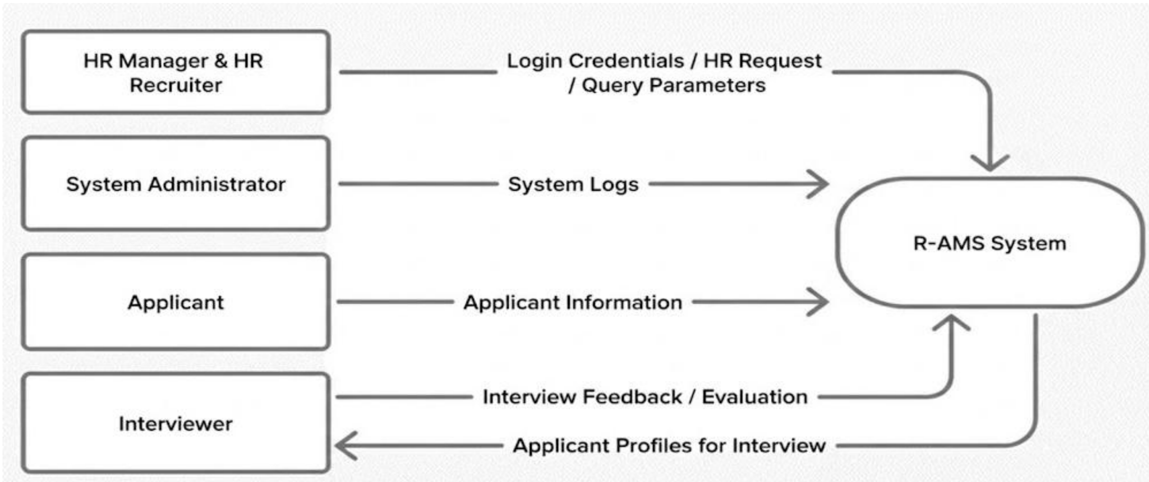


Figure 6: Context Diagram of a developed Web-based R-AMS

Figure 6 shows the Context Diagram for R-AMS, highlighting the interactions among Applicants, HR Recruiters, HR Managers, System Administrators, and the centralized database. Applicants submitted their information via the web form, while HR personnel authenticated via the HR dashboard to view, update, and manage applicant records based on their assigned roles. The Interviewer dashboard allowed the Department manager to view the applicant profile and provide feedback.

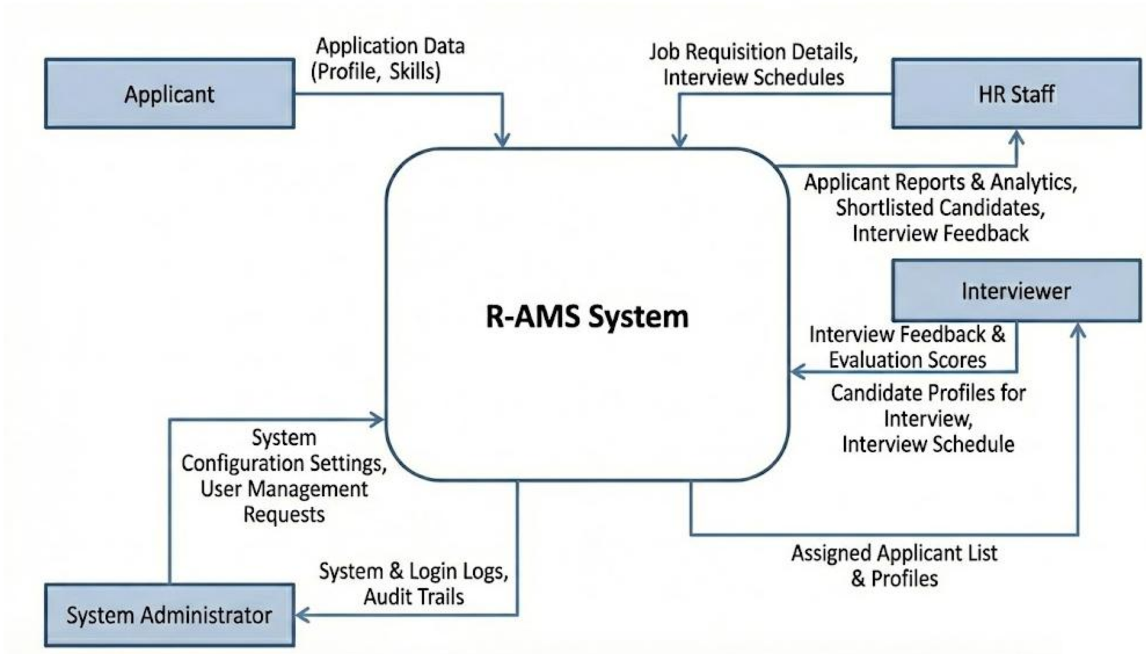


Figure 7: Top level Data Flow Diagram for the development of Web-based R-AMS

Figure 7 illustrates the top-level data flow diagram for the applicants, HR staff, interviewers, and system administrators, which shows their interactions with the centralized recruitment system, application data submissions, and candidate evaluation processes.

Together, these diagrams define the structural and operational design of the R-AMS, showing how users accessed system modules, how data was processed internally, and how each component integrated to support a streamlined and secure recruitment workflow.

Software Design

The Software Design of the Recruitment and Applicant Management System (R-AMS) presented the system's structural and behavioral components, focusing on user interactions, main services, and their permissions (Figure 8). The design centers on a Use Case Diagram that identifies the actors and primary actions the system supports. These diagrams and descriptions guided development and clarified how responsibilities were separated between the applicant-facing interface and the HR administration dashboard.

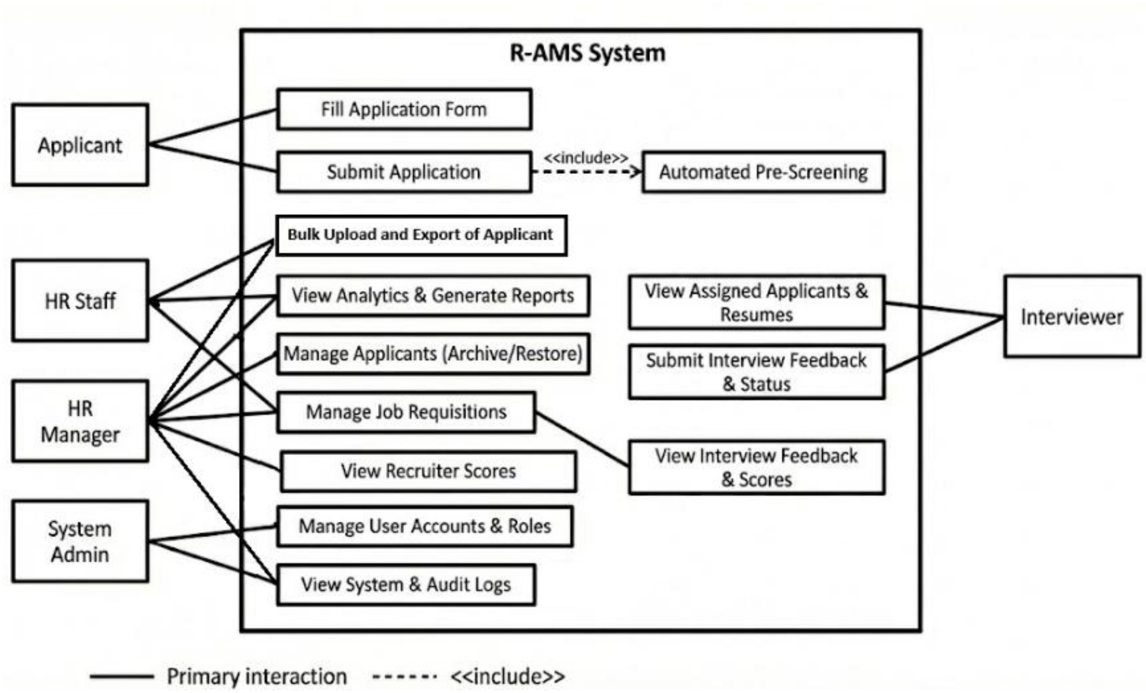


Figure 8: Software design interaction and use cases for R-AMS.

Database Design

The information maintained in the system is illustrated using the MySQL RDBMS (Relational Database Management System) in Figure 9. The RDBMS shows the data of the following: applicant, hr user, and module.

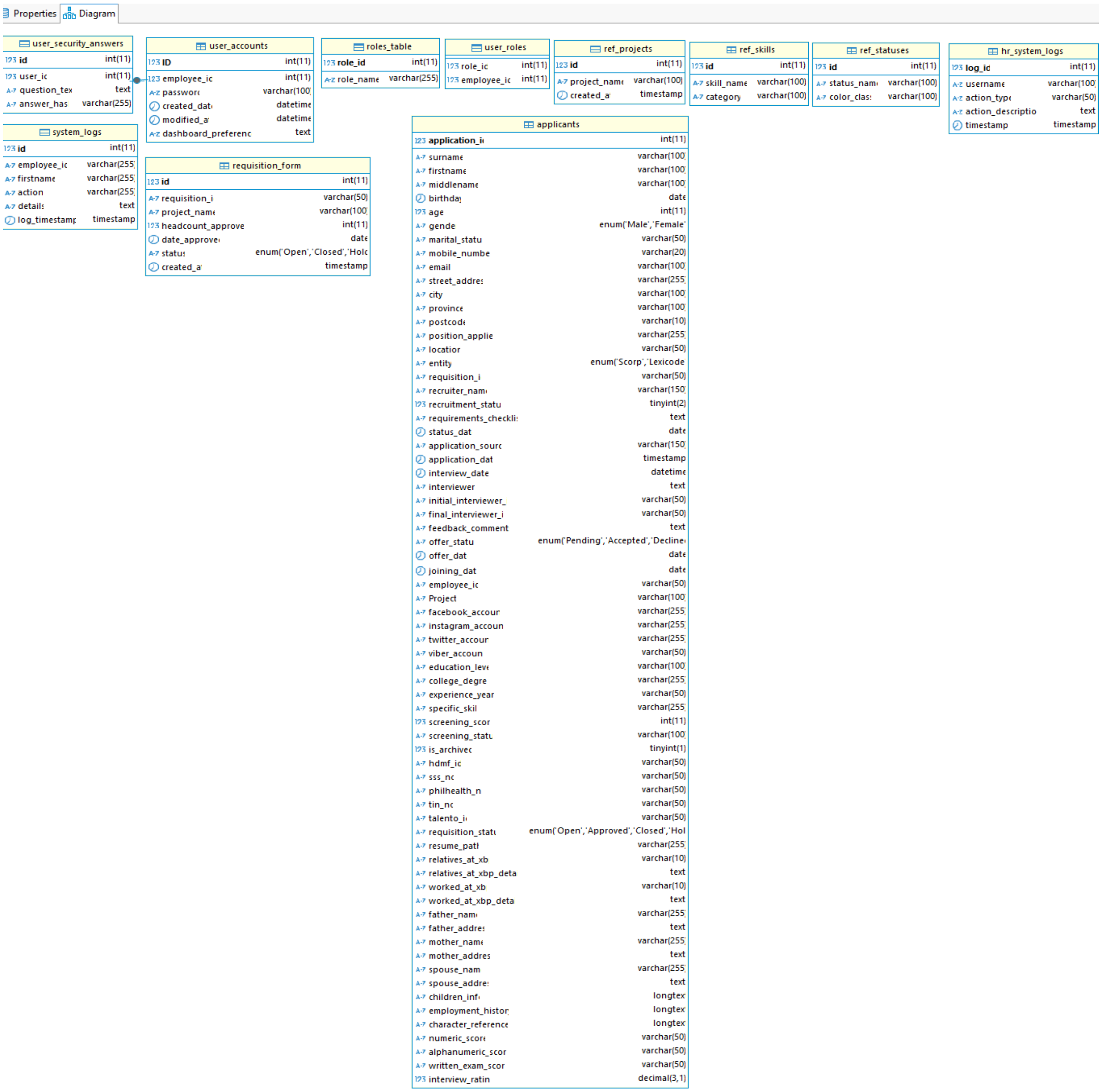


Figure 9: HR User Login, Role permission, other module design for R-AMS

System architecture

The System Architecture Diagram displays the communication interface of the system for the Applicant, Interviewer, HR personnel, and the local server.

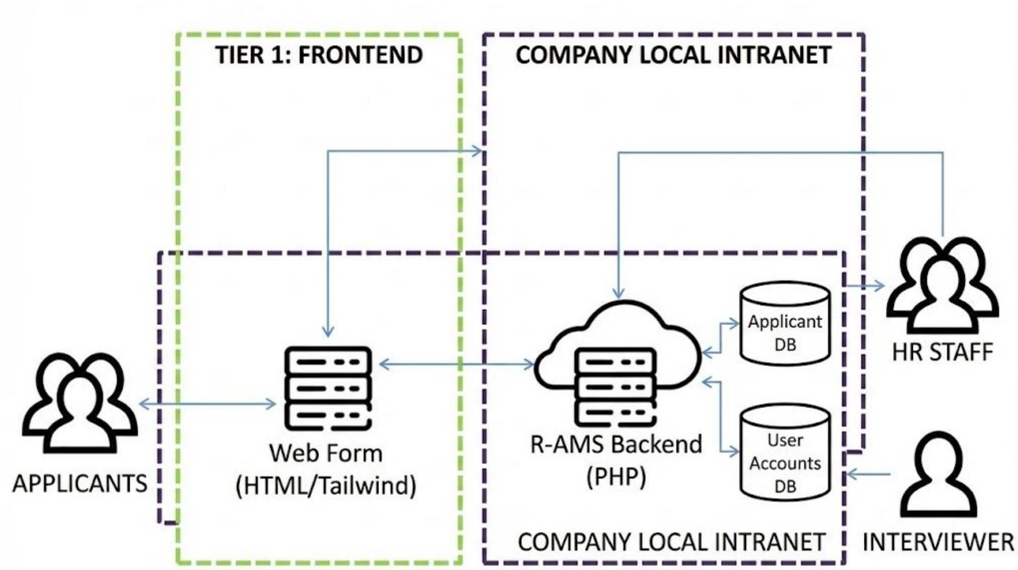


Figure 10: System Architecture Diagram for R-AMS

Desktop UI

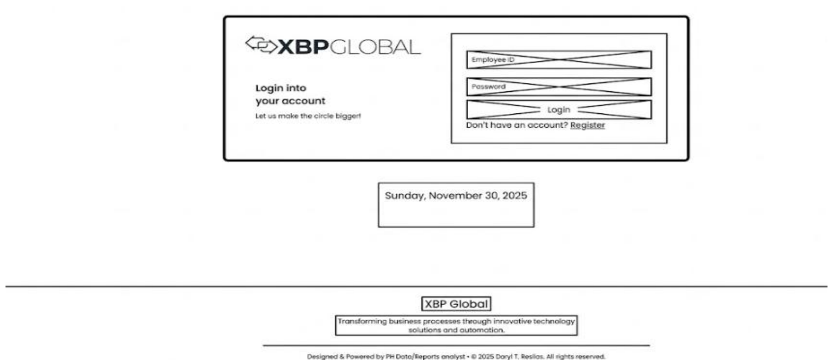


Figure 11: Login on R-AMS

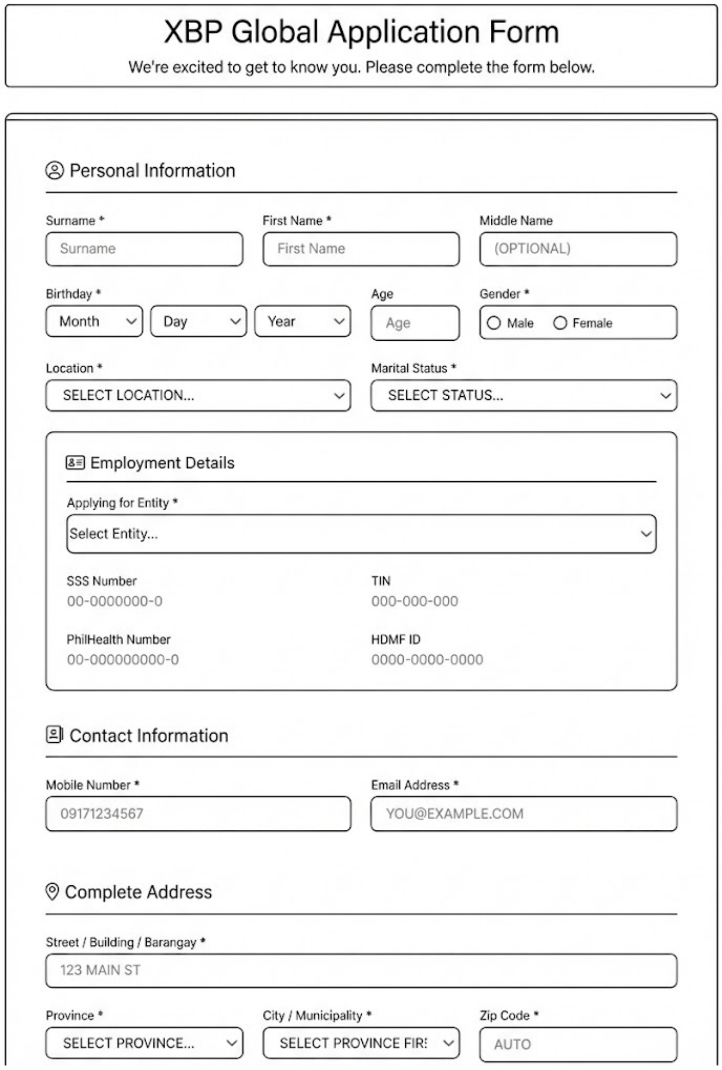


Figure 12: Tier-1 Applicant Form

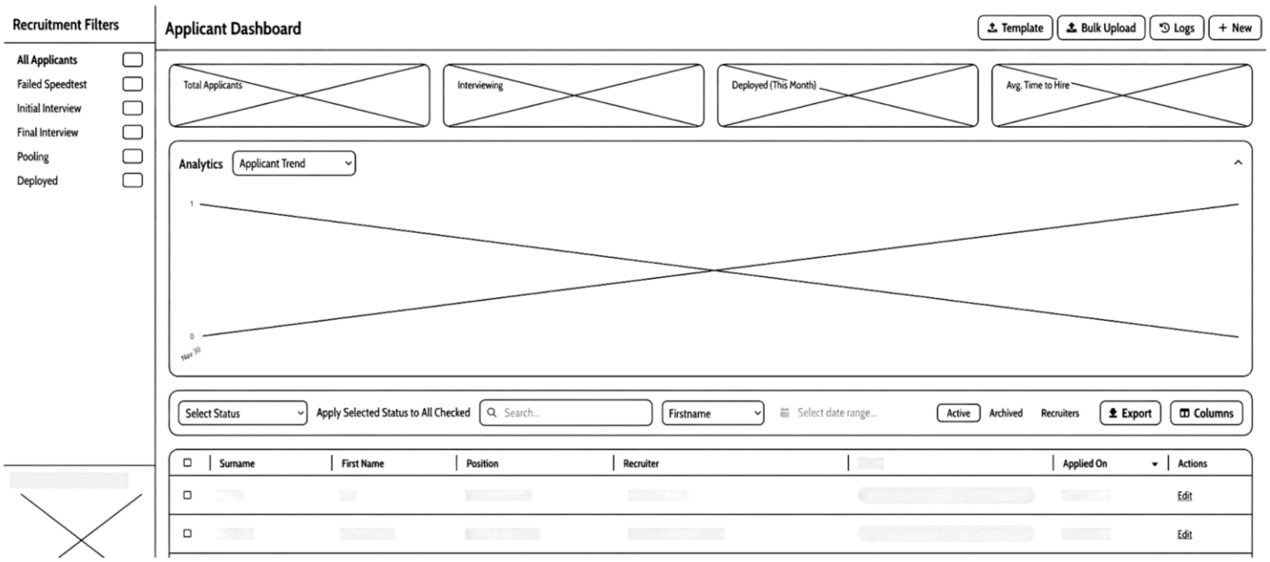


Figure 13: Tier-2 Dashboard Recruitment Active view

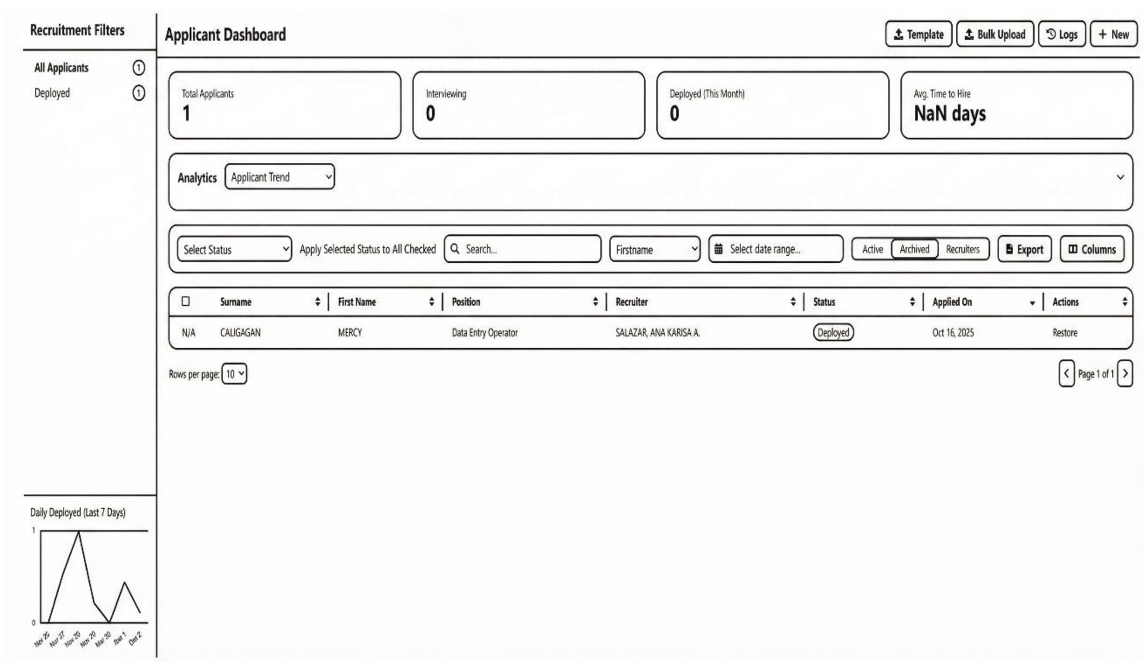


Figure 14: Tier-2 Dashboard Recruitment Active Archived/Restore (chart hidden)

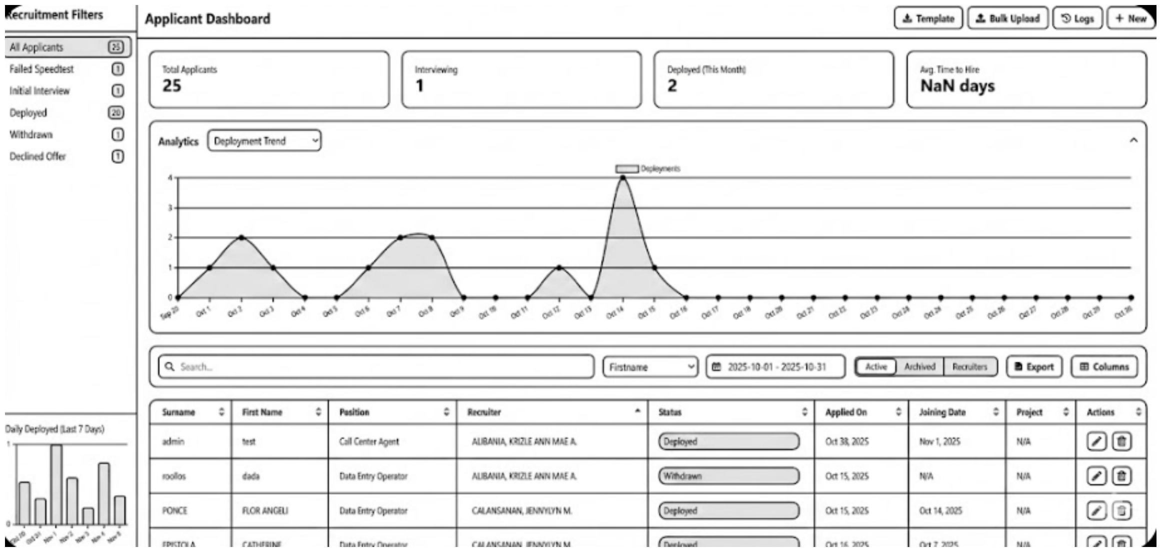


Figure 15: Tier-2 Dashboard Recruitment Active table values

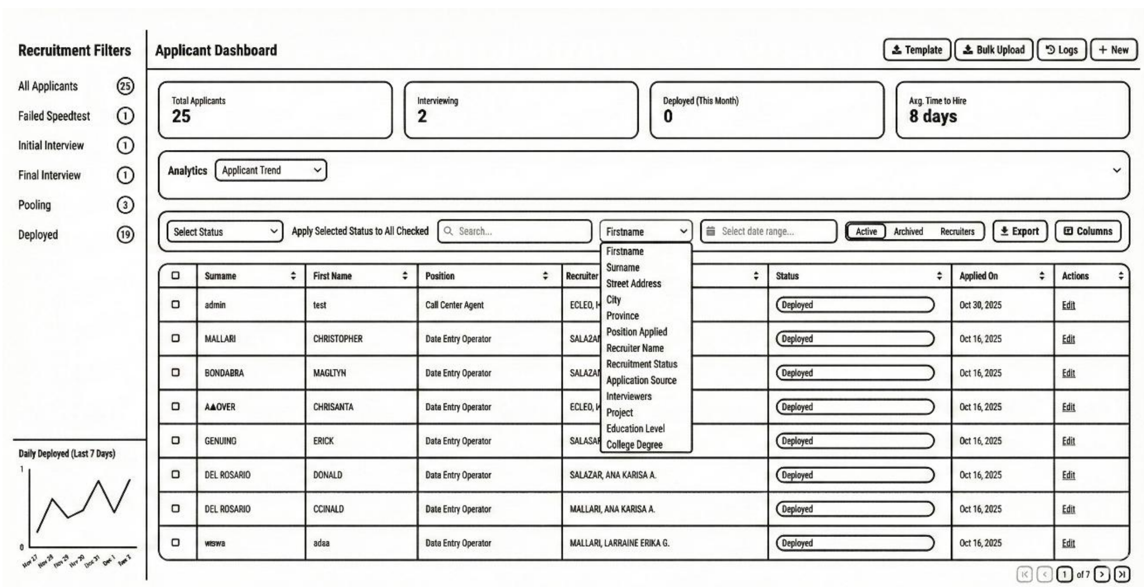


Figure 16: Tier-2 Dashboard Recruitment Active with Table values and search Option dropdown (Chart hidden)

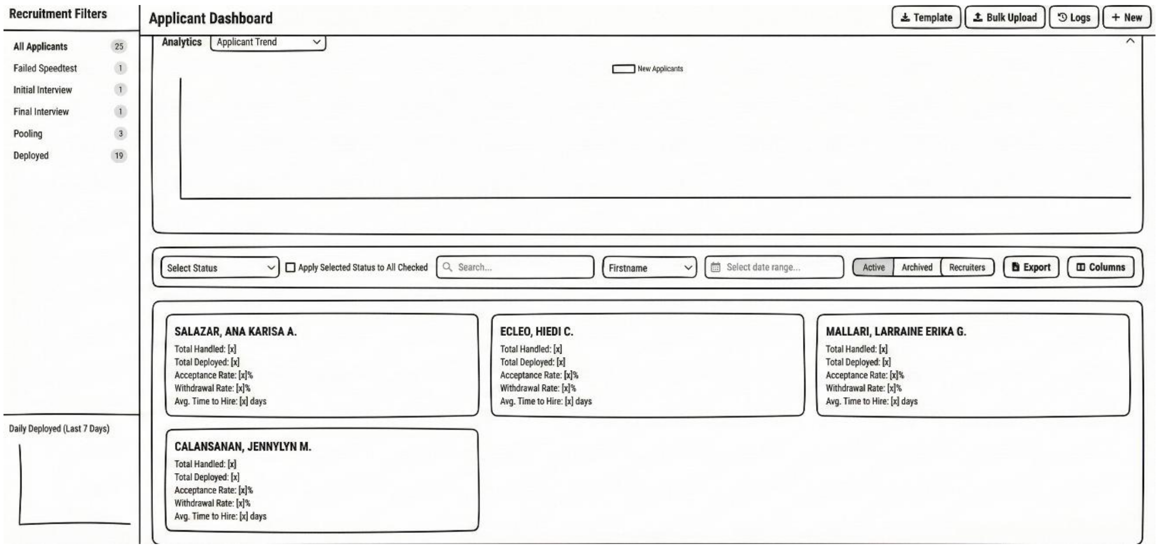


Figure 17: Tier-2 Dashboard Recruitment Recruiter KPI cards

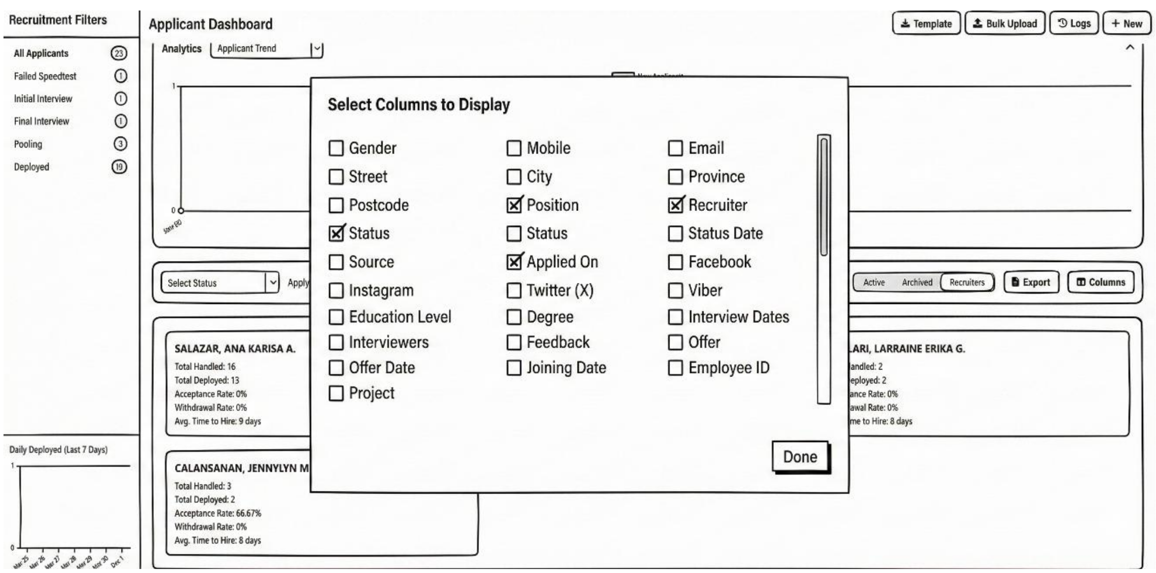


Figure 18: Tier-2 Dashboard Recruitment Column Selection to display

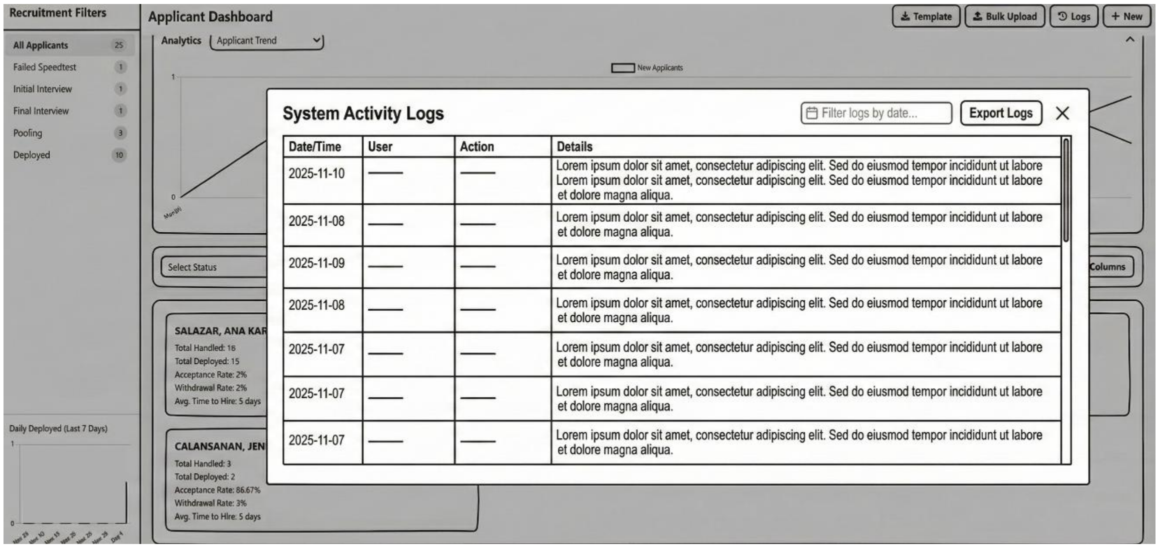


Figure 19: Tier-2 Dashboard Recruitment Activity Logs

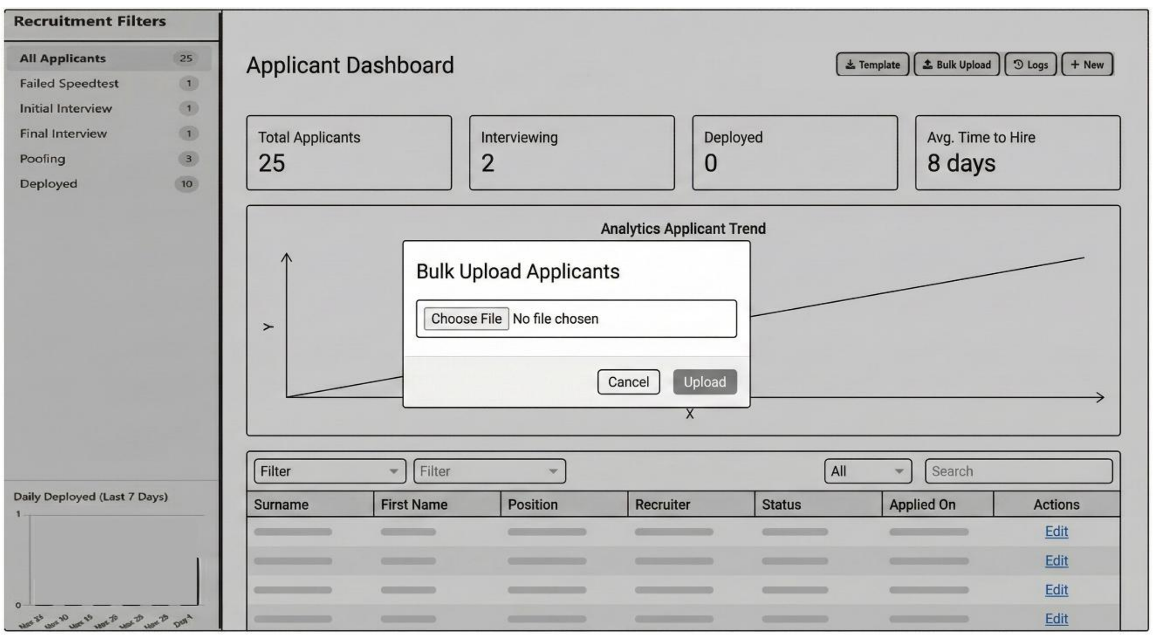


Figure 20: Tier-2 Dashboard Recruitment Bulk Upload Applicants

Figure 21: Tier-2 Dashboard Recruitment Analytics = Resource Efficiency

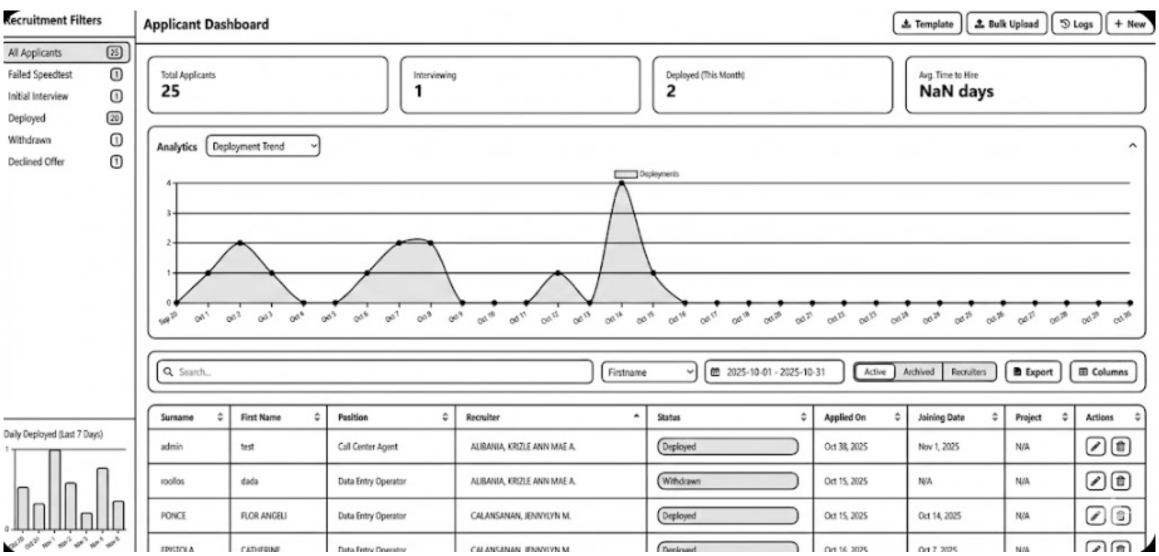


Figure 22: Tier-2 Dashboard Recruitment Analytics = Deployment Trend

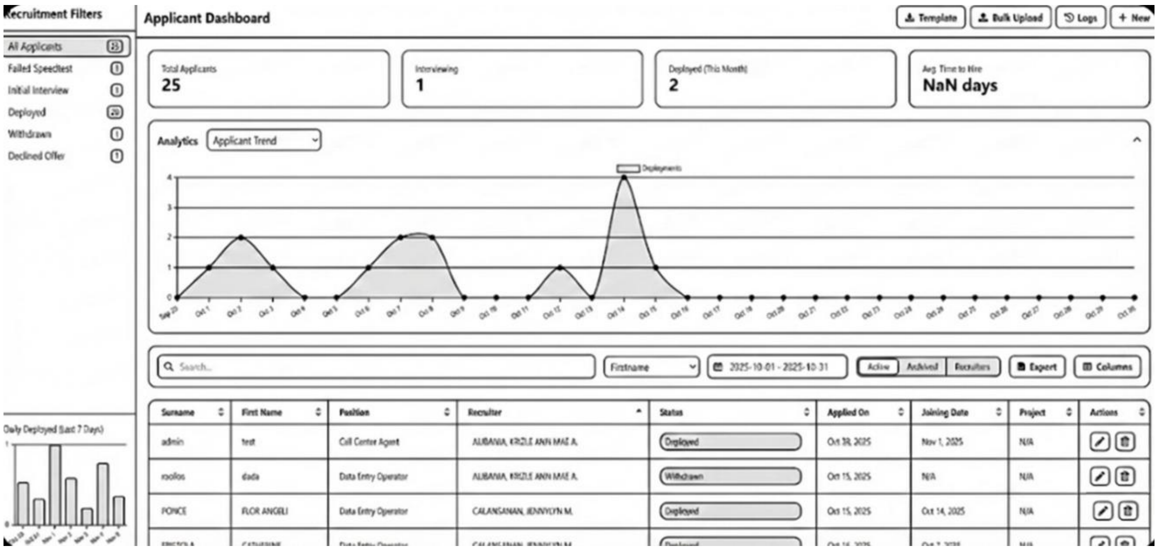


Figure 23: Tier-2 Dashboard Recruitment Analytics = Applicant Trend

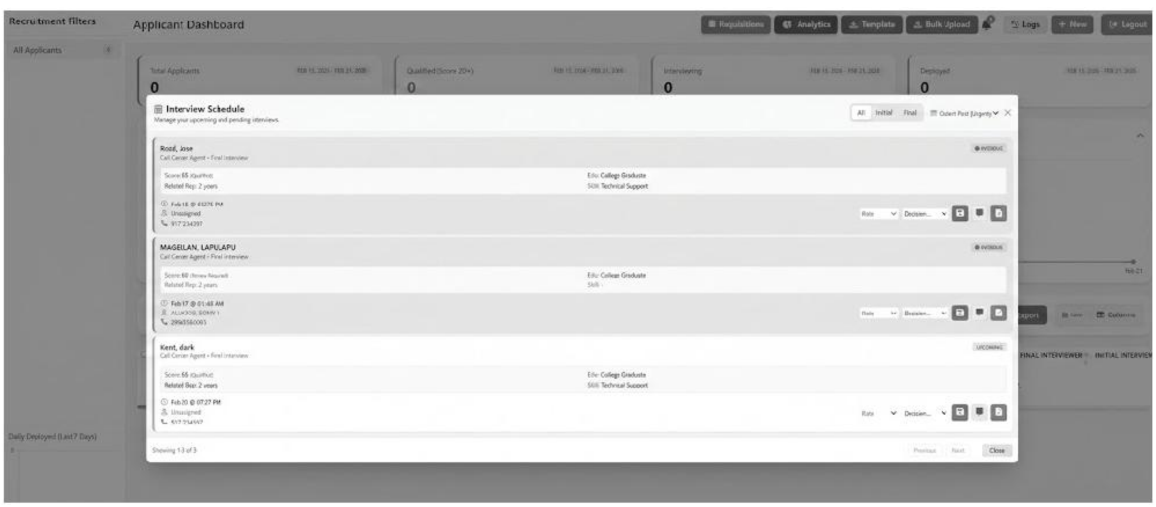


Figure 24: Tier-2 Dashboard Notification for scheduled Interviews

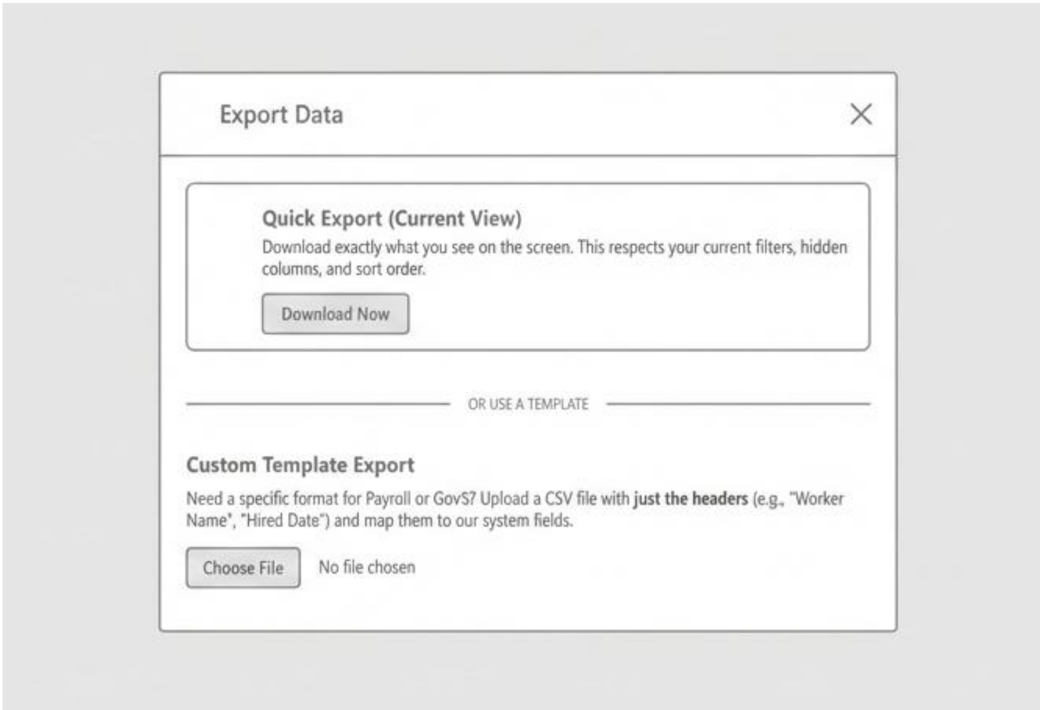


Figure 25: Tier-2 Dashboard Export menu

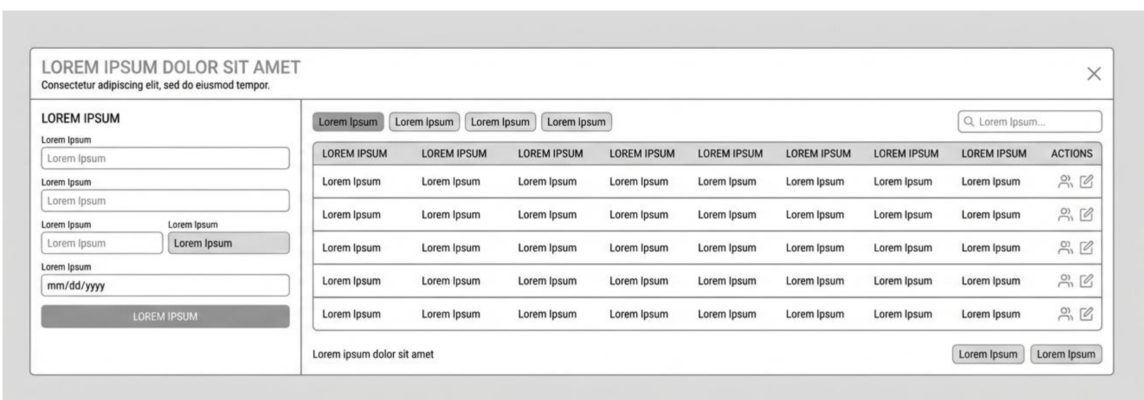


Figure 26: Tier-2 Dashboard Requisition Modal

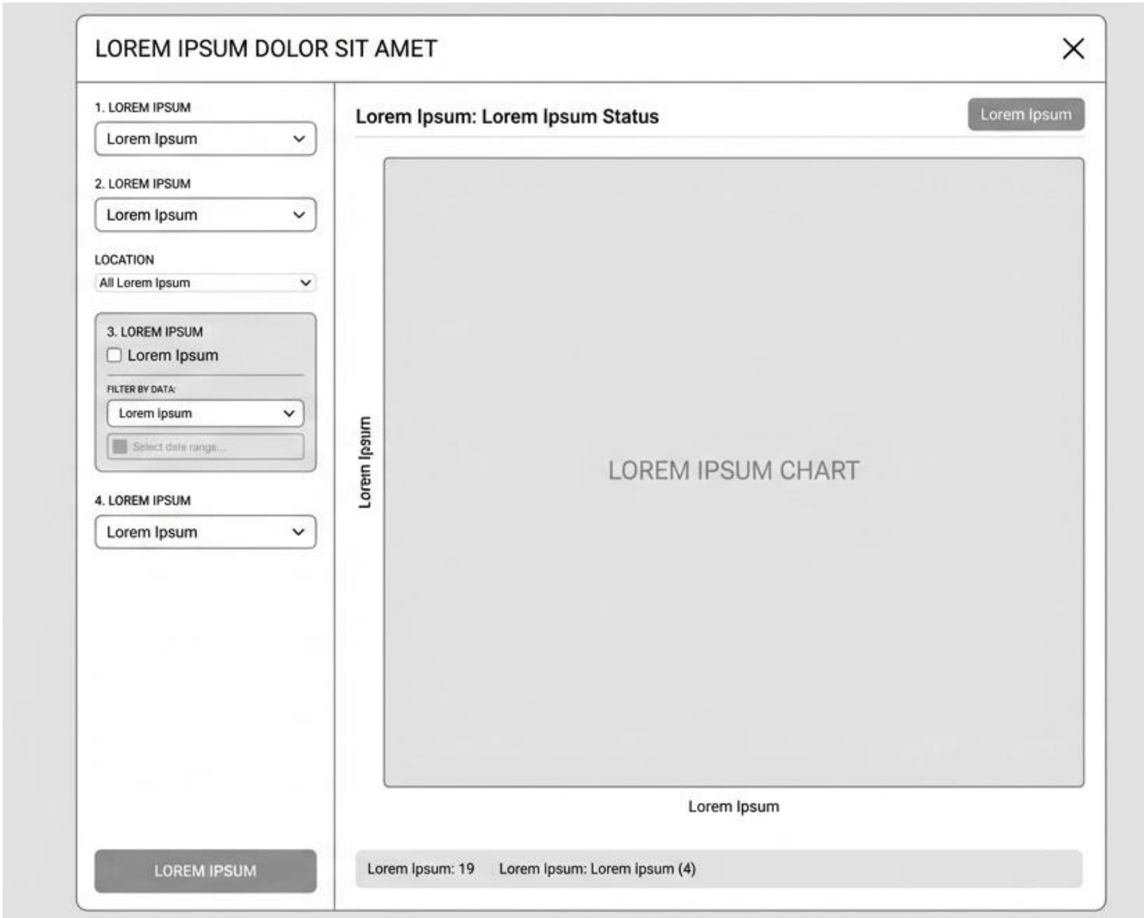


Figure 27: Tier-2 Dashboard Custom Analytics (Bar Chart, Pie Chart, Datasheet)

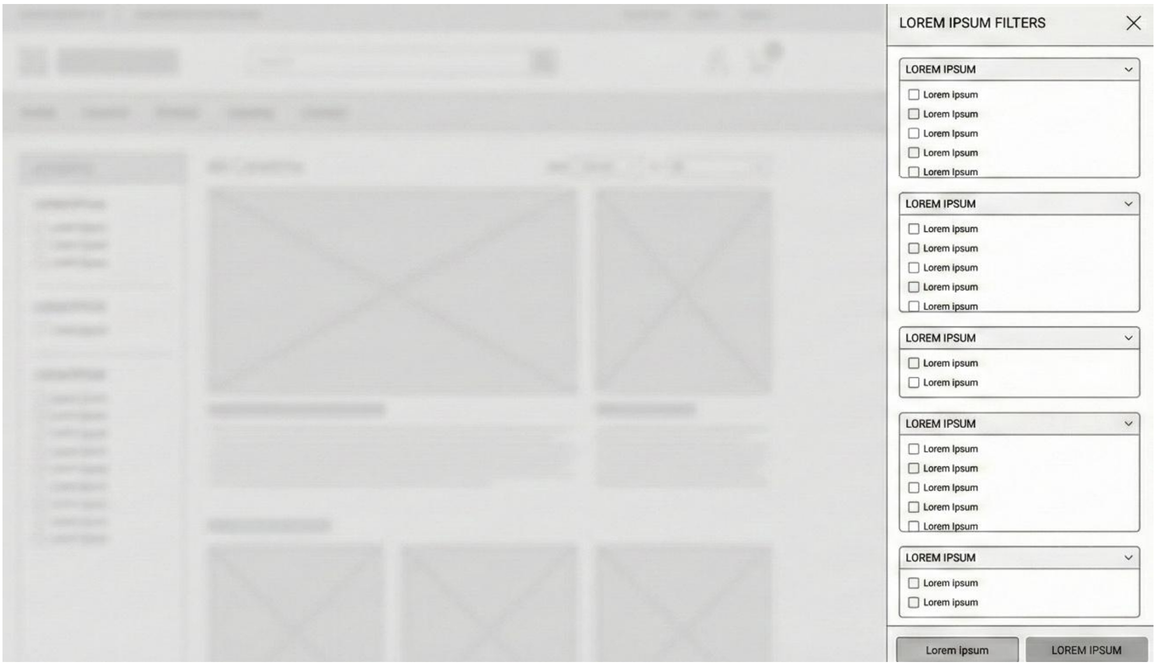


Figure 28: Tier-2 Dashboard Smart Filter

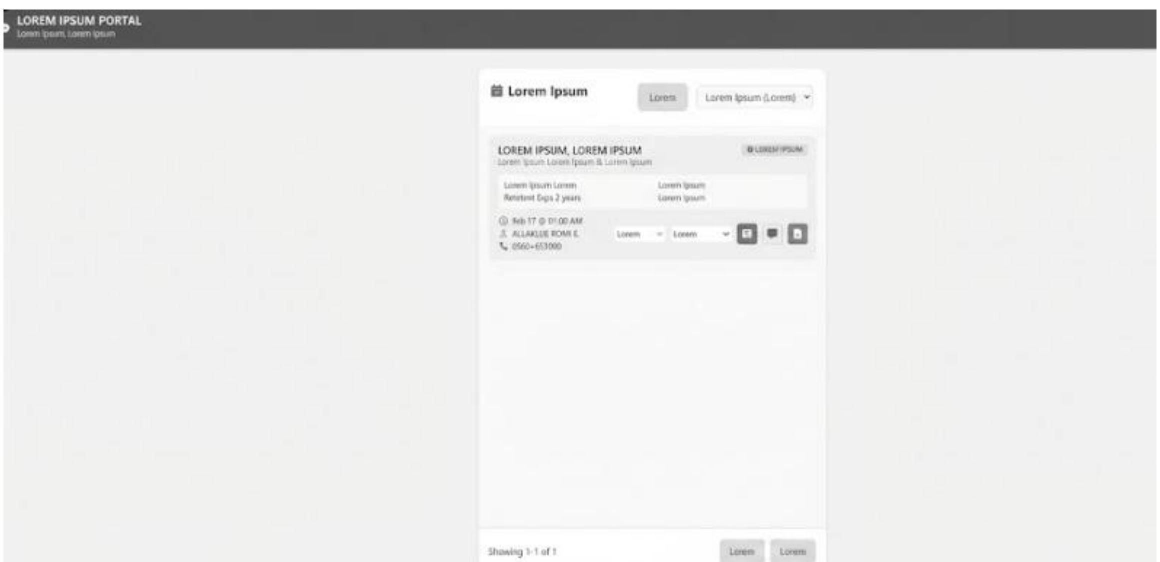


Figure 29: Tier-3 Interviewer Dashboard

Project Development

The Recruitment and Applicant Management System (R-AMS) was developed using the Iterative Waterfall Model, which allowed continuous feedback between phases to improve system quality and correctness. The model consists of Requirement Analysis, System Design, Implementation, Testing, Deployment, and Maintenance, as shown in Figure 30.

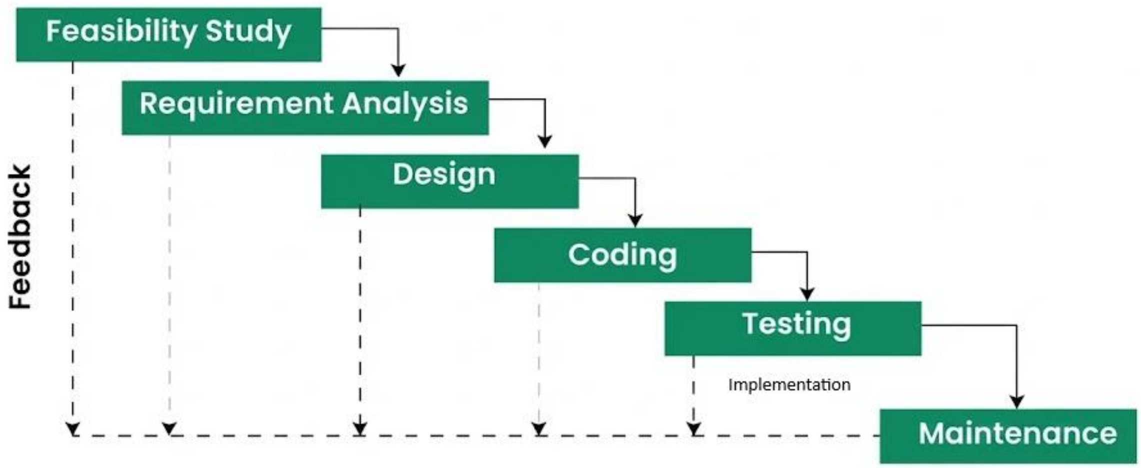


Figure 30: Iterative Waterfall Model

Feasibility Study This phase evaluated the technical, operational, and organizational feasibility of developing R-AMS. It assessed the availability of required technologies, compatibility with the company's intranet environment, and the readiness of HR users to adopt a web-based recruitment system.

Requirement Analysis In this phase, the functional and non-functional requirements of the R-AMS were identified. Existing recruitment workflows were analyzed, and system requirements such as applicant data capture, role-based access control, dashboards, security, and reporting were defined.

Design The Design phase translated the requirements into system architecture and interface specifications. This included the system structure, database design, data flow diagrams, and user interface layouts for the applicant form, HR dashboard, and Interviewer.

Coding During the Coding phase, the system was developed based on the approved design using PHP, JavaScript, HTML, TailwindCSS, CSS, and MySQL. Core modules such as applicant submission, CRUD operations, bulk upload, analytics, and authentication were implemented.

Testing This phase involved testing the system to ensure correctness, usability, performance, and security. Bugs and issues identified during testing were corrected iteratively, guided by ISO/IEC 25010 software quality standards.

Maintenance The Maintenance phase ensured the continued reliability and usability of the R-AMS. It included bug fixes, system updates, performance optimization, and minor enhancements based on user feedback.

Operation and Testing Procedure

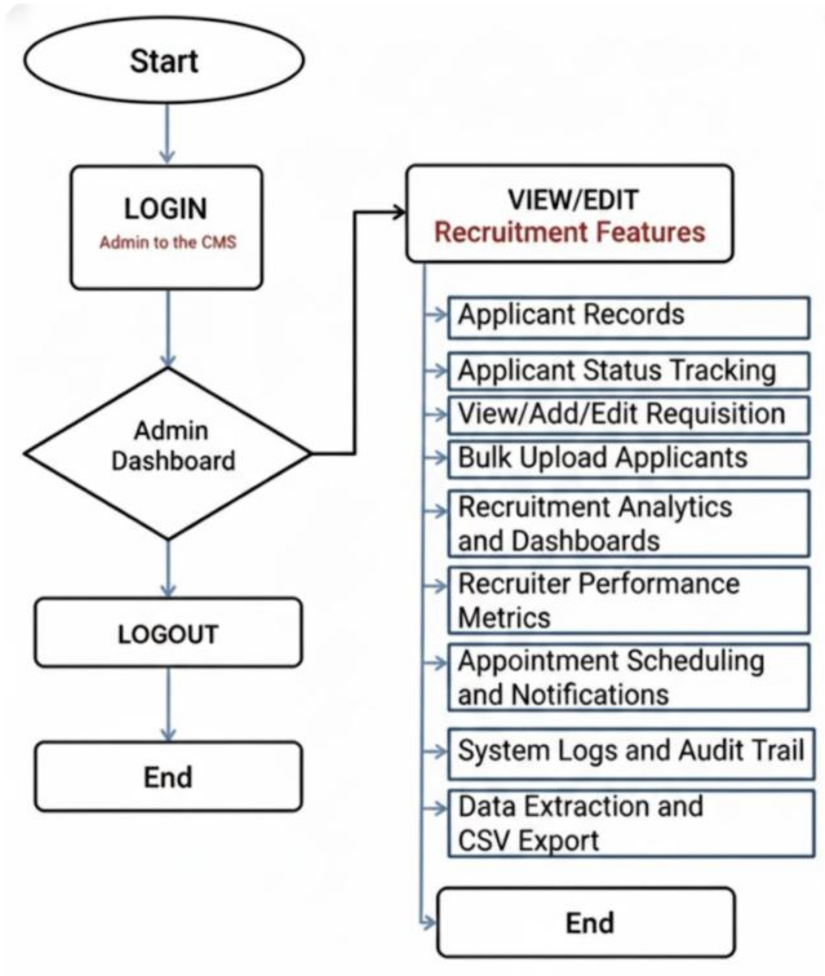


Figure 31: HR Personnel Flowchart

HR Personnel (R-AMS)

1. On the web browser, the user typed the local intranet address (URL) of the Recruitment and Applicant Management System (R-AMS) – Tier 2 (Figure 31).

2. The user entered the username and password to securely access the system.

3. Browse the Admin/HR dashboard after successful authentication.

4. The user could view or manage the following system features:

- **Applicant Records:** Viewed, updated, archived, and restored applicant profiles.
- **Applicant Status Tracking:** Monitored stages such as pre-screened, interviewed, offered, deployed, or archived.
- **View/Add/Edit Requisitions:** Managed job orders, required headcount, and project assignments.
- **Bulk Upload Applicants**: Imported multiple applicant records simultaneously via CSV.
- **Recruitment Analytics & Dashboards:** Viewed real-time charts and visual data on recruitment trends.
- **Recruiter Performance Metrics:** Tracked and evaluated the performance and scores of HR staff.
- **Interview Scheduling:** Managed and assigned interview schedules to designated interviewers.
- **System Logs & Audit Trail:** Monitored user activity, login history, and system changes for security.
- **Data Extraction & CSV Export:** Exported filtered applicant data or reports for external use.

After completing the required tasks, the user logged out of the system to end the session.

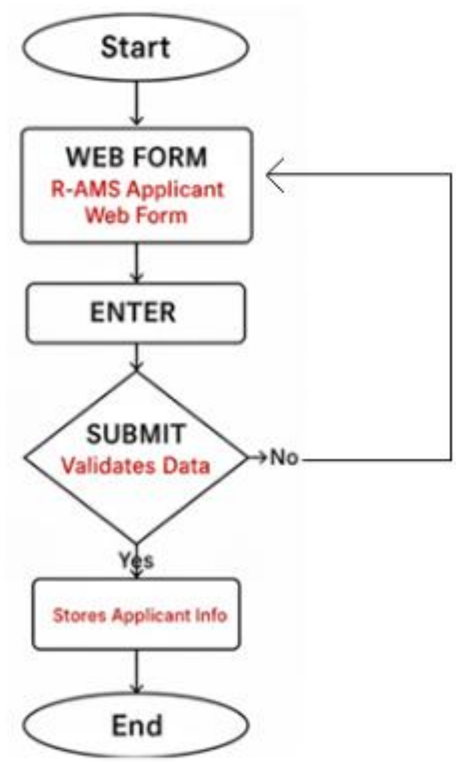


Figure 32: Applicant Flowchart

Applicant

1. On a desktop workstation within the company premises, the applicant accessed the R-AMS Applicant Web Form (Tier-1) through a web browser (Figure 32).
2. The applicant filled in the required information, such as personal details, contact information, educational background, skills, and other required fields.
3. The applicant reviewed the entered information to ensure accuracy and completeness.
4. The applicant submitted the application form.

5. The system validated the data and securely stored the applicant information in the centralized database.
6. The application process ended, and the applicant waited for further communication from the HR department.

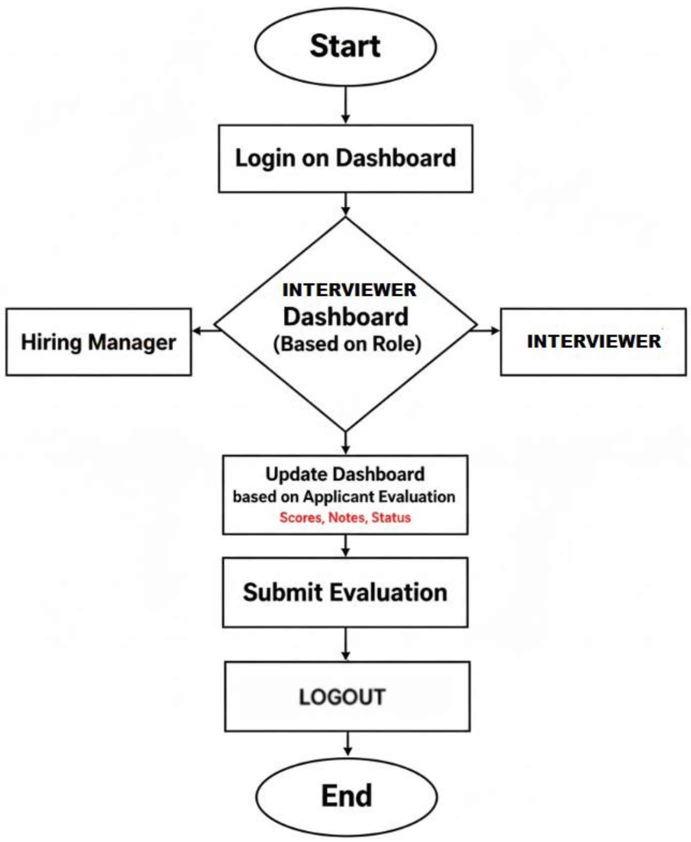


Figure 33: Interviewer Flowchart

Interviewer

1. On a secure desktop workstation within the company premises, the Interviewer accessed the R-AMS Admin Portal through a web browser and logged in using their credentials.

2. Upon successful login, the system redirected the Interviewer to the Interviewer Dashboard, where the list of assigned applicants was displayed.
3. The Interviewer selected an applicant and entered evaluation details, including interview notes and ratings.
4. The Interviewer reviewed the entered evaluation information to ensure accuracy.
5. The Interviewer submitted the evaluation. The system validated the required fields and securely stored the records.
6. The evaluation process ended, and the Interviewer returned to the dashboard to select another applicant or log out of the system.

Functional Testing

This was conducted to verify that all features and functionalities of the (R-AMS) operated according to their specified requirements. This testing ensured that each system module produced the expected output based on defined inputs.

For each development iteration, the following steps were performed:

1. Identified the functions included in each software build.
2. Prepared test input data based on functional specifications.
3. Identified the expected output based on system requirements.
4. Executed the defined test cases.
5. Compared the actual results with the expected outcome.
6. Determined the post-condition of the system after execution.
7. Evaluated whether the test case passed or failed.

The following tables present the functional testing procedures conducted to verify system correctness, portability, and usability.

Portability Testing Procedure

Table 1:

Test Case	Test Environment	Steps Undertaken	Type
Homepage loads correctly	Firefox, Chrome, Edge, Safari	1. Open browser 2. Enter R-AMS URL	Co-existence
Login page access	Firefox, Chrome, Edge, Safari	1. Open browser 2. Navigate to login page(based on role)	Co-existence
Applicant Web Form access	Firefox, Chrome, Edge, Safari	1. Open applicant form URL 2. Load form	Co-existence
Applicant submission	Firefox, Chrome, Edge, Safari	1. Fill out form 2. Click Submit	Co-existence
Interviewer Dashboard	Firefox, Chrome, Edge, Safari	1. Open Interviewer dashboard 2. Access assigned Applicant and able to updates	Co-existence
HR Dashboard access	Firefox, Chrome, Edge, Safari	1. Login as HR user 2. Load dashboard	Co-existence
Applicant Records (CRUD)	Firefox, Chrome, Edge, Safari	1. View applicant list 2. Create/Edit/Archive/Restore record	Co-existence
Bulk Upload (CSV)	Firefox, Chrome, Edge, Safari	1. Upload CSV file 2. Validate data	Co-existence
Analytics Dashboard	Firefox, Chrome, Edge, Safari	1. Open analytics module 2. View charts	Co-existence
Requisition Modal	Firefox, Chrome, Edge, Safari	1. Open Requisition Modal 2. View/Add/Edit Modal	Co-existence
System Logs	Firefox, Chrome, Edge, Safari	1. Open logs module 2. View activity logs	Co-existence
CSV Export	Firefox, Chrome, Edge, Safari	1. Select records 2. Export CSV	Co-existence

Compatibility Testing

Compatibility testing was conducted to ensure that the R-AMS operated consistently across different devices, operating systems, and browsers. This test verified that the system produced the same output under varying environments. Please note that the R-AMS did not include a mobile setup since it required company intranet access.

The following steps were followed:

1. Executed the system on different devices and environments (desktops/laptops).
2. Observed system behavior and recorded any failures.
3. Compared outputs across platforms to ensure consistency.

Compatibility Testing Procedure

Table 2

Test Case	Browser	Steps Undertaken	Type
Access R-AMS Login page	Chrome/Edge/Firefox	Open intranet URL and login	Co-existence
Access R-AMS Dashboard	Chrome/Edge/Firefox	Open intranet URL and login	Co-existence
Access R-AMS Interviewer Dashboard	Chrome/Edge/Firefox	Open intranet URL and login	Co-existence
Access Recruitment Modules	Chrome/Edge/Firefox	Navigate applicant records, dashboards, logs	Co-existence

Usability Testing

Usability testing was conducted to ensure that the R-AMS provided an intuitive, user-friendly, and accessible interface for the applicant, HR personnel, and interviewers. This test evaluated ease of navigation, clarity of content, and overall user experience.

The following steps were performed:

1. Executed the system across multiple browsers and workstations.
2. Observed user interaction and task completion.
3. Identified usability issues affecting navigation or comprehension.

Usability Testing Procedure

Table 3

Test Case	Steps Taken	Expected Output	Type
Easy navigation	1. Open homepage 2. Navigate modules	Sections accessible within 1–2 clicks	User Interface
Clear content display	1. Open forms/pages 2. Review text	Text is readable and structured	Operability
Intuitive layout	1. Browse dashboard	Menus and buttons are easily identifiable	UI Aesthetics
Applicant form usability	1. Fill form 2. Submit	Form is easy to complete	Accessibility

Evaluation Procedure

The evaluation of the R-AMS was conducted using descriptive and quantitative methods. A 4-Point Likert Scale was utilized to assess the system's acceptability based on ISO/IEC 25010 quality criteria.

The following procedure was followed to evaluate the acceptability of the developed system application:

1. Selected HR personnel, IT, and system users as evaluator-respondents.
2. Demonstrated the R-AMS and explained its modules and features.
3. Allowed evaluators to use and explore the system.
4. Distributed the evaluation questionnaires to the respondents once the demonstration and exploration were completed.
5. Instructed the evaluators to rate the system using the ISO/IEC 25010 criteria.
6. Tabulated the results using Microsoft Excel.
7. Interpreted the results using the numerical rating scale.

Table 4

4-Point Likert Scale

Numerical Rating	Adjectival Rating
4	Highly Acceptable
3	Very Acceptable
2	Acceptable
1	Not Acceptable

Chapter 4

RESULTS AND DISCUSSION

This chapter contains the project description, project structure, project capabilities and limitations, and results of evaluation for the project.

Project Description

The study developed a web-based R-AMS designed for company use. The web application, which was developed to assist the company, was particularly for the hiring process.

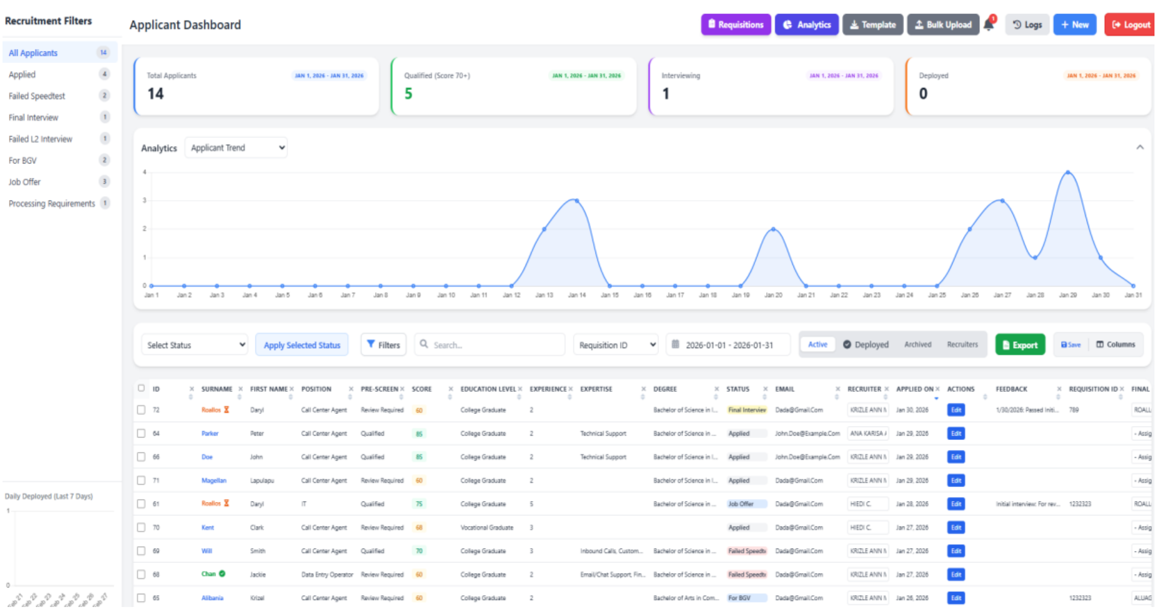


__Figure 34__: Web application Interface

Project Structure

The web application was designed and developed to be accessible only on web browsers that were accessible via the company intranet.

The Web Application

Figure 34 shows the web application, which was the Home page interface where features and options were clearly displayed for the user.

Here were the major features of the web application:

- *Applicant Web form (Figure 35):* This served as the primary gateway for applicants, wherein they could fill out their information (Personal data, Skills, Educational background, Employment history, etc.). It automatically calculated a pre-screening score to help HR recruiters shortlist qualified applicants.

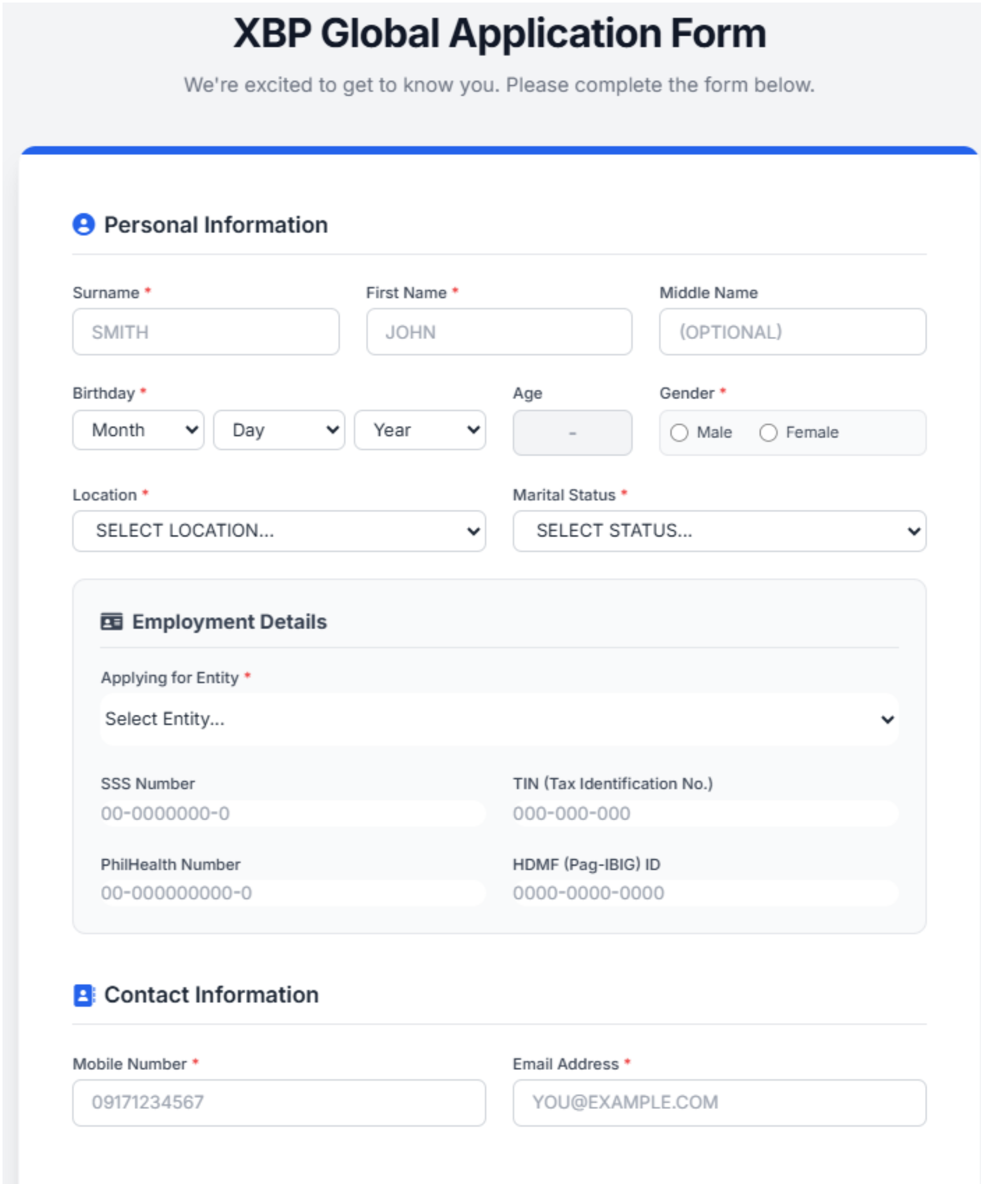


***Figure 35**: Applicant Web Form*

- *Pre-screening Calculation (Figure 36):* The automated scoring evaluated job applicants' submitted forms based on four key metrics: Education, Experience, Skills, and position-specific "Hard Gates". The function had a linear scoring pipeline followed by a validation layer. It calculated a numerical score (out of 100+) and mapped it to a qualitative status.

- The system utilized "Contextual Scoring." Entry-level roles (e.g., Data Entry, Call Center) received higher points for lower education levels than professionals.

Education Level	Professional Roles	Entry-Level Roles
Post Graduate	30 pts	30 pts
College Graduate	25 pts	30 pts
Vocational Graduate	20 pts	28 pts
Some College	15 pts	25 pts
High School Grad	5 pts	20 pts

- Weight was distributed toward tenure; years of experience were the primary predictor of job readiness.

Experience	Points
5+ Years	50 pts
3 Years	40 pts
2 Years	30 pts
1 Year	15 pts
No Experience	5 pts

- Skill Matching (Max 25 points): The system utilized intelligent skill mapping. It compared the applicant's selected skills against a global data option (from the database ref_skills).

- Premium Points (+5) were awarded if the skill category matched the applied position.
- Standard Points (+1) were awarded for general skills.

- Hard Gate Validation ("Auto-Fail" logic): This was a critical feature that ensured compliance with industry standards. Even if an applicant had a high score, they were marked as "Underqualified," and their score was reset to 0 if they failed specific criteria.

Position	**Required Condition**
Nurse	Must be College/Post Graduate + Nursing degree
Accounting	Must be College/Post Graduate + Accountancy degree
Human Resource	Must have HR/Psychology related degree
IT	Minimum Vocational/College + IT/CS/Engineering degree
Medcoder	Requires Medical Coding skill OR medical-related degree
Manager Level	Requires experience (not 0–1 year)
Quality Analyst	Requires prior experience
Procurement	Requires College/Post Graduate
Reports Analyst	Requires experience

ID	SURNAME	FIRST NAME	POSITION	PRE-SCREEN	SCORE	EDUCATION LEVEL	EXPERIENCE	EXPERTISE	DEGREE
94	Aquino	Marz	Data Entry Operator	Low Match	30	Some College	0		
93	De Guzman	Jean	Data Entry Operator	Low Match	35	College Graduate	0		Bachelor of Arts in Communication
92	Alibania	Krizle Ann Mae	Data Entry Operator	Low Match	35	College Graduate	0		Bachelor of Elementary/Secondary Education
91	Garcia	Barbara	Human Resource	Qualified	76	College Graduate	5	Recruitment/Sourcing	Bachelor of Science in Information Technology
90	Lee	Richard	Call Center Agent	Qualified	71	High School Graduate	5	International Voice	Bachelor of Science in Business Administration
72	Roallos	Daryl	Call Center Agent	Review Required	60	College Graduate	2		Bachelor of Science in Information Technology
64	Parker	Peter	Call Center Agent	Qualified	85	College Graduate	2	Technical Support	Bachelor of Science in Computer Science

***Figure 36**: Pre-screening Result*

Registration/Login Form (Figure 37): This served as the registration and login form for the HR staff and interviewers. Once they completed the registration, the system admin updated their role mapping as HR Staff, HR Manager, or Interviewer. Every role had distinct features.

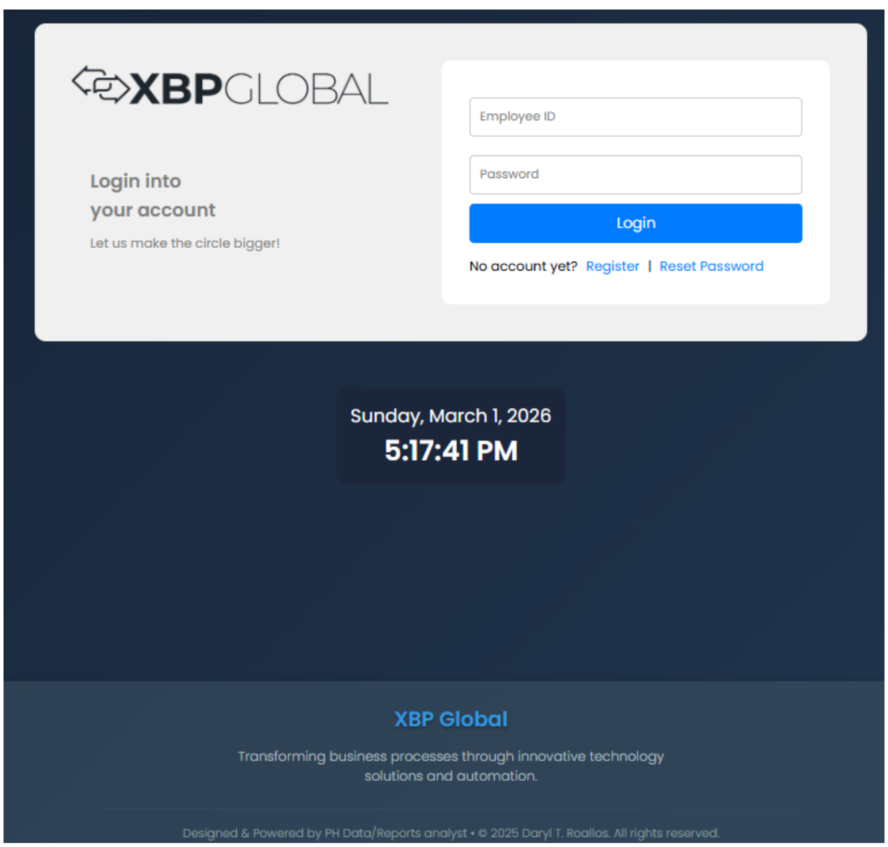


***Figure 37**: Registration/Login form*

Requisition Management Modal (Figure 38): This was a centralized tool for tracking and fulfilling approved headcount requests across various projects. It allowed HR and recruitment teams to create new job requisitions, update statuses (Open, Hold, Closed), and monitor the aging of open roles.

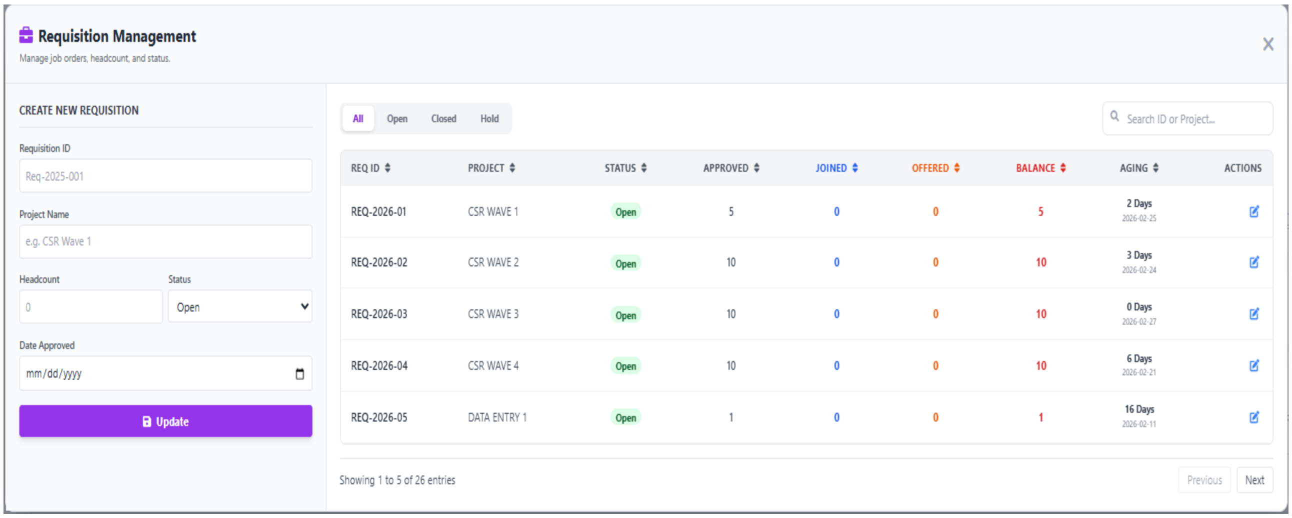


***Figure 38**: Requisition Management Modal*

Interviewer Web page (Figure 39): This served as the Appointment section, where the Department Managers/Interviewers could see all the applicants assigned to them. This area also allowed the interviewer to provide feedback and evaluate the applicant (Passed, Failed, or Rescheduled).

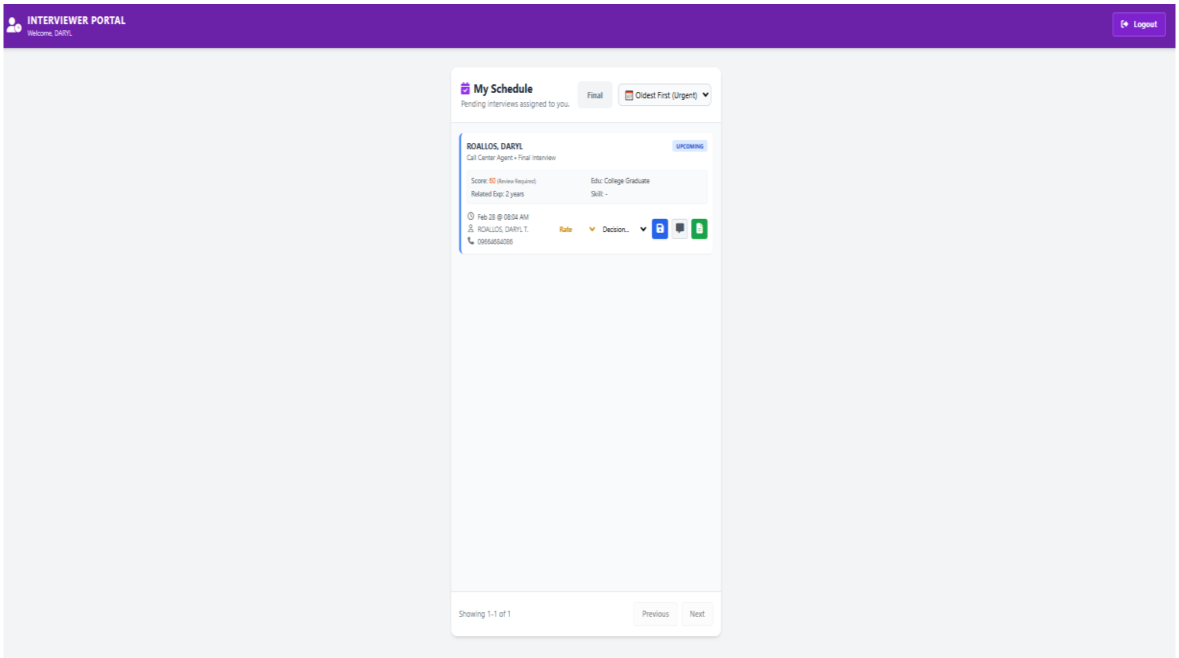


***Figure 39**: Interviewer Web Page*

HR Interview modal viewer (Figure 40): This served as a viewer for all upcoming or due interviews. It acted as a reminder for HR recruiters to prioritize pending interviews. As the initial interviewer, HR could provide feedback and evaluate the applicant before forwarding them to the Department Manager dashboard for the next stage of the interview.

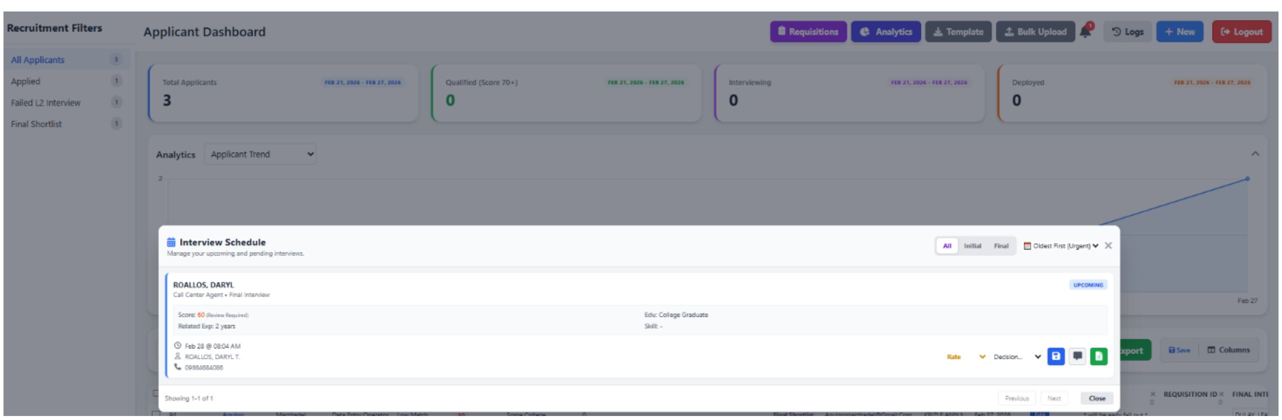


***Figure 40**: HR Interview Modal viewer*

Sidebar Recruitment Filter and HR Dashboard table (Figures 41-42): This worked as an easy filter for HR recruiters to reflect specific statuses on the dashboard table. Upon clicking, it instantly filtered the HR table based on the selected status. It also helped the user identify the number of statuses they wanted to monitor.

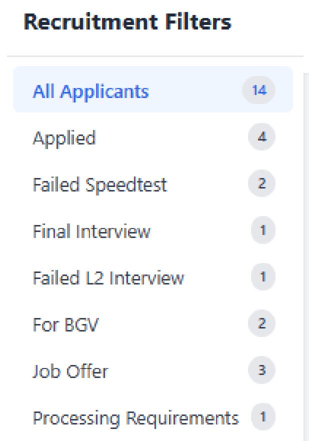


***Figure 41**: Sidebar Recruitment Filter*

ID	SURNAME	FIRST NAME	POSITION	PRE-SCREEN	SCORE	EDUCATION LEVEL	EXPERIENCE	EXPERTISE	DEGREE	STATUS	EMAIL	RECRUITER	APPLIED ON	ACTIONS
72	Roallos	Daryl	Call Center Agent	Review Required	60	College Graduate	2		Bachelor of Science in I...	Final Interviev	Dada@Gmail.Com	KRIZLE ANN N	Jan 30, 2026	Edit
64	Parker	Peter	Call Center Agent	Qualified	85	College Graduate	2	Technical Support	Bachelor of Science in ...	Applied	John.Doe@Example.Com	ANA KARISA A	Jan 29, 2026	Edit
66	Doe	John	Call Center Agent	Qualified	85	College Graduate	2	Technical Support	Bachelor of Science in I...	Applied	John.Doe@Example.Com	KRIZLE ANN N	Jan 29, 2026	Edit
71	Magellan	Lapulapu	Call Center Agent	Review Required	60	College Graduate	2		Bachelor of Science in I...	Applied	Dada@Gmail.Com	KRIZLE ANN N	Jan 29, 2026	Edit
61	Roallos	Daryl	IT	Qualified	75	College Graduate	5		Bachelor of Science in ...	Job Offer	Dada@Gmail.Com	HIEDI C.	Jan 28, 2026	Edit
70	Kent	Clark	Call Center Agent	Review Required	68	Vocational Graduate	3			Applied	Dada@Gmail.Com	HIEDI C.	Jan 27, 2026	Edit
69	Will	Smith	Call Center Agent	Qualified	70	College Graduate	3	Inbound Calls, Custom...	Bachelor of Science in ...	Failed Speedte	Dada@Gmail.Com	KRIZLE ANN N	Jan 27, 2026	Edit
68	Chan	Jackie	Data Entry Operator	Review Required	60	College Graduate	2	Email/Chat Support, Fin...	Bachelor of Science in ...	Failed Speedte	Dada@Gmail.Com	KRIZLE ANN N	Jan 27, 2026	Edit
65	Alibania	Krizel	Call Center Agent	Review Required	60	College Graduate	2		Bachelor of Arts in Com...	For BGV	Dada@Gmail.Com	KRIZLE ANN N	Jan 26, 2026	Edit
60	Dada	Wqeqewq	Facilities	Review Required	55	College Graduate	2		Bachelor of Arts in Com...	For BGV	Dada@Gmail.Com	JENNYLYN M.	Jan 20, 2026	Edit

***Figure 42**: HR Dashboard table*

Smart Filter (Figure 43): This allowed the user to filter multiple criteria based on their preference.

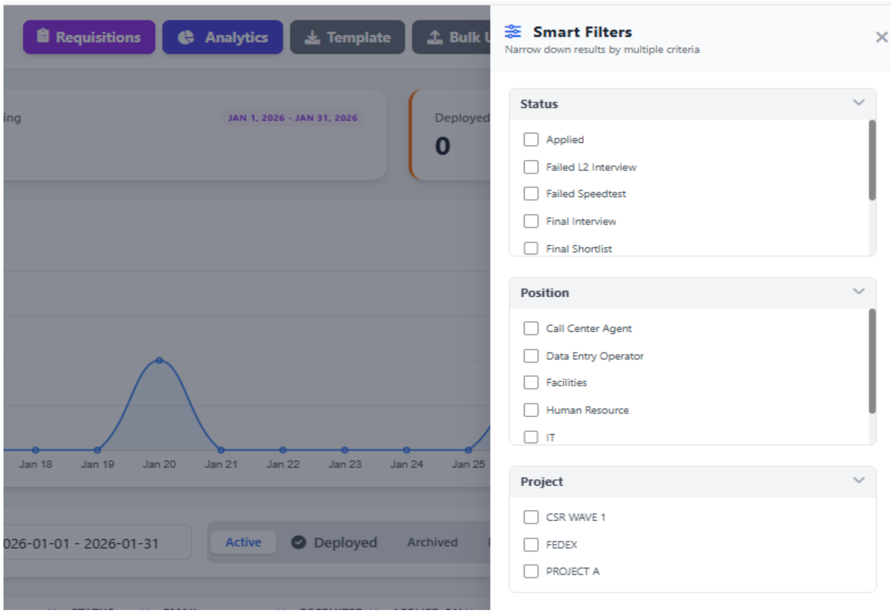


Figure 43*: Smart Filter*

Cards, Datetime Picker, and Quick Analytics (Figure 44): These served as a quick overview for HR recruiters. The cards displayed the count of applicants based on categories (Total Applicants, Qualified, For Interview, and Deployed). The Datetime Picker acted as a filter, allowing users to select specific or predefined dates. Quick Analytics provided a chart that users could toggle on or off, with the ability to change the criteria view.

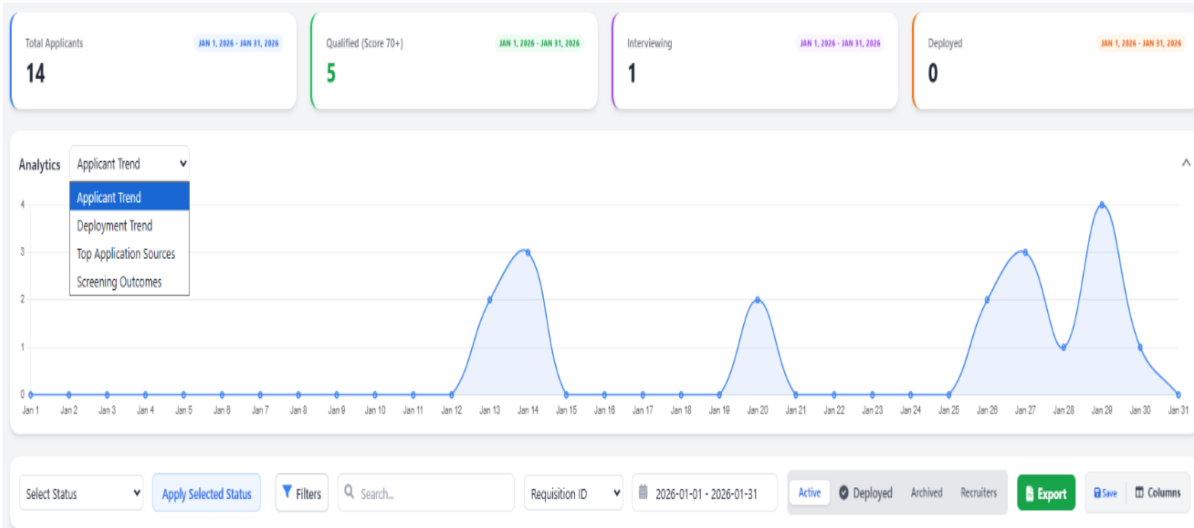
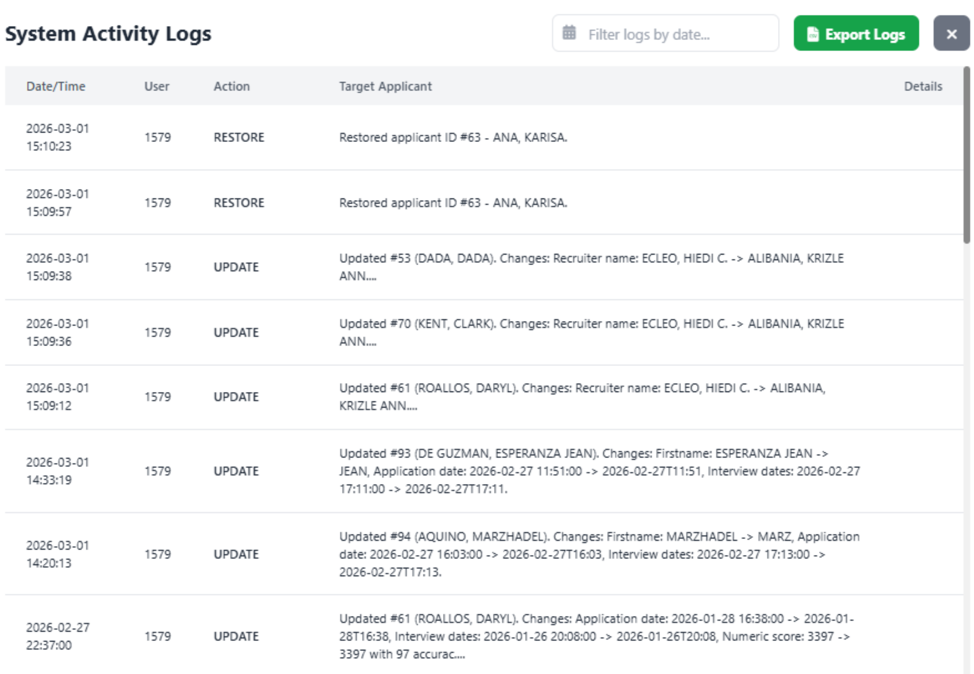


Figure 44: Cards, Datetime Picker and Quick Analytics

System Logs (Figure 45): These served as an audit trail for any actions made on the system by a specific user. This was used to track and maintain user accountability.



System Activity Logs

Filter logs by date... | Export Logs

Date/Time	User	Action	Target Applicant	Details
2026-03-01 15:10:23	1579	RESTORE	Restored applicant ID #63 - ANA, KARISA.	
2026-03-01 15:09:57	1579	RESTORE	Restored applicant ID #63 - ANA, KARISA.	
2026-03-01 15:09:38	1579	UPDATE	Updated #53 (DADA, DADA). Changes: Recruiter name: ECLEO, HIEDI C. -> ALIBANIA, KRIZLE ANN....	
2026-03-01 15:09:36	1579	UPDATE	Updated #70 (KENT, CLARK). Changes: Recruiter name: ECLEO, HIEDI C. -> ALIBANIA, KRIZLE ANN....	
2026-03-01 15:09:12	1579	UPDATE	Updated #61 (ROALLOS, DARYL). Changes: Recruiter name: ECLEO, HIEDI C. -> ALIBANIA, KRIZLE ANN....	
2026-03-01 14:33:19	1579	UPDATE	Updated #93 (DE GUZMAN, ESPERANZA JEAN). Changes: Firstname: ESPERANZA JEAN -> JEAN, Application date: 2026-02-27 11:51:00 -> 2026-02-27T11:51, Interview dates: 2026-02-27 17:11:00 -> 2026-02-27T17:11.	
2026-03-01 14:20:13	1579	UPDATE	Updated #94 (AQUINO, MARZHADEL). Changes: Firstname: MARZHADEL -> MARZ, Application date: 2026-02-27 16:03:00 -> 2026-02-27T16:03, Interview dates: 2026-02-27 17:13:00 -> 2026-02-27T17:13.	
2026-02-27 22:37:00	1579	UPDATE	Updated #61 (ROALLOS, DARYL). Changes: Application date: 2026-01-28 16:38:00 -> 2026-01-28T16:38, Interview dates: 2026-01-26 20:08:00 -> 2026-01-26T20:08, Numeric score: 3397 -> 3397 with 97 accurac....	

Figure 45: System Logs

Customizable Columns (Figure 46): This allowed the user to customize the columns by adding or removing their preferred columns.

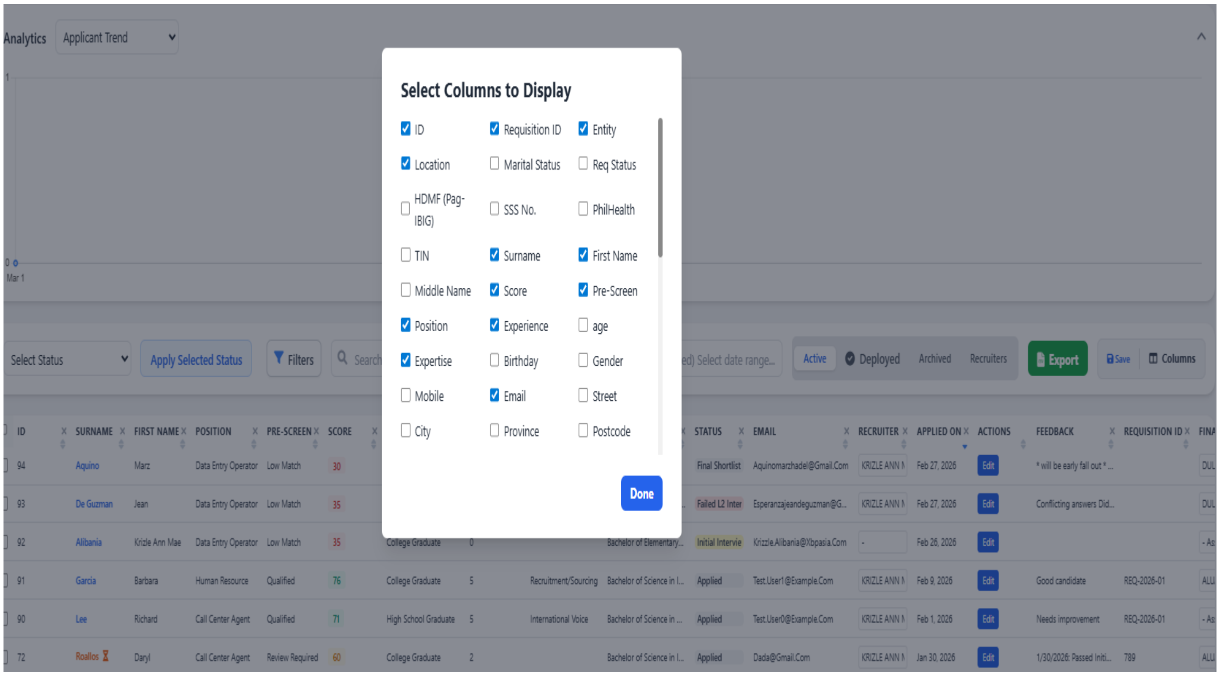


***Figure 46**: Customizable Columns*

Custom Analytics (Figure 47): This allowed the user to view analytics based on their preferred Categories, Metrics, Locations, and Dates. They could also change the visualization view from Bar Chart, Pie Chart, and Data Sheet formats.

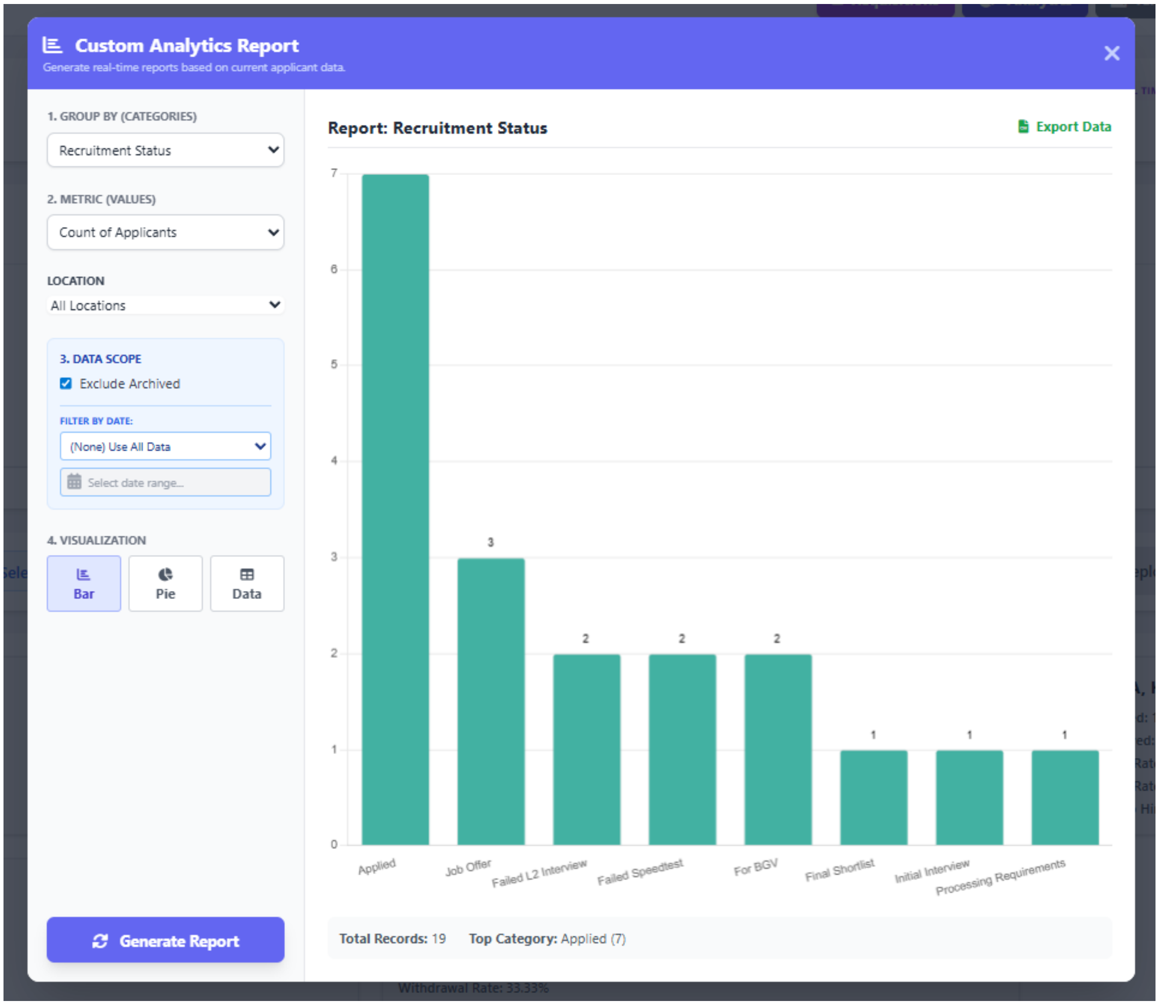


Figure 47*: Custom Analytics Report*

Custom Export (Figure 48): This export function acted as a flexible reporting tool. It allowed users to export reports based on the data displayed in the HR table or generate reports according to their required format. Users simply needed to map their desired headers to the corresponding headers available in the dashboard. They could also save this template to the user's browser cache and use it again whenever they needed it.

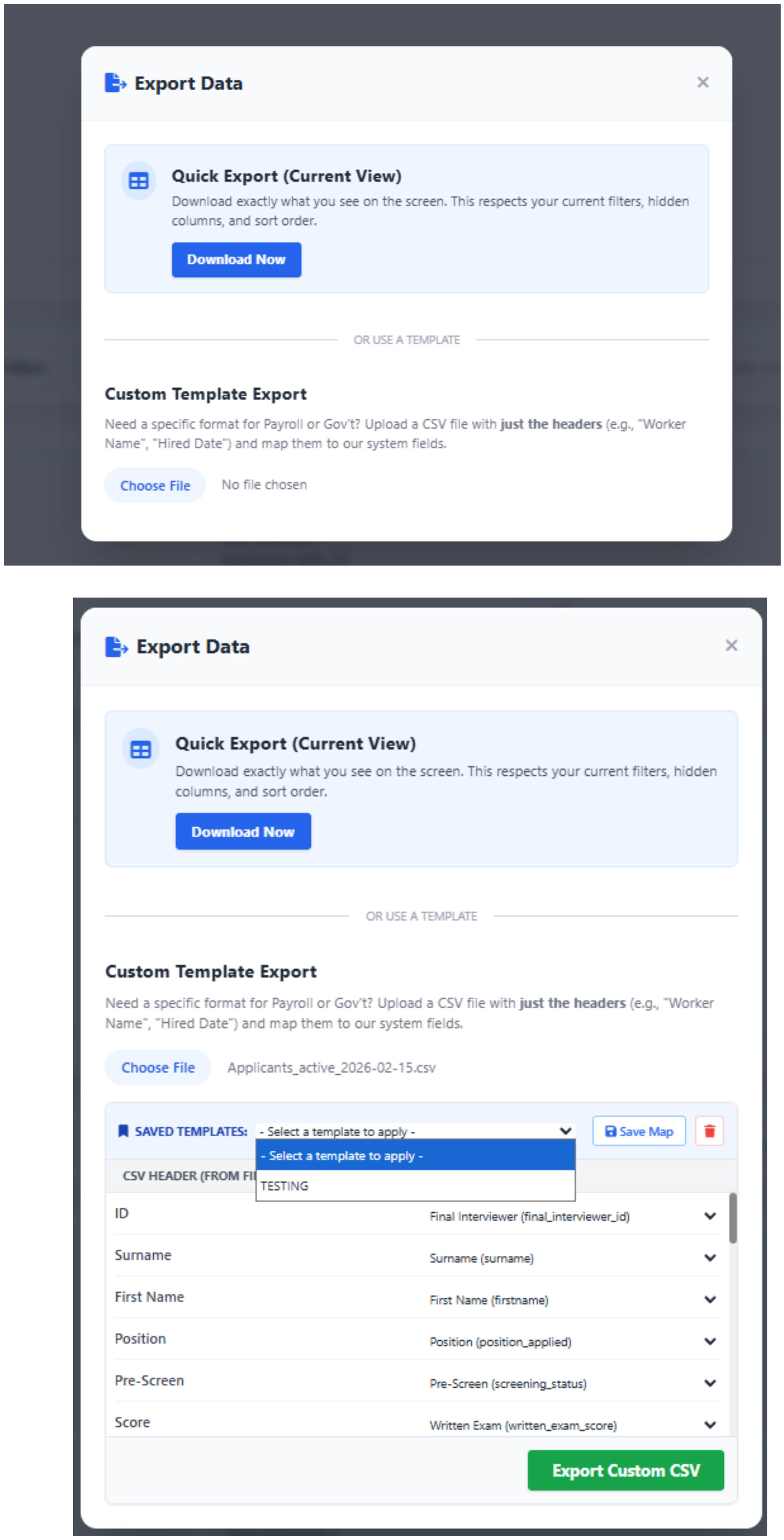


***Figure 48**: Custom Export*

Requirements Checklist (Figure 49): This served as a requirement tracker for the HR Recruiter to monitor which documents had and had not yet been submitted.

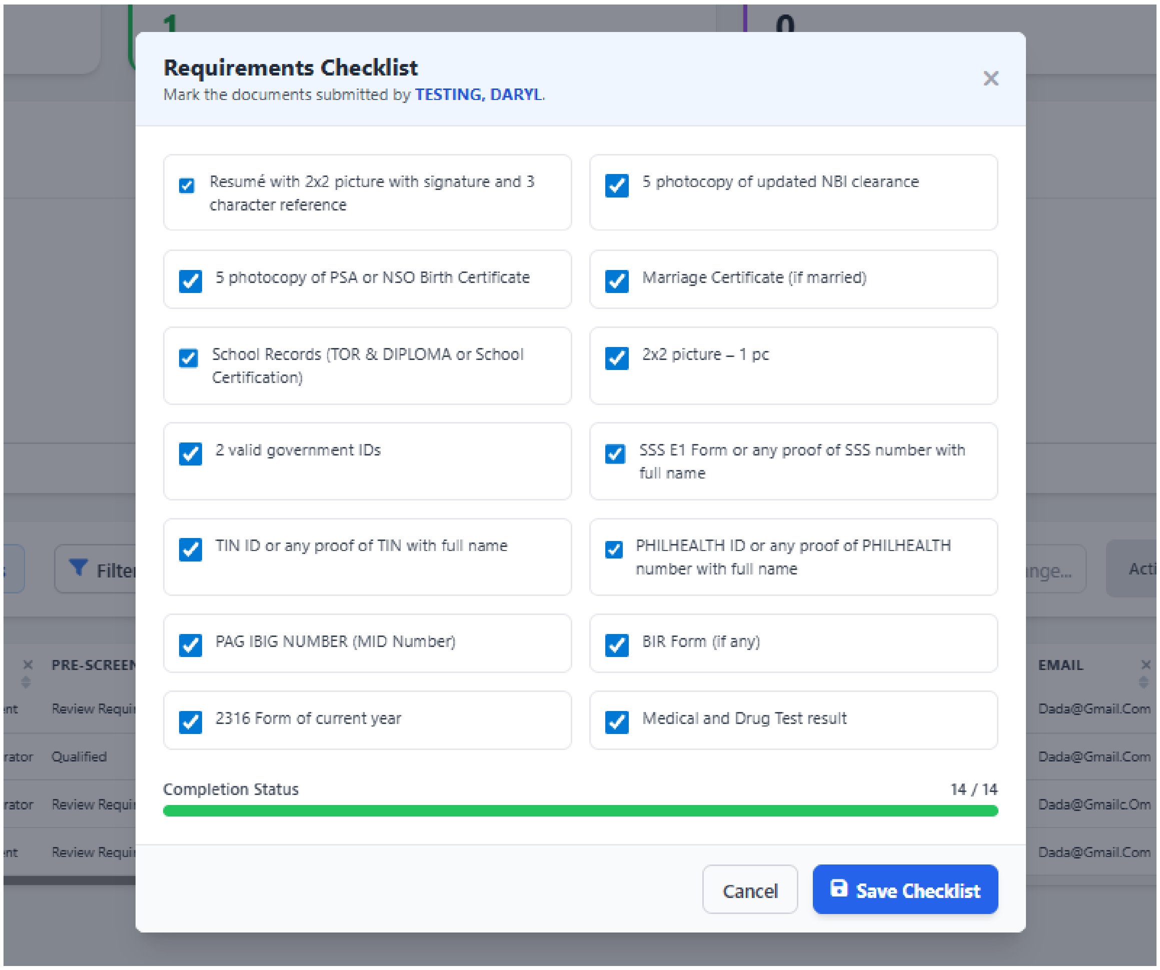


***Figure 49**: Requirements Checklist*

Applicant Edit modal (Figure 50): This served as the editing section for the HR Recruiter. The data fields were initially displayed as greyed out to prevent accidental modifications. If the user needed to update specific details, they had to click the Edit button to enable editing.

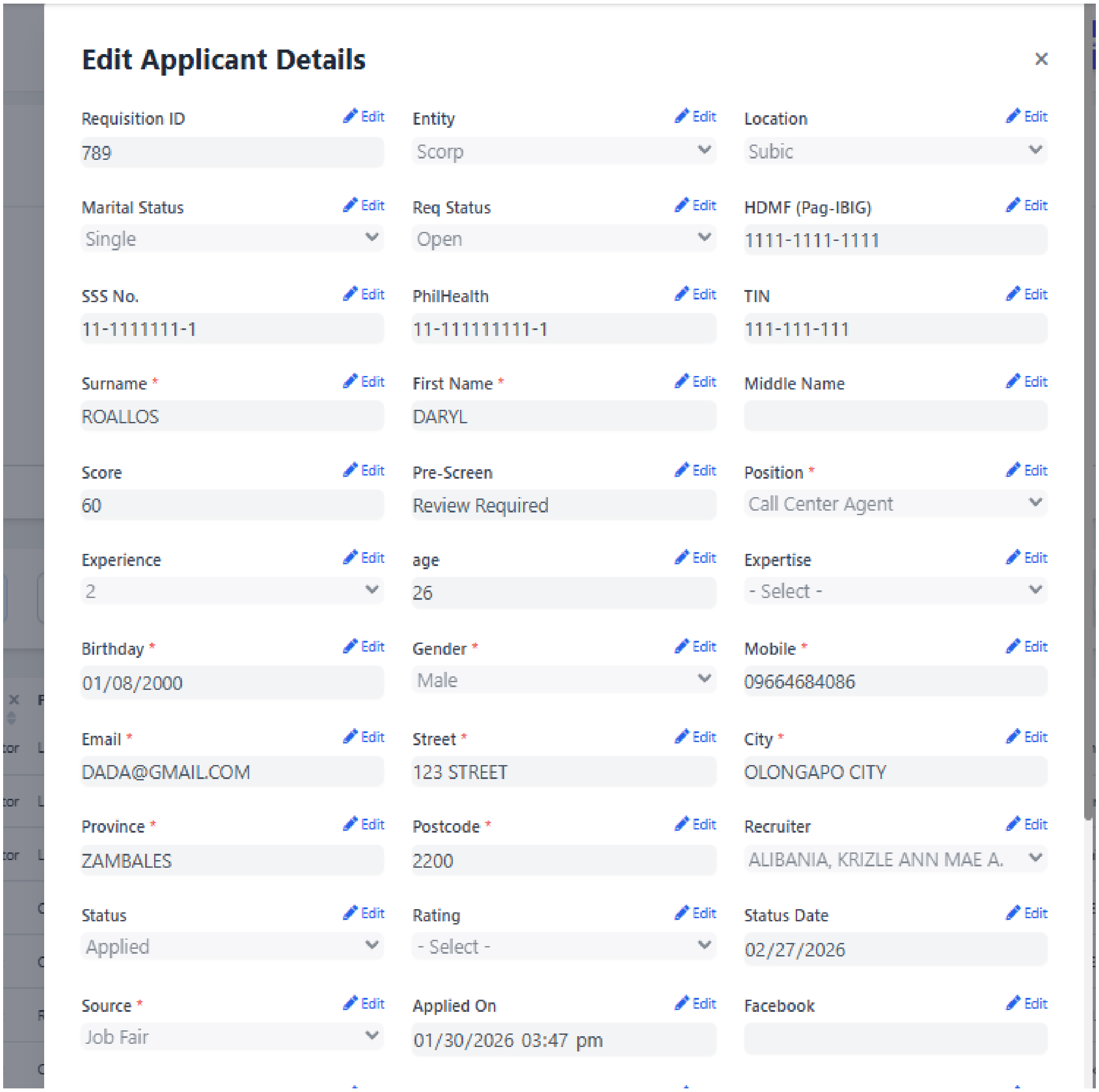


***Figure 50**: Applicant Edit Modal*

Project Capabilities and Limitations

Capabilities of R-AMS:

The R-AMS could efficiently process and evaluate high volumes of applicant data. It supported both individual web form submissions and bulk CSV imports, instantly converting raw data into actionable applicant profiles.

1. It included an advanced search, filter, and analytics feature to help HR staff quickly locate specific applicants and generate real-time reports based on criteria such as recruitment status, location, and application source. A bulk import feature was also included to ensure the HR recruiter could be more efficient.
2. The system provided an intelligent pre-screening and automated scoring engine to help HR shortlist the applicants. It also provided a detailed and organized preview of candidate qualifications to ensure recruitment accuracy and achieve faster hiring times.
3. The interface was intuitive. It ensured ease of use and navigation, featuring drag-and-drop customizable column management, a Requisition modal, and dynamic status updates, with minimal learning required for effective operation.
4. The application implemented role-based access control (RBAC), enabling users to access dedicated dashboards and system functionalities according to their assigned roles.

Limitations of R-AMS

The limitations of the application were as follows:

1. The application required standardized CSV templates during bulk uploads. Unstructured data, highly modified headers, or corrupted date formats could result in import errors.
2. The application only worked on a Local Intranet connection and had no mobile view.
3. The pre-screening score logic was limited to predefined, objective metrics (such as education level and years of experience). It could not evaluate subjective candidate qualities—such as cultural fit, attitude, or communication skills—which still required manual human assessment during interviews.
4. No AI feature was implemented in the application.

Test Results

The web application was created using HTML, PHP, Tailwind/CSS, MySQL, and JavaScript. As such, the web application could run on any web browser only on a desktop. The test results on the compatibility and usability of the developed system are presented in the following tables.

Table 5.

Compatibility Testing Procedure.

Test Case	**Steps Undertaken**	**Observed Results**
(Co-existence) Access the R-AMS Applicant form, HR Dashboard, and Interviewer dashboard from various browsers	1. Open different web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) from a windows PC .	The HR Dashboard, login screen, Applicant Web Form and Interviewer dashboard are displayed correctly and consistently across all modern web browsers.
(Interoperability) The user can interact with complex R-AMS features (e.g., Bulk Uploads, Analytics).	1. Navigate to the HR Dashboard. 2. Upload a formatted CSV file using the Bulk Upload feature. 3. Adjust filters on the sidebar and click "Generate Report" in the Analytics modal.	The web application successfully processes the CSV file into the database, dynamically updates the applicant table, and renders the pie/bar charts interactively based on user inputs.
(Co-existence) The R-AMS Applicant form, HR Dashboard, and Interviewer dashboard	1. Access the HR Dashboard, Applicant web form and Interviewer using a desktop machine.	The system is fully accessible. The HR Dashboard, Applicant Webform and Interviewer

work on the Windows version		dashboard functions optimally only on desktop environments.
(Co-existence) The R-AMS Applicant form, HR Dashboard, and Interviewer dashboard UI respond correctly to various display features.	Click Recruitment filter, Modals, Analytics, and Extract data, Add and remove columns using the desktop machine.	The UI responded correctly to multiple display features. Recruitment filters, modals, analytics, data extraction, and column customization functioned as expected while maintaining consistent table layout and interface stability.

Table 6.

Usability Test Results

Test Case	**Steps Undertaken**	**Observed Results**
(Learnability) [HR Dashboard] Uploading Applicant Data via Bulk CSV	1. Clicked the "Bulk Upload" button on the HR Dashboard. 2. Selected a standard CSV file containing applicant data. 3. Clicked the "Upload" button to process the batch.	The file upload, data parsing (including the automatic date/time fixer), and import functionalities were intuitive and easy to navigate. System responses and progress indicators were consistent.
(Operability) [HR Dashboard] Explore the Applicant Filtering and Search Features	1. Opened the R-AMS HR Dashboard. 2. Used the sidebar filters to select specific statuses (e.g., "Deployed"). 3. Used the search bar to locate an applicant by name or specific skill.	The user successfully applied filters and quickly located specific applicant records. The table dynamically updated without requiring a page reload.

(User Interface Aesthetics) [HR Dashboard] View Analytics and Generated Reports	1. Opened the R-AMS HR Dashboard. 2. Clicked the "Analytics" button to open the reporting modal. 3. Selected criteria (e.g., Application Source, Location) and clicked "Generate Report." 4. Toggled between Bar, Pie, and Datasheets visualizations.	The Analytics modal displayed clear, visually appealing charts (using Chart.js) and data tables. The interface cleanly presented complex data, such as top applicant sources and status distributions.
(Accessibility) [HR Dashboard] Managing Job Requisitions	1. User opened the R-AMS HR Dashboard. 2. Fill out the "Requisition Management" module. 3. Created a new requisition, setting a target headcount. 4. Toggled the status of the requisition between Open, Hold, and Closed.	R-AMS provided an accessible, centralized modal to track approved headcount requests against actual deployed candidates, with color-coded badges indicating current capacity.

(Appropriateness/Recognizability) [HR Dashboard] Processing Individual Applicants via the Edit Modal	1. Open the R-AMS HR Dashboard. 2. Click the "Edit" button on a specific applicant row. 3. Review the candidate's personal data.	The application provided a feature to allow the user to edit applicant information if needed.
(User Error Protection) [HR Dashboard] Display Notification Messages (Login)	1. Navigate to the R-AMS Login Page. 2. Input an invalid Employee ID or incorrect password. 3. Click the Login button.	The app prevented access and displayed a prompt containing details of the invalid action ("Incorrect Password" or "Employee ID Not Found").
(User Error Protection) [HR Dashboard] Preventing Accidental Data Loss during Edits	1. User opened the R-AMS HR Dashboard. 2. Clicked "Edit" on an applicant who applied via a custom Application Source (e.g., "TikTok Ads" from a CSV upload). 3. Modified the applicant's status and saved.	The system successfully preserved the custom source data by dynamically injecting it into the dropdown options, preventing the user from accidentally erasing non-standard data.

(Learnability) [Applicant Form] Familiarization with the UI	1. Applicant navigated to the public-facing R-AMS Applicant Web Form URL. 2. Scrolled through the form sections (Personal Details, Employment, References).	Form sections were logically grouped and clearly labeled. Required fields were distinctly marked, making it easy for applicants with no prior experience to understand the interface.
(Operability) [Applicant Form] Submitting a Complete Application	1. Filled in all required personal and professional information. 2. Selected the position and dynamic skills from the dropdowns.	The submission process was smooth. The system displayed a loading state and successfully showed a confirmation upon completion. Data instantly routed to the HR Dashboard.

(User Error Protection) [Applicant Form] Mandatory Background Check	1. Filled in all required fields on the application form. 2. Intentionally left the "Background Check Consent" checkbox unchecked. 3. Clicked "Submit Application."	The browser natively blocked the form submission and highlighted the mandatory checkbox with a requirement prompt, effectively preventing incomplete legal consents.
(Learnability) [Interviewer Dashboard] Viewing Assigned Applicants	1. Logged into R-AMS using Interviewer credentials. 2. Navigated to the Interviewer Dashboard. 3. Verified the list of applicants displayed on the main table.	The dashboard cleanly filtered out unrelated data, displaying only the specific applicants assigned to that interviewer's ID.
(Operability) [Interviewer Dashboard] Submitting Interview Feedback	1. Clicked on an assigned applicant's record. 2. Input textual feedback and selected a numerical interview rating (1-5 stars).	The interface allowed the interviewer to easily input evaluation metrics and transition the candidate's status in a few clicks.

(User Error Protection) [System Access] Role-Based Restrictions	1. Logged in as an Interviewer. 2. Attempted to access HR-specific modules by typing the HR dashboard URL directly into the browser.	The system successfully recognized the user's role_id and prevented access, redirecting the user back to their designated dashboard or showing an unauthorized error.

Evaluation Results

The R-AMS Application was evaluated to determine its acceptability. The respondents consisted of system beneficiaries like our HR Team, company IT professionals, company Interviewers/Managers, and external HR personnel who worked on the same system as ATS. Table 7 shows the summary of ratings.

Table 7.

Summary of evaluation ratings.

Criteria	Mean	Adjectival Rating
A. Functional Suitability		
1. Functional Completeness	3.71	Highly Acceptable
2. Functional Correctness	3.71	Highly Acceptable
3. Functional Appropriateness	3.71	Highly Acceptable
Sub-Mean	**3.71**	**Highly Acceptable**
B. Performance Efficiency		
1. Time Behavior	3.71	Highly Acceptable
2. Resource Utilization	3.86	Highly Acceptable
3. Capacity	3.43	Highly Acceptable
Sub-Mean	**3.67**	**Highly Acceptable**

C. Compatibility		
1. Co-existence	3.57	Highly Acceptable
2. Interoperability	3.71	Highly Acceptable
Sub-Mean	**3.64**	**Highly Acceptable**
D. Usability		
1. Appropriateness Recognizability	3.57	Highly Acceptable
2. Learnability	3.71	Highly Acceptable
3. Operability	3.86	Highly Acceptable
4. User Error Protection	3.57	Highly Acceptable
5. User Interface Aesthetics	3.71	Highly Acceptable
6. Accessibility	3.71	Highly Acceptable
Sub-Mean	**3.69**	**Highly Acceptable**
E. Reliability		
1. Availability	3.57	Highly Acceptable
2. Fault Tolerance	3.43	Highly Acceptable
Sub-Mean	**3.5**	**Highly Acceptable**
F. Maintainability		
1. Reusability	3.71	Highly Acceptable
2. Modifiability	3.71	Highly Acceptable
3. Testability	3.71	Highly Acceptable
Sub-Mean	**3.71**	**Highly Acceptable**
G. Portability		
1. Adaptability	3.71	Highly Acceptable
2. Installability	3.71	Highly Acceptable
Sub-Mean	**3.71**	**Highly Acceptable**
Grand Mean	**3.66**	**Highly Acceptable**

In Table 7, the web application obtained its highest sub-mean ratings under "Functional Suitability," "Maintainability," and "Portability," all tying with a sub-mean of 3.71 or "Highly Acceptable”. Specifically, the individual criterion of "Operability" under Usability achieved the absolute highest score overall at 3.86, indicating that the application was easy to control and navigate for the end-users.

The web application obtained its lowest rating under "Reliability" with a sub-mean rating of 3.50, which was still classified as "Highly Acceptable". The individual criteria of "Capacity" and "Fault Tolerance" scored the lowest at 3.43, suggesting minor areas for future scalability optimization, though they comfortably met current operational standards.

Overall, the system obtained an exceptional grand mean rating of 3.66 or "Highly Acceptable," which meant that the developed Recruitment Applicant Management System (R-AMS) successfully addressed its intended requirements and was highly acceptable and beneficial to its intended stakeholders in streamlining the recruitment workflow.

Qualitative Feedback Summary: Evaluators consistently praised the system's intuitive design and direct impact on workflow efficiency. IT and HR professionals highlighted the clean, smooth interface and the straightforward nature of the data/file upload modules. Management and HR end-users specifically commended the system's "roll-out readiness," noting that the searchable database, unique requirements checklist, and automated interview reminders significantly saved time, reduced human error, and streamlined the overall recruitment process.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

This chapter presents the summary of findings, the conclusions drawn from the findings, and the corresponding recommendations for further enhancement of the project.

Summary of Findings

Based on the evaluation results, the developed web application entitled "Recruitment Applicant Management System (R-AMS)" enabled HR professionals, recruiters, and interviewers to efficiently manage, track, and evaluate high volumes of applicants in a centralized platform. The following is the summary of findings using the web application:

- **Functional Suitability:** The evaluators rated the web application as highly acceptable regarding the application's core functions, such as the automated screening score, bulk CSV data parsing, requisition tracking, and the overall responsiveness of established recruitment objectives.
- **Performance Efficiency:** The evaluators rated the web application as highly acceptable regarding how the application processed heavy tasks such as rendering real-time Analytics charts and processing bulk uploads.
- **Compatibility:** The evaluators rated the web application as highly acceptable regarding its interoperability, effectively handling bulk uploads and platform/browser responsiveness.
- **Usability:** The evaluators rated the web application as highly acceptable regarding how the framework was appropriate for the users' needs. Features like the drag-and-drop table

columns, smart filters, visual dashboards, and recruitment sidebar filter made the system easy to learn, operate, and control.

- **Reliability:** The evaluators rated the web application as highly acceptable regarding its data availability and fault tolerance, particularly how its role-based access control reliably restricted access to sensitive HR data based on user credentials.
- **Maintainability:** The evaluators rated the web application as highly acceptable regarding its modular codebase, ensuring that changes or updates to specific components (such as dropdown logic or table structures) had minimal impact on the continued operations of the rest of the system.
- **Portability:** The evaluators rated the web application as highly acceptable regarding how seamlessly the public-facing Applicant Web Form scaled for walk-in applicants, while the complex administrative HR Dashboard and interviewer dashboard ran optimally on standard computer web browsers.

Conclusions

The following conclusions were derived from the above findings:

1. The web application was successfully designed to have the following features:
    - Allowed HR staff and Interviewers to navigate an easy-to-use, centralized dashboard tailored to their specific roles.
    - Provided a seamlessly responsive experience for applicants using company-dedicated desktops.
    - Delivered fast loading times for complex data processing, including real-time Chart.js analytics and CSV batch imports.

- Featured an intelligent data-cleaning architecture capable of handling unstructured data and standardizing applicant records automatically.
- Worked properly on major browsers like Chrome, Firefox, Safari, and Edge with consistent behavior.

2. The developed framework was successfully built using PHP, JavaScript (Vanilla, SweetAlert2, Chart.js), Tailwind CSS, and a MySQL Database to ensure a fast, modern, and secure environment.
3. The test results showed that the web application was fully compatible and highly usable according to the strict requirements of HR stakeholders, recruiters, and applicants.
4. The developed web application was evaluated to be Highly Acceptable in all software quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Maintainability, and Portability.

Recommendations

The following are suggestions for further enhancement of the developed web application:

1. **Improve Interview/Notification System:** Integration with calendar applications (such as Google Calendar or Microsoft Outlook APIs). This would allow the system to notify the interviewer via email instead of only sending a notification on the Web app, which they still need to log in to see.
2. **AI-Powered Resume Parsing with Resume Upload Capability:** Integration of an OCR (Optical Character Recognition) feature that automatically scans uploaded PDF resumes to pre-fill the applicant web form, saving candidates time during application.

3. **Integration of an Assessment Module (Paperless Initiative):** To further support the organization’s sustainability and paperless initiatives, future iterations of the system could include written exams as part of the R-AMS tier. Instead of traditional physical paper-and-pen tests, applicants could take timed digital assessments through a dedicated web portal. The system would be programmed to automatically grade these examinations and instantly map the results directly to the candidate's profile on the HR Dashboard.

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Appendix A

USER MANUAL

HR Personnel (HR Manager & HR Staff)

1. On the web browser, type the local intranet address (URL) of the Recruitment and Applicant Management System (R-AMS).
2. Input your Employee ID and password to securely log in.
3. The system will automatically authenticate the user role and redirect it to the main HR Dashboard.
4. Browse, view, or manage the following recruitment features:
   - **Applicant Records:** View, edit, archive, or restore individual applicant profiles.
   - **Smart Filters & Table:** Filter applicants by specific statuses, positions, or recruiters, and customize which columns are displayed.
   - **Requisition Management:** Create new job orders, update statuses (Open, Hold, Closed), and monitor approved headcounts.
   - **Bulk Upload:** Import multiple applicant records simultaneously using a standardized CSV file.
   - **Analytics Dashboard:** View real-time charts (Applicant Trend, Deployment Trend, Resource Efficiency) and generate custom visual reports.
   - **Data Extraction:** Map specific headers and export filtered applicant data to CSV formats.
   - **System Logs:** Review the audit trail of recent user activities and data changes.

5. After completing the required tasks, click "Logout" to end the session securely.

Interviewers (Department Managers)

1. On the web browser, type the local intranet address (URL) of the R-AMS.
2. Input your Employee ID and password to log in.
3. The system will recognize your specific role and redirect you to the restricted Interviewer Dashboard.
4. View your pending and upcoming assigned interview schedules.
5. Click on an assigned applicant to securely review their pre-screening score, profile details, and uploaded resume.
6. Input your evaluation details, including specific interview feedback, and a 1-5 star competency rating.
7. Submit the evaluation and update the applicant's final decision status (e.g., Passed, Failed, Rescheduled).
8. After completing your evaluations, click "Logout" to end the session.

Applicants

1. On the company workstation, type the URL of the R-AMS Applicant Web Form in the browser.
2. Fill out the required information, including Personal Details, Contact Information, Educational Background, and Specific Skills.
3. Check the mandatory consent box acknowledging the background verification terms.
4. Review the entered information carefully to ensure accuracy and completeness.

5. Click "Submit Application".
6. Wait for the success prompt; the system will instantly calculate your pre-screening score and securely route your information to the HR department.

Appendix B

COMPATIBILITY TEST RESULTS

Desktop View (HR/Interviewer Login form)

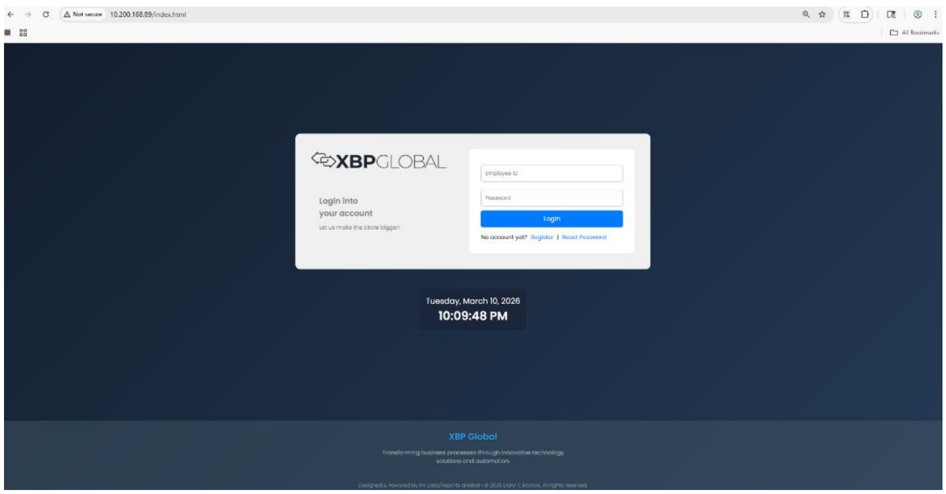


Figure 51: *Google Chrome*

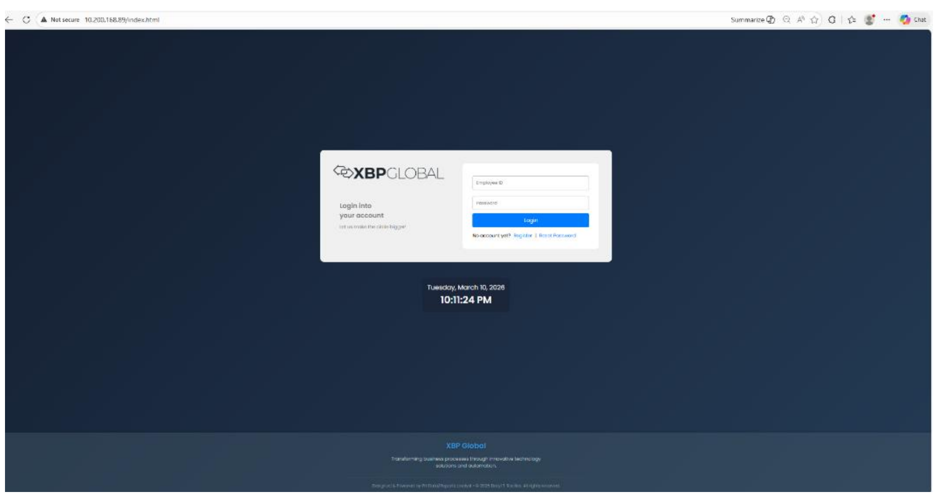


Figure 52: *Microsoft Edge*

Desktop View (HR Dashboard)

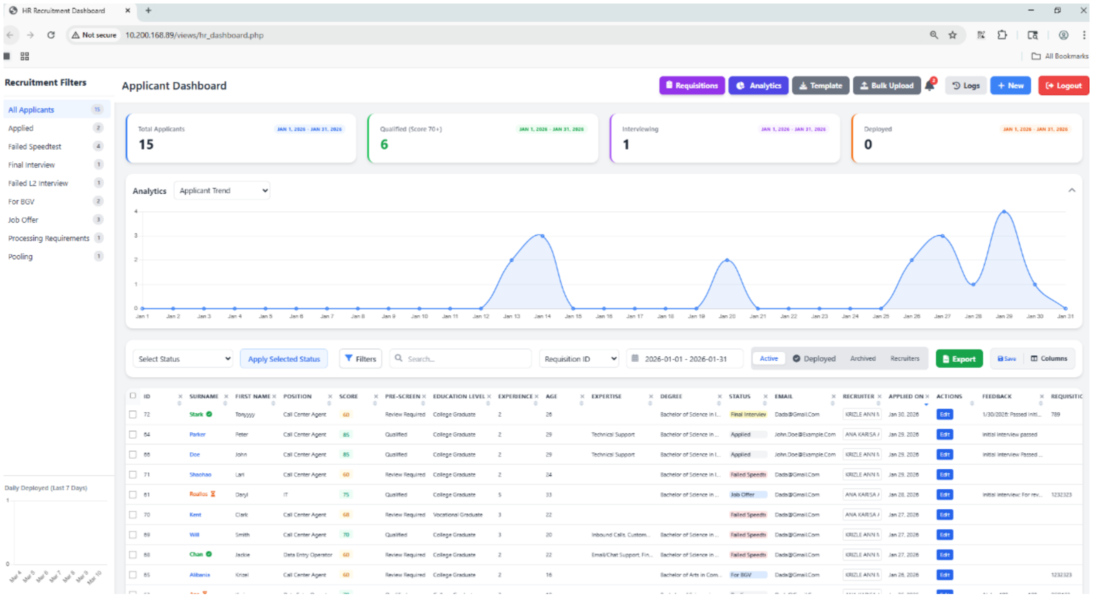


Figure 53: *Google Chrome*

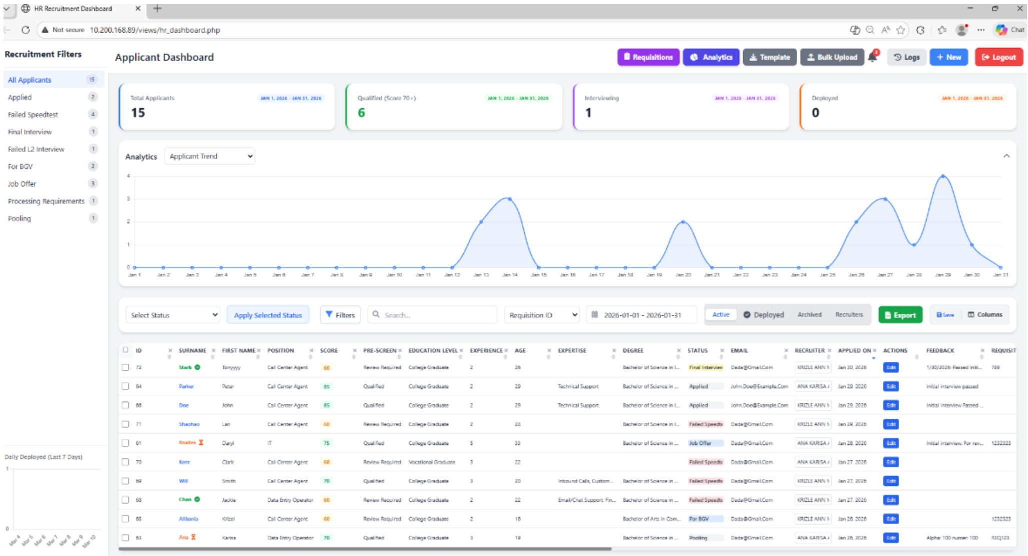


Figure 54: *Microsoft Edge*

Desktop View (Interviewer Dashboard)

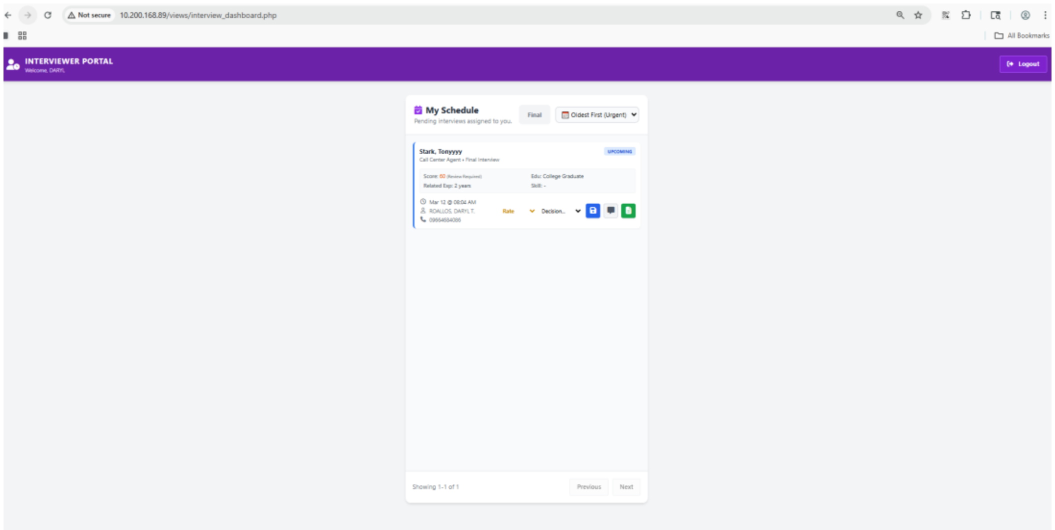


Figure 55: *Google Chrome*

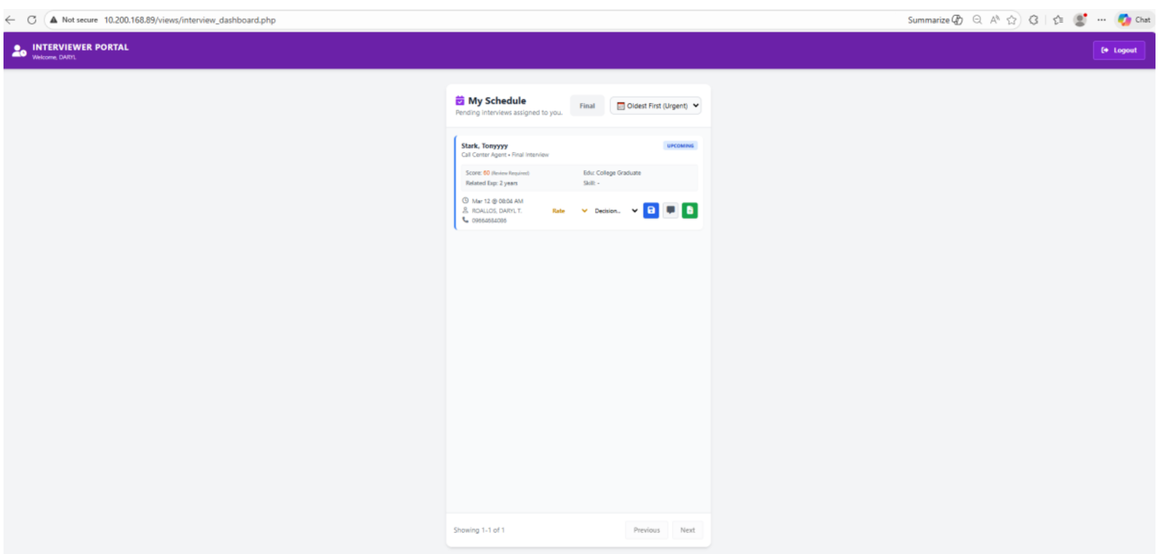


Figure 56: *Microsoft Edge*

Desktop View (Applicant Userform)

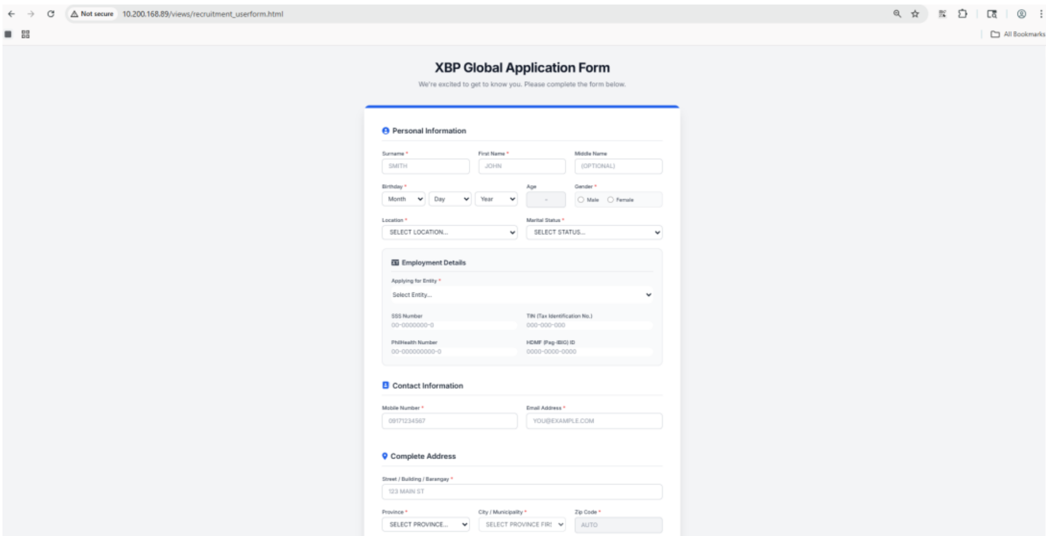


Figure 57: *Google Chrome*

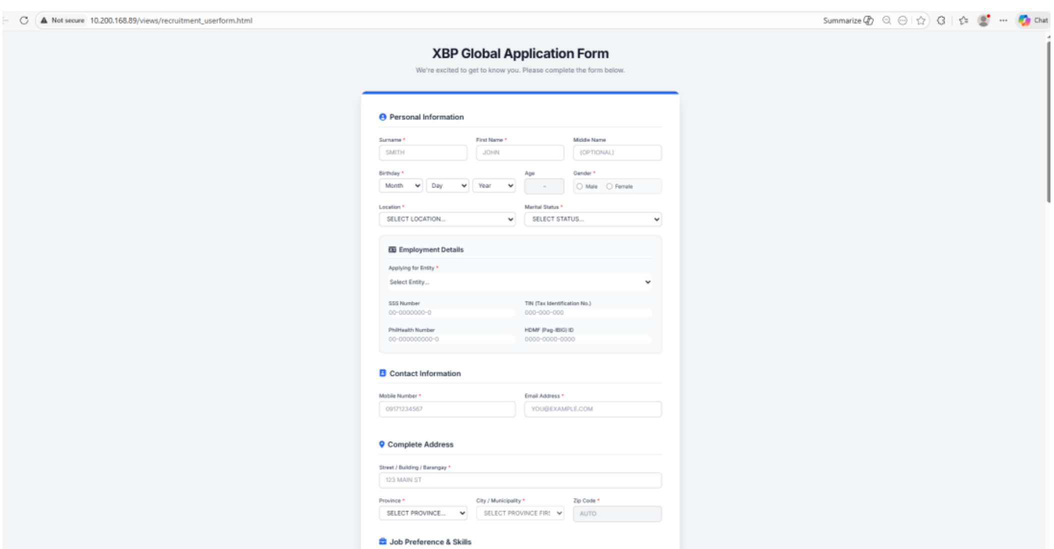


Figure 58: *Microsoft Edge*

Appendix C

SAMPLE EVALUATION SHEET

Technological University of the Philippines

College of Science

Computer Studies Department

Name (Optional): ______________________________________**Occupation:**________________**Date:**_________

Direction: Please encircle the appropriate number of your rating to evaluate the project entitled "DEVELOPMENT OF TRANSCRIPTIONISTA (TRANSCRIBE, DATA CAPTURE OPTION TO TEXT APP)" using the below:

4-Highly Acceptable 3-Very Acceptable 2-Acceptable 1-Not Acceptable

Criteria	Rating			
A. Functionality Suitability				
1. Functional Completeness. The Application functions complete all the established objectives	4	3	2	1
2. Functional Correctness. The Application delivers correct results	4	3	2	1
3. Functional Appropriateness. The Application functions allow the accomplishment of the established time.	4	3	2	1
B. Performance Efficiency				
1. Time Behavior. The Application processes the task and response according to the established time.	4	3	2	1
2. Resource Utilization. The Application utilizes its resources efficiently.	4	3	2	1
C. Compatibility				
1. Cross-Browser Co-existence. The Application operates across multiple browsers (Chrome, Firefox, etc.) without performance degradation or resource conflicts.	4	3	2	1
2. Web Interoperability. The Application operates across multiple browsers (Chrome, Firefox, etc.) without performance degradation or resource conflicts.	4	3	2	1
D. Usability				
1. Appropriateness Recognizability. The Application can be recognized as appropriate for the users' needs.	4	3	2	1
2. Learnability. The effective and efficient usage of the Application is easy to learn.	4	3	2	1
3. Operability. The Application has features that make it easy to operate and control.	4	3	2	1

4. User Error Protection. The Application protects the user from making errors.	4	3	2	1
5. User Interface Aesthetics. The Application interface allows good and satisfying interactions with the users.	4	3	2	1
6. Accessibility. The Application can be used by anyone easily.	4	3	2	1
D. Reliability				
1. Availability. The Application design makes it easy to operate whenever required during its use.	4	3	2	1
2. Fault Tolerance. The Application can resume working in the event of an interruption or a failure.	4	3	2	1
F. Maintainability				
1. Analyzability. The Application allows easy diagnosis of deficiencies or causes of failures.	4	3	2	1
2. Modifiability. The Application can be effectively modified without creating defects.	4	3	2	1
3. Testability. The Application is structured to allow for effective testing (manual or automated) across different browsers to verify that all functional requirements are met.	4	3	2	1
G. Portability				
1. Installability. The Application can be installed easily.	4	3	2	1

Comments/Suggestions:

___

___

___

Appendix D

SURVEY QUESTIONNAIRE AND RESULTS SHEETS

R-AMS

Recruitment and Applicant Management System (R-AMS)

Good day!

I am Daryl Roallos, a BS Information Systems graduate and currently an ETEEAP student enrolled in the Bachelor of Science in computer studies program at the Technological University of the Philippines – Manila.

I would like to respectfully request a brief portion of your time to participate in the evaluation of my thesis project entitled **"Recruitment and Applicant Management System (R-AMS)"**

This study aims to develop and present the R-AMS (Recruitment and Management System) designed to improve and streamline the recruitment and human resource management processes. A detailed explanation of the system, including its features and functionalities, is provided in the video in the link below.

Drive Link:
https://drive.google.com/file/d/1int_yLC9yjGFSTYIRigRA8lsQwvdnRsr/view?usp=drivesdk

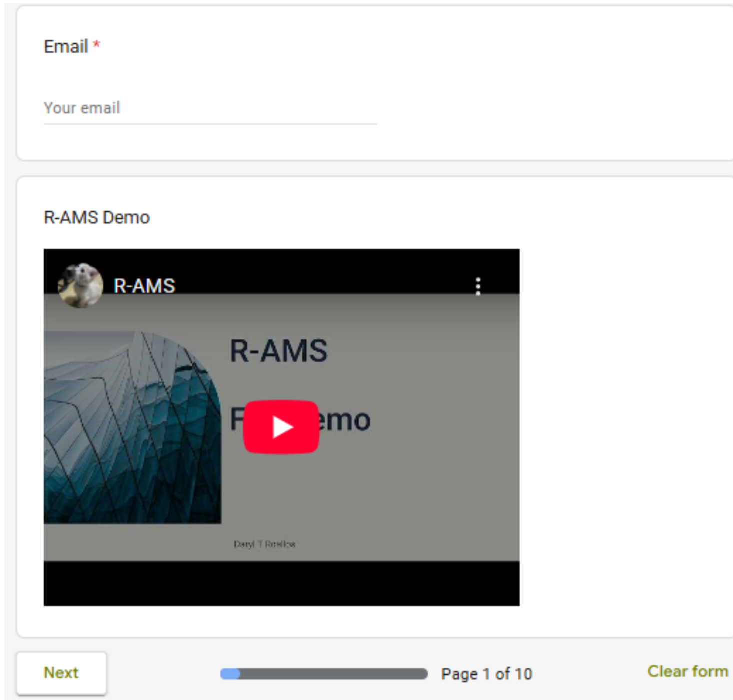
All data collected through this form will be used solely for research and results analysis purposes. The information provided will be handled in accordance with Republic Act No. 10173, also known as the Data Privacy Act of 2012.
Should you have any comments, suggestions, or questions, you may include them at the end of the survey form, or you may contact me directly through this email as the researcher of the study: relient2195@gmail.com

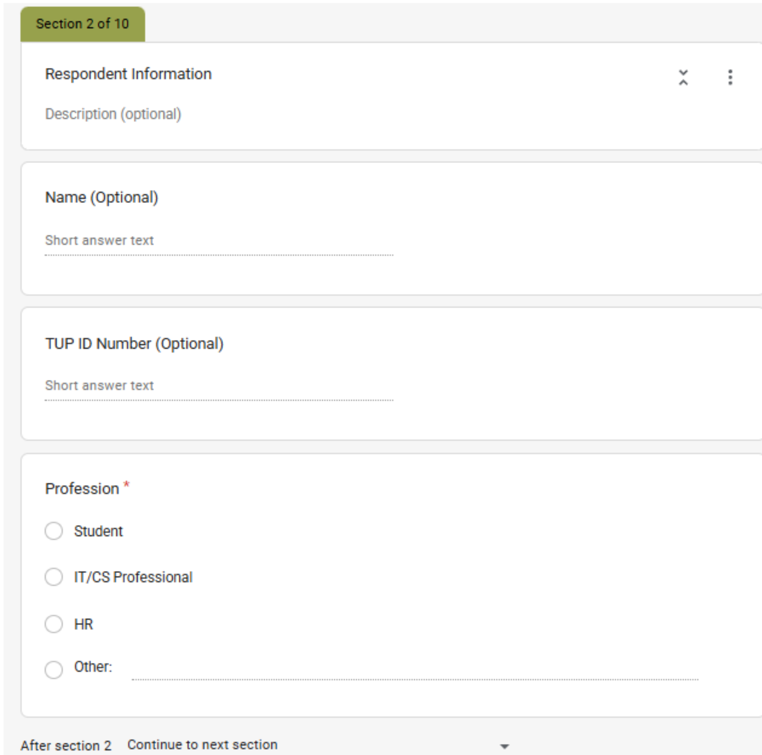
Thank you for your cooperation, have a nice day!

Very truly yours

Daryl T. Roallos

* Indicates required question


Email *
Your email
R-AMS Demo
R-AMS
R-AMS
Next
Page 1 of 10
Clear form


Section 2 of 10
Respondent Information
Description (optional)
Name (Optional)
Short answer text
TUP ID Number (Optional)
Short answer text
Profession *
Student
IT/CS Professional
HR
Other:
After section 2
Continue to next section

Section 3 of 10

Functional Suitability

This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.

Functional Completeness *

The software's functionalities cover all specified tasks and user objectives

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Functional Correctness *

The software provides results that are accurate based on the industry standard applications and mathematics.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Functional Appropriateness *

The software's functionalities can be used for accomplishment of specified tasks and objectives.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Section 4 of 10

Performance Efficiency

This characteristic represents the performance relative to the amount of resources used under stated conditions.

Time Behavior *

Time taken for the software to process data and give responses to its users.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Resource Utilization *

Software utilization and management of resources when performing its functions.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Capacity *

Maximum limitations of the software are enough to perform its functions.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Section 5 of 10

Compatibility

Degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions while sharing the same hardware or software environment.

Co-existence *

System can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Interoperability *

Two or more systems can exchange information and use the information that has been exchanged.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Section 6 of 10

Usability

Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.

Appropriateness recognizability *

Degree to which users can recognize whether a product or system is appropriate for their needs.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Learnability *

Degree to which the user can learn to operate the software with efficiency and effectiveness.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Operability *

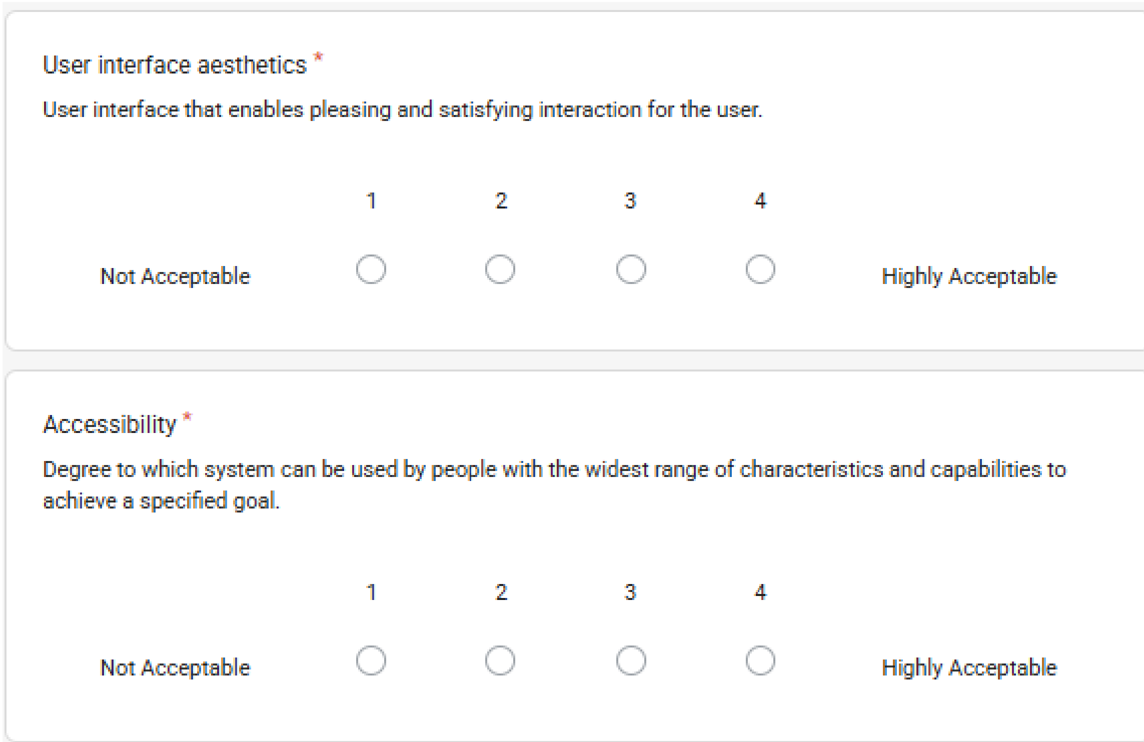
Software design that affects the easiness of operating the system.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

User error protection *

System that protects users against making errors.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable



User interface aesthetics *

User interface that enables pleasing and satisfying interaction for the user.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Accessibility *

Degree to which system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Section 7 of 10

Reliability

Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.

Availability *

Software design that affects the easiness of operating the system.

	1	2	3	4	
Not Acceptable	◯	◯	◯	◯	Highly Acceptable

Fault Tolerance *

The system operates as intended despite the presence of hardware or software faults.

	1	2	3	4	
Not Acceptable	◯	◯	◯	◯	Highly Acceptable

Section 8 of 10

Maintainability

This characteristic represents the degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment, and in requirements.

Reusability *

The software can be used to generate results that can be reused for other projects.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Modifiability *

The software can be modified efficiently without introducing defects or degrading existing product quality.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Testability *

Software's effectiveness and efficiency can easily be tested through different test criteria.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Section 9 of 10

Portability

This characteristic represents the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.

Adaptability *

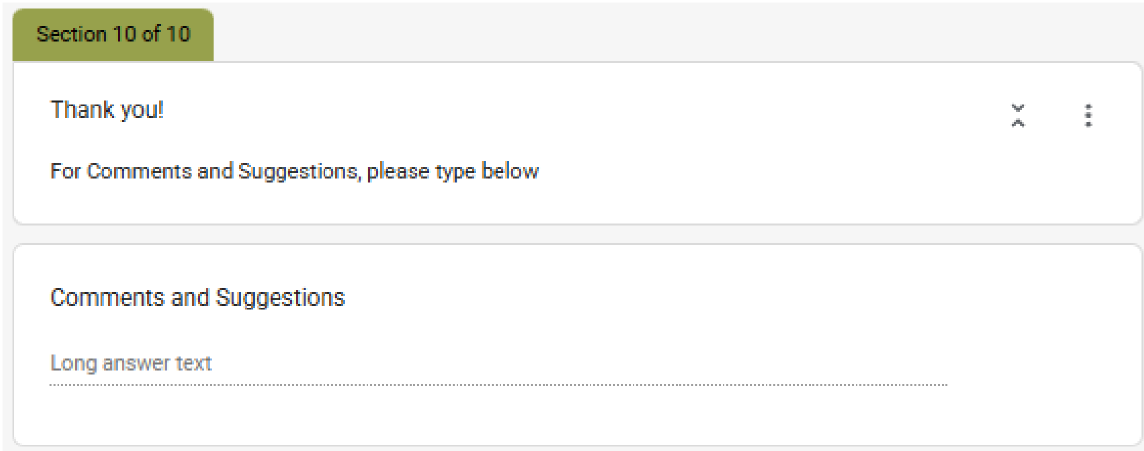
The application can be installed easily.

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

Installability *

The system can run on different operating environments

	1	2	3	4	
Not Acceptable	○	○	○	○	Highly Acceptable

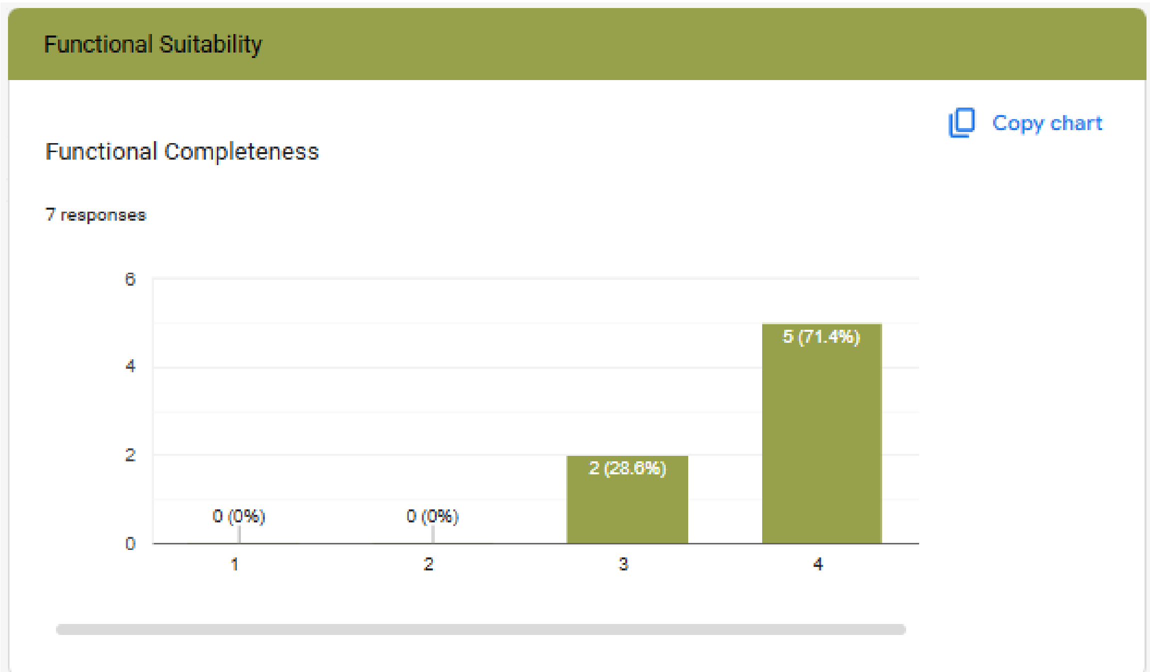

Section 10 of 10

Thank you!

For Comments and Suggestions, please type below

Comments and Suggestions

Long answer text


Functional Suitability
Functional Completeness
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7 responses
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4
2
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5 (71.4%)
2 (28.6%)
0 (0%)
0 (0%)
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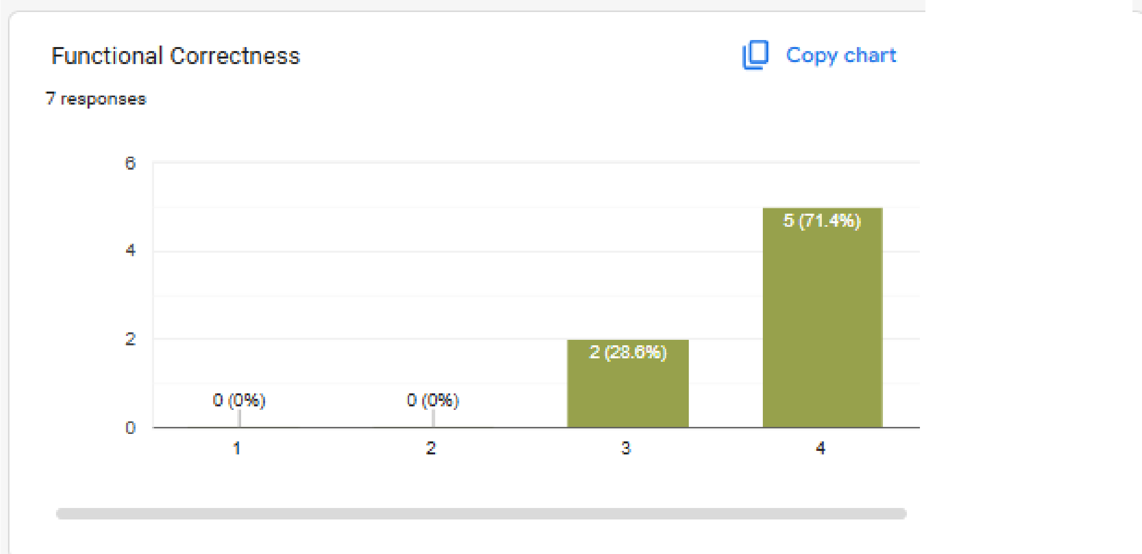
Functional Correctness
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0 (0%)
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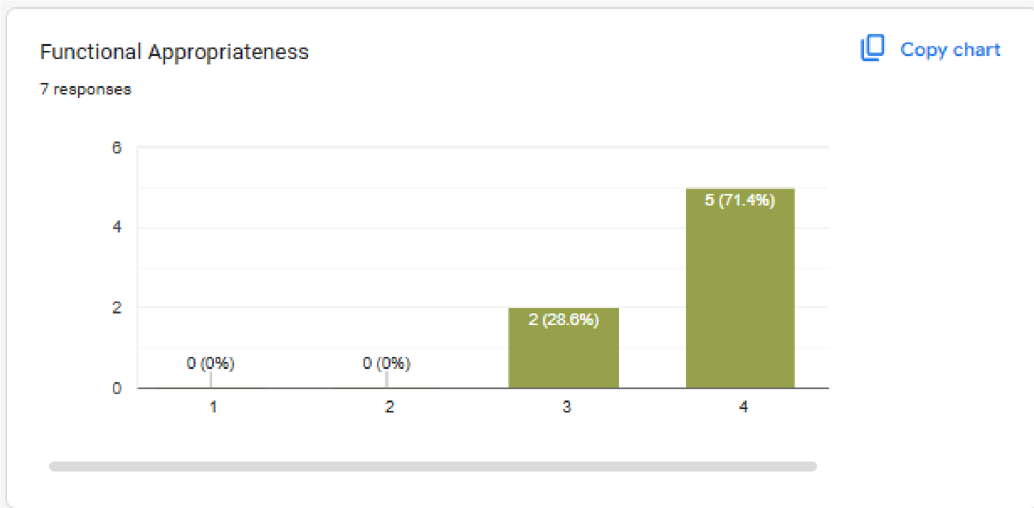
Functional Appropriateness
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5 (71.4%)
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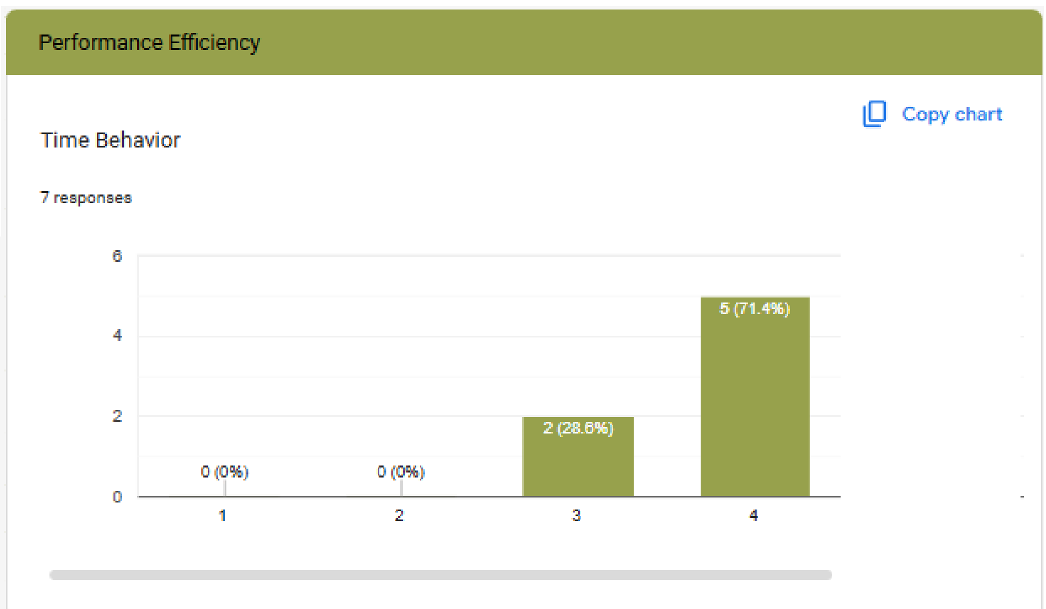
Performance Efficiency
Time Behavior
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4
2
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2 (28.6%)
5 (71.4%)
1
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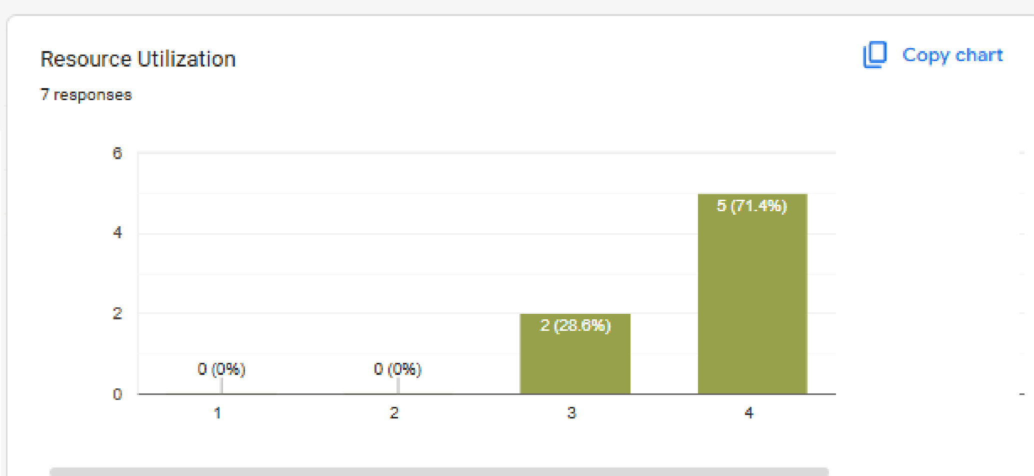
4


Resource Utilization
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7 responses
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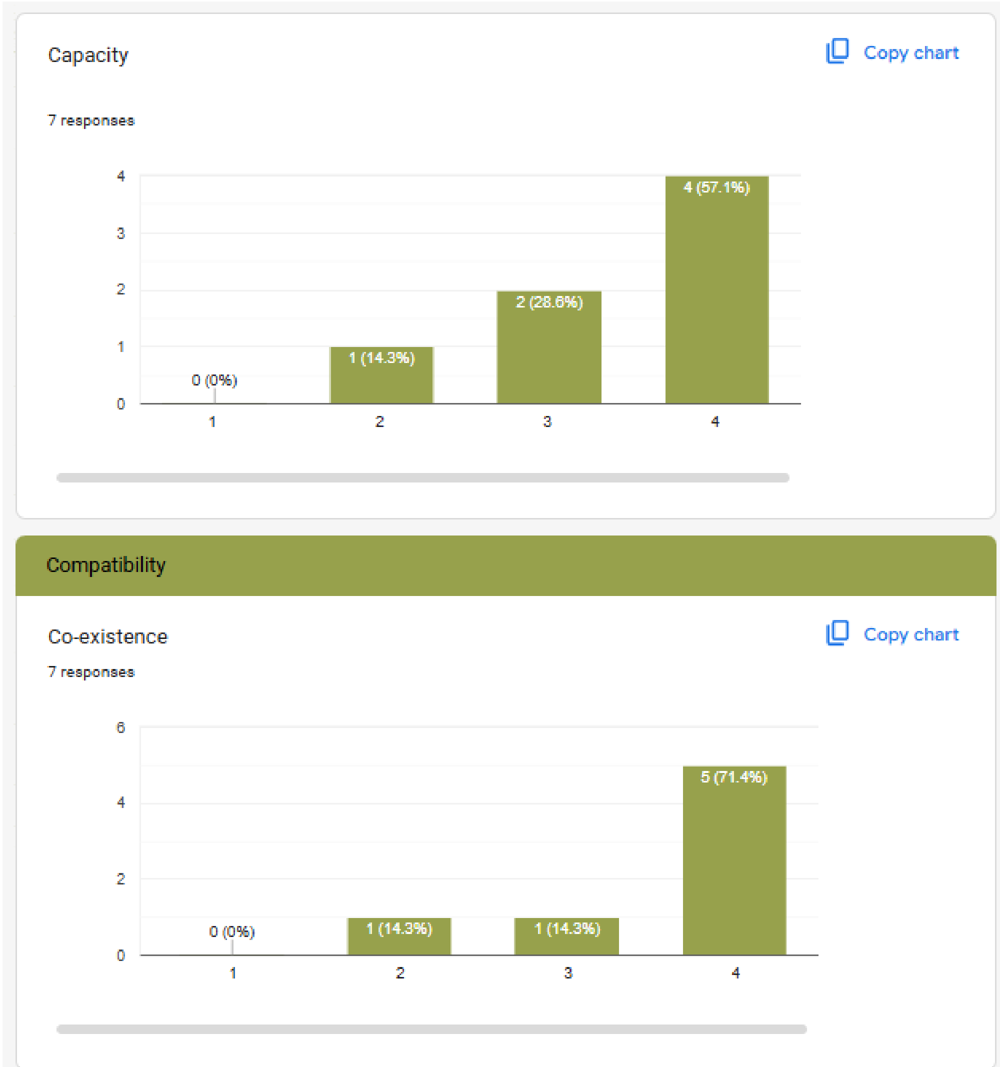
0 (0%)
2 (28.6%)
5 (71.4%)
1
2
3
4


Capacity
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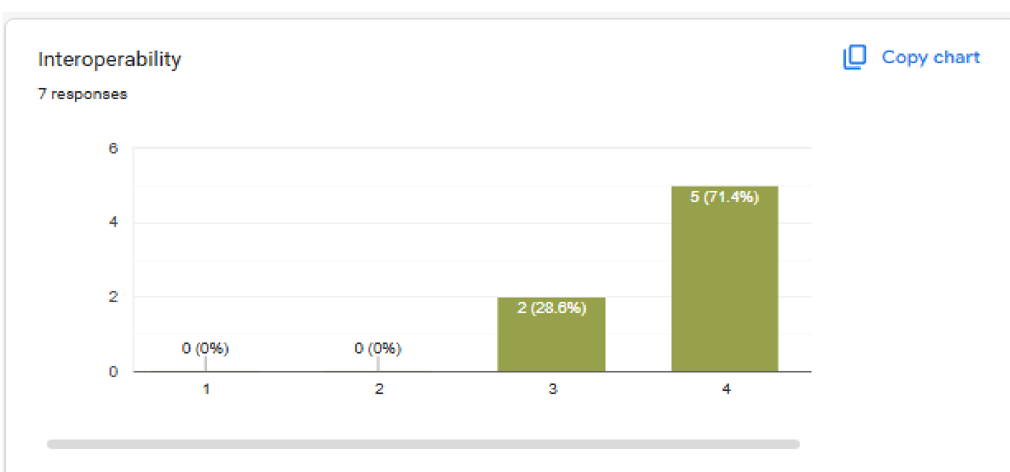
7 responses
4
3
2
1
0
4 (57.1%)
2 (28.6%)
1 (14.3%)
0 (0%)
1
2
3
4
Compatibility
Co-existence
Copy chart
7 responses
6
4
2
0
5 (71.4%)
0 (0%)
1 (14.3%)
1 (14.3%)
1
2
3
4


Interoperability
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6
4
2
0
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2 (28.6%)
0 (0%)
0 (0%)
1
2
3
4


Usability
Appropriateness recognizability
7 responses
Copy chart
6
4
2
0
0 (0%)
1 (14.3%)
1 (14.3%)
5 (71.4%)
1
2
3
4
Learnability
7 responses
Copy chart
6
4
2
0
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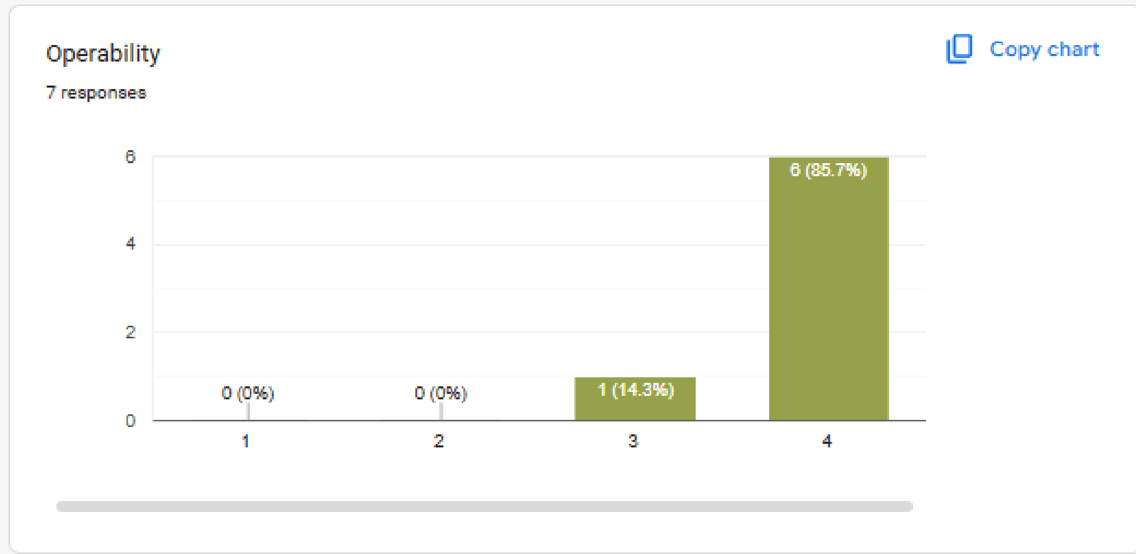
0 (0%)
2 (28.6%)
5 (71.4%)
1
2
3
4


Operability
7 responses
Copy chart
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4
2
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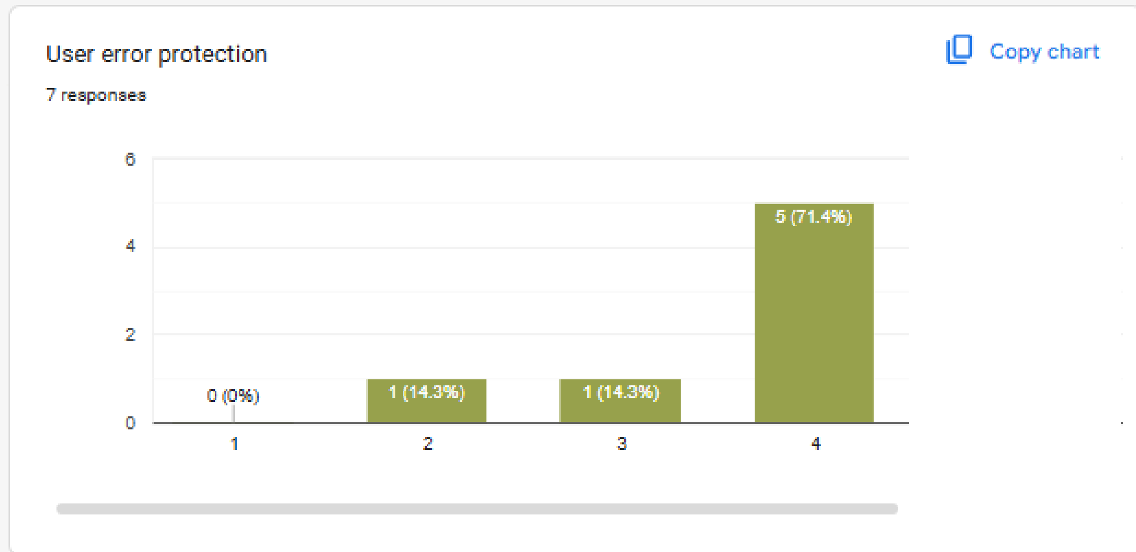
1 (14.3%)
6 (85.7%)
1
2
3
4


User error protection
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4
2
0
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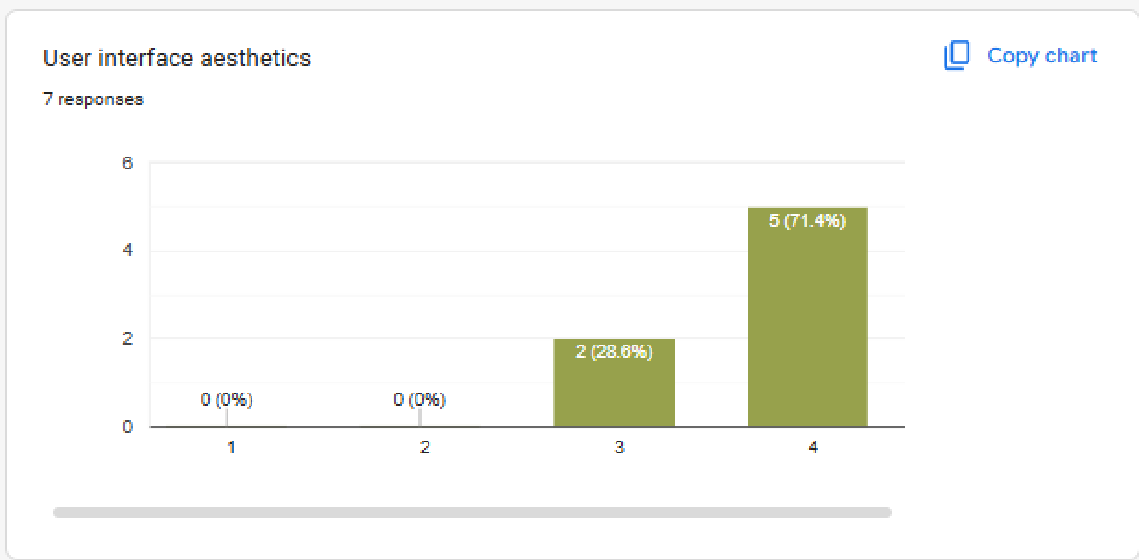
1 (14.3%)
5 (71.4%)
1
2
3
4


User interface aesthetics
7 responses
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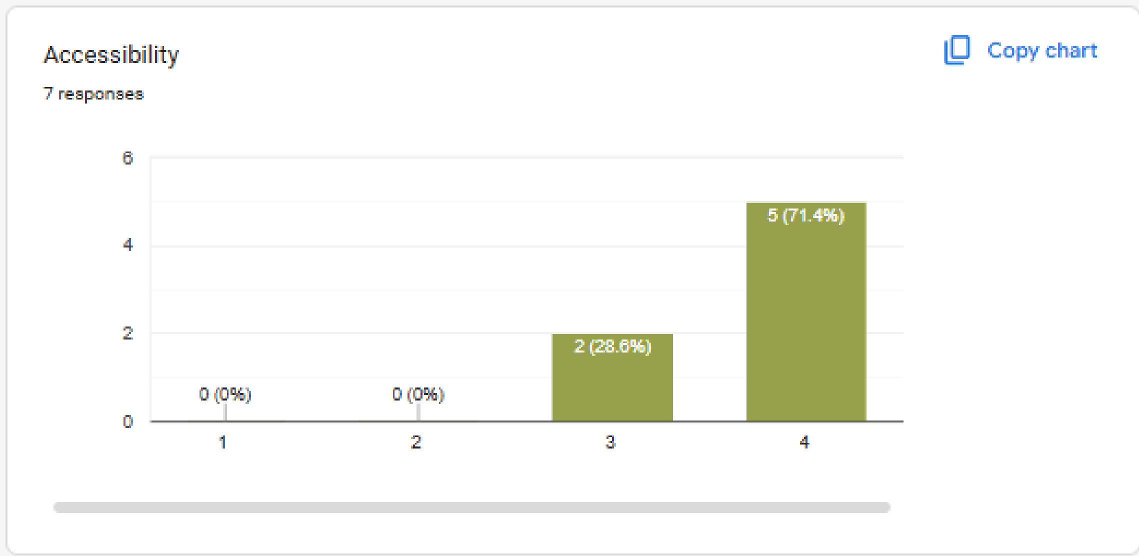
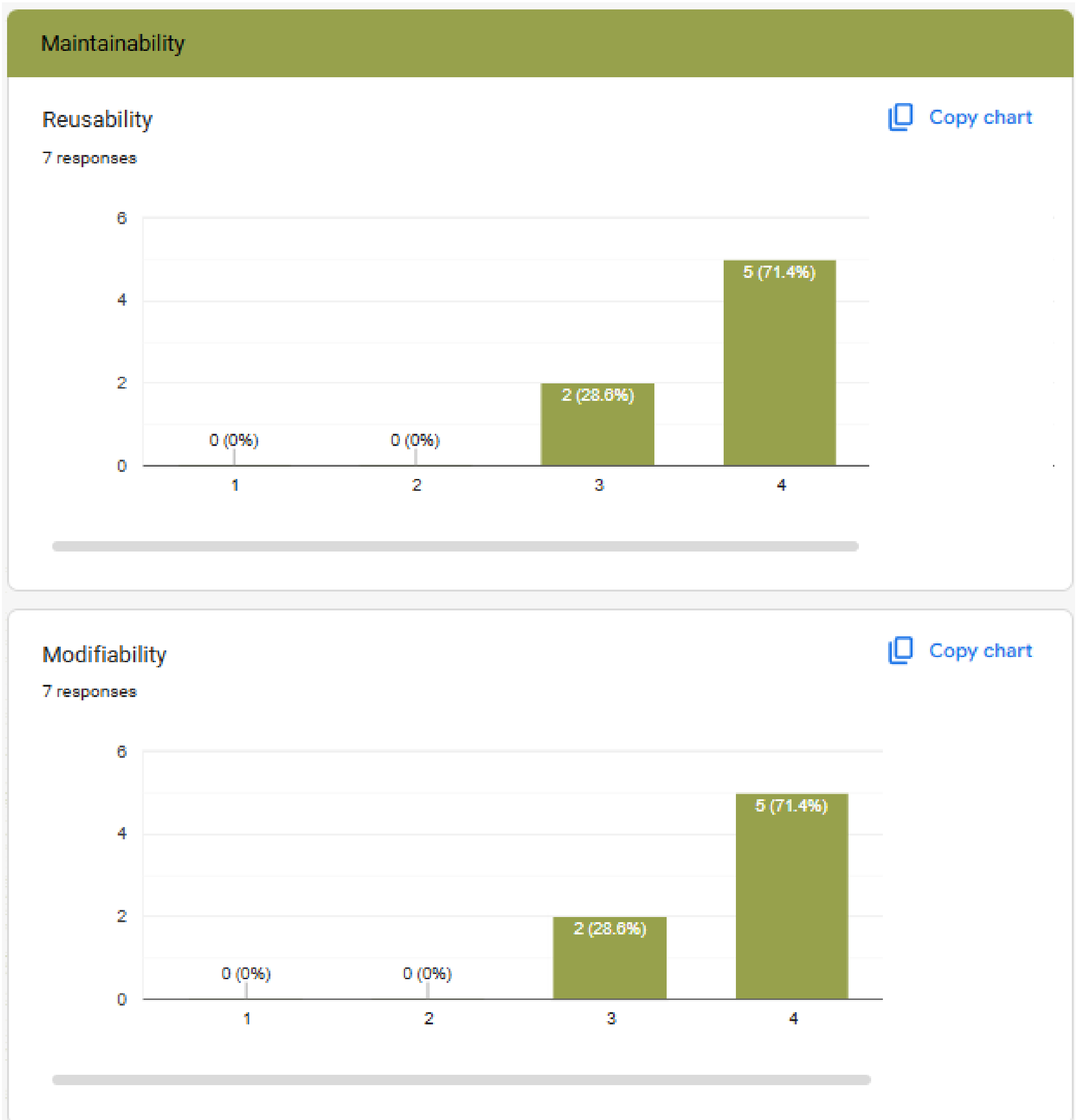
3
4


Accessibility
7 responses
Copy chart
6
4
2
0
0 (0%)
0 (0%)
2 (28.6%)
5 (71.4%)
1
2
3
4


Reliability
Availability
Copy chart
7 responses
4
3
2
1
0
0 (0%)
0 (0%)
3 (42.9%)
4 (57.1%)
1
2
3
4
Fault Tolerance
Copy chart
7 responses
4
3
2
1
0
0 (0%)
1 (14.3%)
2 (28.6%)
4 (57.1%)
1
2
3
4


Maintainability
Reusability
7 responses
Copy chart
0 (0%)
0 (0%)
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5 (71.4%)
Modifiability
7 responses
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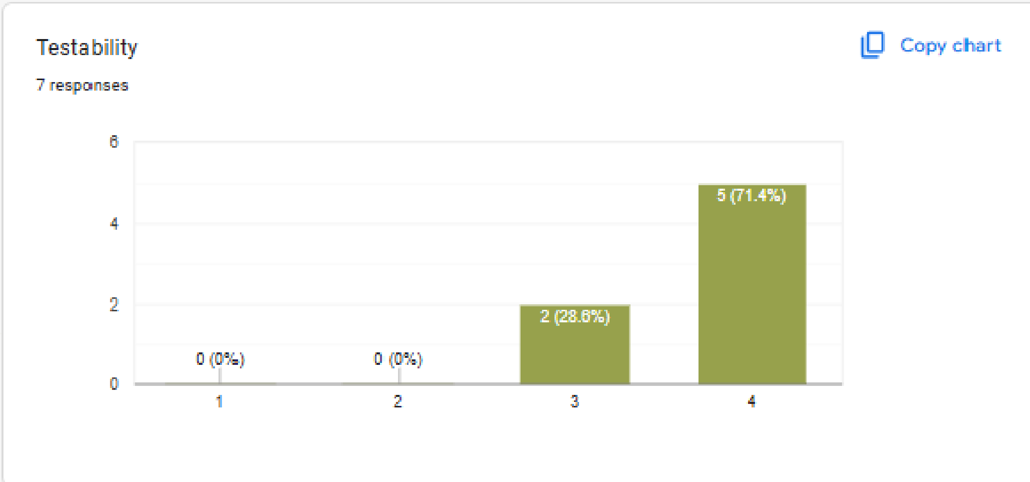
2 (28.6%)
5 (71.4%)


Testability
7 responses
Copy chart
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2 (28.6%)
5 (71.4%)
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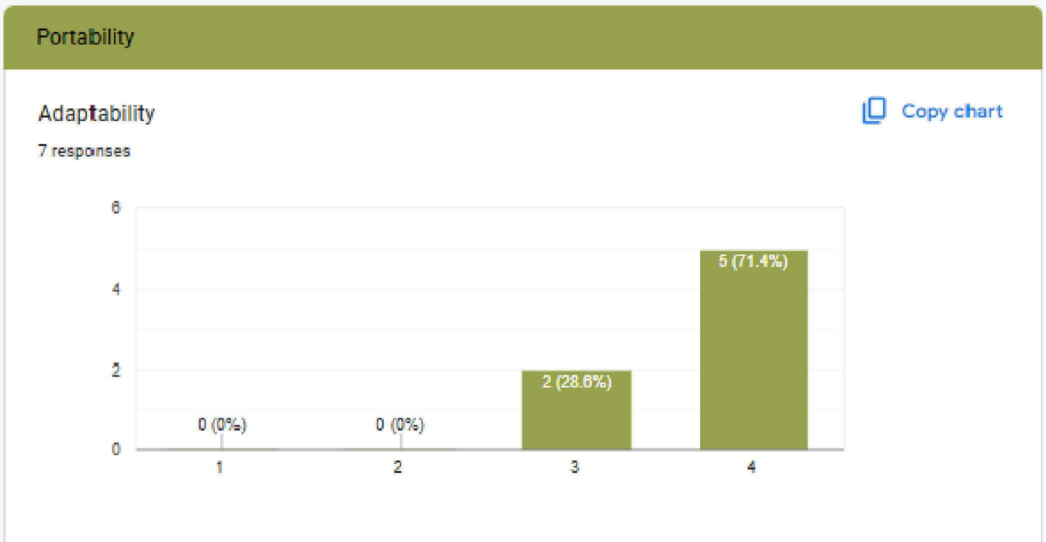
Portability
Adaptability
7 responses
Copy chart
6
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5 (71.4%)
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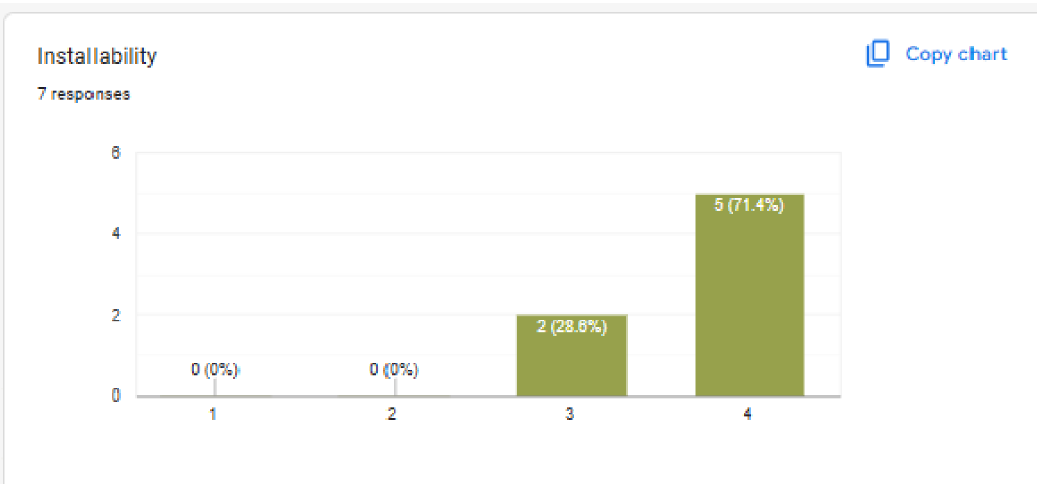
Installability
7 responses
Copy chart
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Thank you!

Comments and Suggestions

4 responses

Summarize responses

Good for roll-out

What I like most about this Recruitment and Applicant System-- it makes finding candidates very easy because the database is searchable and well organized... the checklist of requirements will be very helpful for recruiters and not something I have seen in other systems... It allows the recruiters to track the candidates easily.. I also like that it sends interview reminders for recruiters, so they dont have to manage scheduling manually in a calendar.. overall, these features save a lot of time, reduce mistakes, and make the recruitment process much smoother and more efficient...

The system interface is clean and smooth, and all the functions are working properly in a timely manner. additionally uploading data and files is straightforward and efficient.

Daryl's project really helps us a lot in Recruitment. It help us save time and maximize the effort of my team members in recruitment.

USER PROFILE

DARYL T. ROALLOS

Objective: A motivated individual seeking to engage in a career that will allow for progress in terms of expertise, socio-economic development, and innovation through exposure to new ideas for professional growth, as well as the growth of the company.

Contacts:

12 Murphy St. Pag-asa, Olongapo City, Zambales
09664684086
relient2195@gmail.com

Skills

- P&L and PLM workings
- Project costing allocation
- Ability to create PLM dashboard/trends to help management to come up with a decision
- Ability to create a productivity and bonus performance dashboard for employees
- Develop Decision making and problem-solving skills
- Knowledge in using Microsoft Office, especially MS Excel, VBA, Macros, SQL and MS Access
- Proven initiative to work on minimal supervision and has multitasking capabilities
- Knowledgeable with VB.net application programming (web automation & desktop application)
- Time management and good communication skills
- Excellent analytical skills and keen to details
- Enthusiastic, hardworking, and eager to learn

Education and Achievement

Columban College, Inc – Olongapo City

- Graduated at College of Engineering – Industrial Technology Center
- 2008 - 2010

Character Reference

- Richard Paul Azarcon ()919 990 2507)
  Sr. FP&A Manager

PERSONAL INFORMATION

- Age : 34 years old
- Gender : Male
- Date of Birth : February 2, 1992
- Place of Birth : Olongapo City
- Civil Status : Single
- Citizenship : Filipino
- Religion : Catholic

WORK EXPERIENCE

FP & A

Scorp Phils / October 2020 – Present

- Handled Local P&L and PLM cost allocation – about 30 projects being handled with about 700 employees per Cost Center
- Handled India site PLM workings, Costing allocation per project with about 1.8k employees.
- Supporting Upper Management data request and requirement for them to come up with decision making.
- Experienced in providing analysis and recommendation to help the Team to reduce the financial cost
- Effectively perform and review detailed procedures in both low risk and high risk areas.
- Experienced creating financial dashboard to visualize cost trending.

Reports/Data Analyst

Scorp Phils / April 2017 – October 2020

- Track Employees productivity performance in a day-to-day basis.
- Publishing Productivity dashboard to help their Team leaders and Managers assess individual performance.
- Ability to review and derive employee Task goals which will be fit to every user.
- Data extraction and Maintaining databases (MS Access)
- Ability to create data automation (automated web extraction - .Net/Macros)
- Creating Bonus performance computation and dashboard
- Supporting management for any data request such as historical data to track and assess previous performance

Quality Analyst

Scorp Phils / September 2014 – April 2017

- Set Dashboard to visually track, analyzes and display (KPI), other metrics and data points to monitor the business process.
- Handling Project Quality Status: Provide Root Cause Analysis and set measurable action plans.
- Knowledge in doing Audits to track if everyone is on the same page and ensure they are prioritizing SLA.
- Setting up Quality calls to collaborate with other sites. Provide feedback and set action plans.
- Provide Statistical trends & charts to visually monitor the positive and negative factors that affect the business.
  •Develop test cases and plans that will help the team to be more efficient.

Data Entry / SME

Scorp Phils / August 2010 – September 2014

- Support and Audit employees' process document
- Verify data by comparing it to source documents